Climate Vulnerability Assessment 2.0

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1 Introduction

1.1 History of the Climate Vulnerability Assessment

The Climate Vulnerability Assessment (CVA) was originally performed by the Northeast Fisheries Science Center in 2015. This original work, which was later replicated by all the National Marine Fisheries Service (NMFS) regions, used expert opinions to rank how target species would be impacted by future climate change using both exposure and sensitivity attributes. For the original CVA in the northeast US, experts ranked how much change species would be exposed to using large glocal ocean model forecasts and how sensitive they would be to these changes. This resulted in a single exposure and sensitivity score for each species, which was combined to give an overall vulnerability score. Experts were also asked to rank the directionality of climate impacts. The original methods are described in Morrison et al 2015 and Hare et al 2016, and the original results for the northeast shelf can be found in Hare et al 2016. CVA results for all regions can be explored using the NOAA CVA Tool.

1.2 Why CVA2.0?

The original CVA produced single metrics for each species across their entire range. However, in the past decade, it has become clear that climate change is not uniform worldwide. Therefore, species or stocks may be more or less vulnerable in different portions of their range. As a result, there is a need for a spatially explicit CVA. In addition, the development of high resolution regional ocean models such as the MOM6 model also allow changes to be predicted on fine spatial and temporal scales. The availability of the MOM6 hindcast and forecast on both decadal and long-term scales, plus new information on life history attributes for species, will allow us to update the existing CVA.

1.3 How to Use this Book

The purpose of this book is to document the methods and capture the decisions made in the development of the CVA2.0 for the Northeast Shelf, so that these methods could be broadly applied to other NMFS regions or management bodies. This book will also outline the R package that is actively in development to assist in reproducability across regions. Automatic

reporting methods and any future RShiny application development to assist in gathering expert opinons will also be outlined here.

Part I

Methods Overview

Overview

The CVA2.0 will take a semi-quantiative approach, using both quantitative and qualitative methods.

Quantitative Methods

Species distributions will be quantified using ensemble species distribution models, forced by environmental covariates from the MOM6 hindcast (1993 - 2023) and driven by fisheries dependent and independent datasets. These species distribution models will then be projected into the future using the MOM6 decadal and long-term forecasts to quantify changes in species distribution and their drivers.

Qualitative Methods

Species sensitivity will be estimated following similar methods to the original CVA. We will perform a literature review to update our understanding of important life history attributes and solict expert feedback on the most important life history attributes.

R Package

All analysis was conducted in R. An R package is under active development to ensure methods can be replicated. Vignettes will be available to describe the functions. Workflows are also being documented and will be provided in their own GitHub repository.

2 Data

2.1 Fisheries Data

We utilized two different data types for this work - fisheries independent and dependent data.

2.1.1 Fisheries Independent Data

Fisheries independent data comes from sources outside of the fishing industry. These often come in the form of regular surveys run by federal or state-level management bodies. Traditionally, these surveys occur in the spring and fall. For the CVA2.0 in the Northeast, we used the following surveys:

- Northeast Fisheries Science Center (NEFSC) Bottom Trawl
- NEFSC Northern Shrimp Survey
- NEFSC Gulf of Maine Bottom Long Line Survey
- NEFSC Atlantic Surfclam and Ocean Quahog Survey
- Northeast Area Monitoring and Assessment Program Nearshore Trawl
- Maine-New Hampshire Inshore Trawl Survey
- Massachusetts Division of Marine Fisheries Bottom Trawl Survey
- Long Island Sound Trawl Survey
- New York Nearshore Trawl Survey
- New Jersey Ocean Stock Assessment Survey
- Delaware Trawl Surveys

Additional daily positions from tag data were provided by the Highly Migratory Species (HMS) group for smooth dogfish.

2.1.2 Fisheries Dependent Data

Fisheries dependent data is data collected by commercial and recreational fishers. In the United States, commercial fisheries data is collected by fisheries observers in the National Observer Program, Gillnet Observer Program, and Pelagic Observer Program. These are trained individuals who accompany fishing boats on their trips to collect data on the species caught. Data from recreational fisheries is collected using post-effort in person or phone surveys

at different docks via the Southeast Fisheries Science Center's Logbook and the National Marine Fisheries Services' Large Pelagics Survey programs.

2.1.3 Combining Fisheries Independent and Dependent Data

Fisheries independent and dependent data are collected using different methods - they could fish for different lengths of time, at different speeds, or use different equipment. This is especially true for fisheries dependent data, which may used specialized equipment depending on the species being targeted. Therefore, we cannot compare the number of individuals caught across these methods. As a result, we will only examine whether species are present or absent and will not model the number, or abundance, of species. While different fishing methods could impact the ability to catch a particular species, thus impacting our results, we believe that this approach will account for most differences and will assume that all datasets are sampling for long enough periods of time and appropriate equipment to sample all species.

2.2 Environmental Data

We will be using species distribution modeling to quantify the relationships between species presence and their environment. We are using the Modular Ocean Model (MOM) 6 hindcast data to drive these models. The MOM6 is developed by the Changing Fisheries and Ecosystem Initiative and includes both a physical ocean model and a coupled biogeochemical model.

We are using the hindcast timeseries from 1993-2023 from the MOM6 for the Northwest Atlantic to build the species distribution models. We will then project these models into the future using the decadal projections from the MOM6.

Covariates used to build the species distribution models include:

- Sea Surface Temperature & Salinity
- Bottom Temperature & Salinity
- Surface pH
- Bottom Aragonite Solubility
- Mixed Layer Depth
- Diazotroph, small, medium, and large phytoplankton primary productivity (integrated in the top 100 m)
- Small, medium, and large zooplankton biomass (integrated in the top 100 m)
- Downward particulate organic carbon (POC) flux
- Water column net primary productivity (NPP)

Covariates will be normalized to long term (1993 - 2023) averages and matched to species based on their habitat and prey preferences. For example, the species distribution model

for summer flounder will be built using bottom temperature and salinity, bottom aragonite solubility, downward POC flux, and NPP.

3 Species Distribution Modeling

The first step in the CVA2.0 is the construction of species distribution models (SDMs). SDMs are statistical models that describe the relationships between species presence or abundance and environmental attributes or covariates. There are many different types of SDMs that use different data types and statistical methods. There is an abundance of scientific literature that both describe and compare different model types. We will provide a brief overview below and describe the methods selected for the CVA2.0.

3.1 Presence/Absence vs Abundance Data

Species distribution models can be built with primarily two different kinds of species data - presence/absence or abundance data. Presence/absence data describes when a species was present or absent, usually in binary with 0s representing absences and 1s representing presences. SDMs built with presence/absence data predict how likely it is that a species will be found there - often called habitat suitability - but does not consider how many individuals will be present.

Abundance data on the other hand uses count or species density estimates. SDMS built with abundance data can predict the abundance or density of species in a given area or at a given time, but must deal with often irregular data.

3.1.0.1 In the CVA2.0...

For the CVA2.0, we use presence/absence data from a variety of sources. Using presence/absence data will allow us to combine multiple datasets across fisheries independent and dependent surveys.

3.2 Statistical Modeling

There are many statistical models that we could have chosen for our SDMs. Options included generalized linear models (GLMs), generalized additive models (GAMs), uni (single) or multivariate state-space models, machine learning methods, decision tree methods, and more! Many models have been used to describe species distributions along the Northeast Shelf. Each model

type has pros and cons, and different models use different assumptions, statistical methods, etc to define habitat preferences.

To account for some of these differences between models, model ensembles can be generated by combining different model outputs together. Ensembles are usually built by averaging the outputs of different component models. Weighted averages can also be used to weight the averages based on model performance.

3.2.0.1 CVA2.0 Approach

We use an ensemble modeling approach for the CVA2.0, following the work of the Alaska Fisheries Science Center (AFSC)'s Groundfish program, recently implimented an ensemble modeling framework for their Essential Fish Habitat work. Here, we use the following models to build our ensemble (henceforth referred to as **component models**):

1. Generalized Additive Model (GAM) using methods in EFHSDM
2. MAXimum ENTropy (MAXENT) using methods in EFHSDM
3. Boosted Regression Trees (BRTs) using methods from Braun et al 2023
4. Spatio-temporal Generalized Linear Mixed Models (GLMMs) with sdmTMB

All of these models accept presence/absence data; can be made spatially and temporally explicit, which was important for our analysis; and combine more "traditional" SDM methods such as GAM and MAXENT included in the AFSC's ensemble with machine learning/decision tree-based models such as BRTs. sdmTMB models are similar to other SDM methods common in fisheries work like VAST and tinyVAST. These models use spatial and temporal random fields to account for spatial and random variation in species distributions that the environmental variables cannot describe. In addition, these or similar models have been used on the Northeast Shelf in previously published works.

We will generate our ensemble model by performing a weighted average on our components' outputs. Weights will be calculated using the same methods as the AFSC. However, we will use Area under the Curve (AUC) rather than root mean squared error (RMSE) because AUC is a better accuracy metric for presence/absence models than RMSE. AUC will be scaled from 0-1, rather than its native 0.5 - 1 range, using the Gini Index to help better illustrate the spread of component model performance.

4 Species Exposure

Species exposure is quantified using the forecasts from the MOM6 model. The CVA2.0 follows similar methods to the original analysis while also integrating the species distribution model results.

Exposure of each variable is calculated as a z-score using monthly averages across future (2025-2034) and present time periods (2014-2023). Global (not spatially or temporally explicit) averages and standard deviations were calculated from the present time series and used to calculate the z-score on monthly averaged future data to produce a spatially and temporally explicit exposure.

Exposure values are then averaged across time and space each month to produce an exposure map and monthly timeseries for each variable, using predicted species distributions to perform weighted averages so that only expsoure within the species ranges are considered and habitat use by the species is taken into account.

Total exposure is calculated using the same logic rule implimented in the CVA 1.0 in two different ways: 1) counting all ecologicaly relevant variables and 2) only counting variables that have relative weights over 10% in the species distribution models. This will be done across both time and space to produce both a total timeseries and map of exposure for each species.

5 Sensitivity

5.1 Overview

Species sensitivity to change will be measured qualitatively as in the original CVA. Species experts will rank the impacts of different life history attributes on species sensitivity.

5.1.1 A Note on Sensitivity vs Adaptability

The Boyce et al 2022 and 2024 papers seperated species sensitivity and adaptability attributes. In the original CVA, sensitivity and adaptive attributes were grouped together, as removing one or the other significantly changed the results. Furthermore, Morrison et al 2015 considered sensitivity and adaptivity inverses of each other (ie higher sensitivity would mean that a species is less adaptive) and decided to group them.

5.2 Sensitivity Attributes

The sensitivity attributes used in the CVA2.0 were similar to those used in the Northeast US for the original. A handful were updated to reflect new insights into the impacts of climate change on species life histories since the original analysis.

The following sensitivity attributes were used in the CVA2.0 analysis for the Northeast US. Updated attributes are **bolded**.

- **Juvenile Habitat Requirements**
- Prey Specificity
- **Relationship to Calcified Organisms**
- **Complexity in Reproductive Strategy**
- **Historical Variability in Temperature**
- **Egg and Larval Survival and Settlement Requirements**
- Stock Size/Status
- **Other Stressors**
- Population Growth Rate
- Dispersal of Early Life Stages
- Adult Mobility

- Spawning Cycle

Full text of the attributes will be made available once finalized.

5.3 Expert Scoring

Experts across NOAA, research, and industry were asked to score how species life history traits may make them sensitive to climate change, following the original methods in Morrison et al 2015. Experts were provided basic life history information relevant to the sensitivity attributes. The species profiles from the original analysis were updated and reviewed for consistency by 2 NOAA scientists. Experts were asked to score species both within and outside their field of expertise. Each species was scored by five experts, with at least two scorers per species considered experts on that species guild.

Briefly, experts assign a score based on four scoring bins (low, moderate, high, and very high) for each attribute using their expert judgement and life history information. Each expert had five tallies to distribute across the four bins to help quantify certainty. If an expert was certain about a score, they could put all five tallies in a single bin. Conversely, if they were uncertain, the tallies could be distributed across multiple bins. Certainty was quantified using a bootstrap analysis by resampling tallies with replacement and recalculating the attribute and sensitivity scores.

Experts were asked to score species independently, and then met for three virtual discussion workshops to discuss scores with other experts. They were then given the chance to finalize their scores based on these discussions. We asked experts to **not** consider the scores from the original analysis to reduce bias in the results.

6 Vulnerability & Directionality

6.0.1 Vulnerability

The final vulnerability scores will be estimated using the same methods as Hare et al 2016. Final vulnerability scores will be the product of exposure and sensitivity. Spatially and temporally explicit products will be available, as well as a total vulnerability score for the species, as well as individual scores at the stock levels as appropriate.

6.0.2 Directionality & Predicted Habitat Change

In the original CVA, in addition to ranking how sensitive species may be to future climate change, experts also estimated if species would be positively or negatively impacted by climate change. We will follow the same protocols here to estimate directionality, specifically in terms of species abundance, in the CVA2.0.

The original CVA also used a subset of expert scores to estimate the likelihood that species distributions would shift. For the CVA2.0, we will use the species distribution models to quantify how species distributions will shift using the species distribution models.

Part II

RPackage Overview

An R package is undergoing active development to document the workflows and codes used to contruct the new Climate Vulnerability Assessment.

The package will have 4 major components:

- Building the ensemble species distribution models
- Estimating exposure
- Summarizing sensitivity from expert scores
- Calculating final vulnerability scores

The following pages provide a tutorial for the different portions of the package.

7 Installation & Setup

7.1 Installation

Installation instructions will come once the R package is finalized.

7.2 Set Up

7.2.1 Necessary Files

7.2.1.1 Key Files

If you wish to subset environmental covariates used in the models when making the data frames, you will need files that 1) list the species and their habitat and feeding guilds and 2) define the covariates to keep for each feeding and habitat guild, referred to as key files.

The species list should at least include a column with the species name and then columns for the associated feeding and habitat guild. While it is not necessary, this file is also a good place to list alternative species names (common names, scientific names, any variations on either) that can be used to make sure that observations of the same species are combined across data types.

```
           Common.Name             COM_NAME           Scientific.Name
1         Atlantic cod         ATLANTIC COD               Gadus morhua
2     Atlantic croaker     ATLANTIC CROAKER   Micropogonias undulatus
3     Atlantic halibut     ATLANTIC HALIBUT Hippoglossus hippoglossus
4     Atlantic herring     ATLANTIC HERRING           Clupea harengus
5    Atlantic mackerel    ATLANTIC MACKEREL          Scomber scombrus
6 Atlantic sea scallop ATLANTIC SEA SCALLOP  Placopecten magellanicus
     Alternate.Name                  SCI_NAME            SCI_NAME_ALT
1       Cod Atlantic              GADUS MORHUA
2  CROAKER ATLANTIC   MICROPOGONIAS UNDULATUS MICROPOGONIAS UNDULATES
3  Halibut Atlantic HIPPOGLOSSUS HIPPOGLOSSUS        HALIBUT ATLANTIC
4  Herring Atlantic           CLUPEA HARENGUS
5 Mackerel Atlantic          SCOMBER SCOMBRUS       MACKEREL ATLANTIC
```

```
6       Scallop Sea  PLACOPECTEN MAGELLANICUS PLACOPECTEN MAGELANICUS
            SCI_NAME_ALT2 Managing.Body Feeding.Guild Habitat.Guild
1                                 NEFMC     Piscivore    Groundfish
2 MICROPOGONIUS UNDULATUS         ASMFC    Benthivore    Groundfish
3                                 NEFMC     Piscivore    Groundfish
4                                 NEFMC   Planktivore       Pelagic
5                                 MAFMC   Planktivore      Pelagic
6                                 NEFMC       Benthos       Benthic
```

Key files should have seperate columns, for each feeding and habitat guild with names that match the different feeding and habitat guilds in the species list. The entries in each column should be a column name associated with that guild.

```
  Planktivore Piscivore Benthos Benthivore Apex.Predator
1      diazPP  smallZoo  intNPP     intNPP        intNPP
2     smallPP mediumZoo     POC        POC
3    mediumPP  largeZoo
4     largePP
```

```
  Groundfish   Benthic   Pelagic Pelagic.Migratory
1    bottomT   bottomT  surfaceT          surfaceT
2    bottomS   bottomS  surfaceS          surfaceS
3   bottomO2  bottomO2 surfacepH         surfacepH
4  bottomArg bottomArg       MLD               MLD
```

It is **critical** that the column names in the key files and the different habitat/feeding guild names in the species key are identical (remember: 'Pelagic' and 'Pelagic' are technically two different character strings in R) or else they will not be matched correctly.

7.2.1.2 Fisheries Data

This workflow is designed to accept both fisheries independent and dependent data sources. Data standardization functions have the ability to pull NEFSC survey (using survdat) and observer data using ROracle, but can also accept CSV files. CSV files will need records of all efforts, regardless of if the target species is collected or not, to correctly document absences and have a time column that can be converted with the POSIXct function.

7.2.1.3 Environmental Data

Functions are provided to pull hindcast or forecast data from MOM6 models via the CEFI portal. Other environmental data can be supplied, but should be on regular grids and are ideally netcdf files that can be read using raster if fisheries presence/absence rasters are desired on the same grid.

7.2.2 Directory

This R package is designed to work within a single working directory with folders for the Exposure and SDM results. To use the wrapper functions provided, the working directory should include a folder for each target species, a `Data` folder, and a `logs` folder. Key files can be added to the SDM working directory.

Within the `SDM` directory, the `Data` subdirectory should have subfolders for `csv` and `MOM6` (or other environmental) data, as well as an object containing a list of static environmental covariates (bathymetry, distance to shore, etc). The CSV folder should contain `raw` and `standardized` subfolders to hold the raw csv files from different data sources, and their standardized counterparts from the standardize_data function. The MOM6 data should contain the raw, averaged, standard deviation, and normalized outputs from the MOM6 functions. Each species folder should contain three subfolders: 1) `input_rasters`, 2) `model_output`, and 3) `output_rasters`. The `input_rasters` folder contains all the individual rasters from each data source and the combined raster. The `output_rasters` folder contains all the predicted rasters from each model and the final ensemble model. The `model_output` folder contains the following folders: 1) `models`, 2) `cvs`, 3) `preds`, 4) `eval_metrics`, and 5) `importance`. Each of these folders will contain the output from their respective functions for each model component, and as necessary, the ensemble model.

The `Exposure` directory has a similar structure to the `SDM` directory, where each species specific folder contains seperate folders for `Data` and `Figures`. It is recommended that you use seperate subfolders within each of these if you are calculating exposure across multiple time frames to keep them seperate. The `RawExposure` folder contains the raw and ranked exposure data and figures.

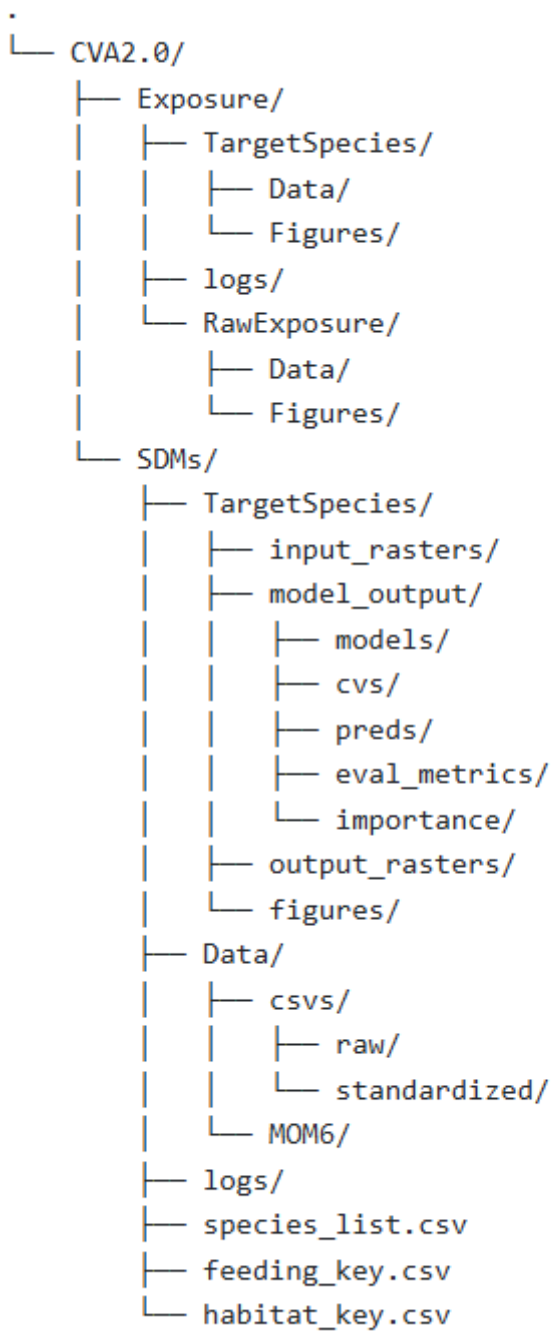


Figure 7.1: Example directory

8 Building & Predicting Ensemble Species Distribution Models

8.1 Overview

The first step in the CVA2.0 workflow is to build the species distribution models. These will be the basis of the exposure, directionality, and possibly additional indicators from the CVA. These are what allow the calculations to be spatially explicit and account for species habitat use.

Within this workflow, there are three steps:

1. Data preparation
2. Building & Predicting Species Distribution Models
3. Building & Predicting the Ensemble

The functions outlined below are designed to be run either sequentially or in parallel with the R parallelization method of your choice (`doParallel` and `furrr` are recommended) and can be run in loops across a list of species.

8.2 Data Preparation

8.2.1 Fisheries Data

The function `standardize_data` can be used to standardize provided csv files or pull NEFSC survey or observer data from NEFSC databases via ROracle. This can also be accomplished in R outside of the package, but `standardize_data` provides a single function to format data in a way that will be accepted by downstream functions. If you are pulling together data from a variety of sources, you will likely still need to manipulate data outside of this function to make sure that data are in a single csv file and that dates and times are in the correct format.

To pull NEFSC survey or observer data, a channel from the package `dbutils` must be provided:

To standardize CSV data, a file path to the csv file, as well as column names for station IDs; position (longitude/latitude); date; the column corresponding to the presence/absence, or count of the species; and species name.

This function results in a dataframe being transformed from this:

```
  X      ID TRIP_NUMBER.x HAUL_NUMBER.x AREA_FISHED SET_BEGIN_DATE SET_END_DATE
1 1 44473.1         44473             1        3674      4/11/2010    4/11/2010
2 2 44473.2         44473             2        3674      4/11/2010    4/11/2010
3 3 44473.3         44473             3        3674      4/11/2010    4/11/2010
4 4 44473.5         44473             5        3674      4/11/2010    4/11/2010
5 5 44473.5         44473             5        3674      4/11/2010    4/11/2010
6 6 44473.5         44473             5        3674      4/11/2010    4/11/2010
  HAUL_BEGIN_DATE HAUL_END_DATE SET_BEGIN_LAT_CONV SET_BEGIN_LONG_CONV
1       4/11/2010     4/11/2010             36.068             -74.921
2       4/11/2010     4/11/2010             36.050             -74.909
3       4/11/2010     4/11/2010             36.042             -74.942
4       4/11/2010     4/11/2010             36.002             -74.975
5       4/11/2010     4/11/2010             36.002             -74.975
6       4/11/2010     4/11/2010             36.002             -74.975
  HAUL_BEGIN_LAT_CONV HAUL_BEGIN_LONG_CONV HAUL_END_LAT_CONV HAUL_END_LONG_CONV
1              36.063               74.921            36.065              74.936
2              36.049               74.909            36.050              74.913
3              36.044               74.941            36.044              74.945
4              36.007               74.975            36.004              74.975
5              36.007               74.975            36.004              74.975
6              36.007               74.975            36.004              74.975
  SET_DURATION HAUL_DURATION_HOURS GEAR_SOAK_HOURS  startDate VESSEL_ID
1         0.07                0.90            3.08 2010-04-11    926397
2         0.07                0.32            0.80 2010-04-11    926397
3         0.03                0.18            0.48 2010-04-11    926397
4         0.05                0.27            0.50 2010-04-11    926397
5         0.05                0.27            0.50 2010-04-11    926397
6         0.05                0.27            0.50 2010-04-11    926397
  TRIP_NUMBER.y HAUL_NUMBER.y         SPECIES_NAME NUM_FISH SPECIES_ON_LIST
1         44473             1             BLUEFISH      363               1
2         44473             2             BLUEFISH       64               1
3         44473             3             BLUEFISH        7               1
4         44473             5             BLUEFISH       14               1
5         44473             5 SHARK DOGFISH SMOOTH        3               0
6         44473             5  SHARK DOGFISH SPINY        6               0
```

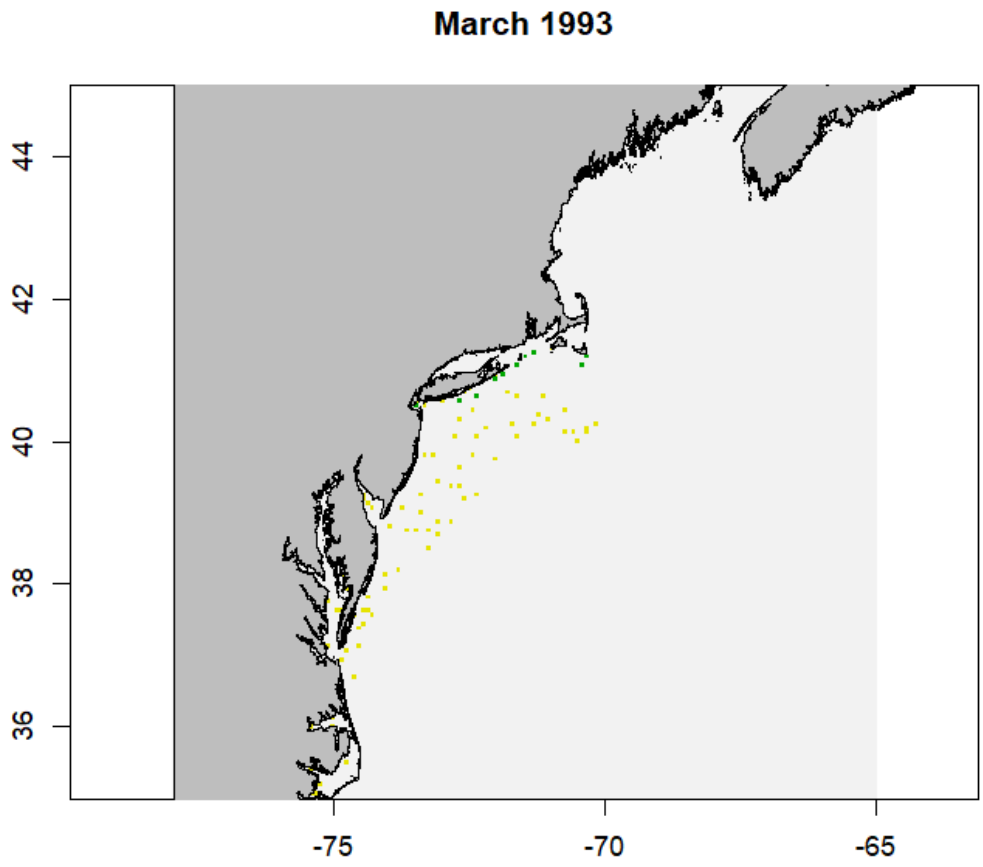
To this:

```
  X       time month year   towID     lon    lat       date count
1 1 2010-04-11     4 2010 44473.1 -74.921 36.068 2010-04-11   363
2 2 2010-04-11     4 2010 44473.2 -74.909 36.050 2010-04-11    64
3 3 2010-04-11     4 2010 44473.3 -74.942 36.042 2010-04-11     7
4 4 2010-04-11     4 2010 44473.5 -74.975 36.002 2010-04-11    14
5 5 2010-04-11     4 2010 44473.5 -74.975 36.002 2010-04-11     3
6 6 2010-04-11     4 2010 44473.5 -74.975 36.002 2010-04-11     6
                   name
1              BLUEFISH
2              BLUEFISH
3              BLUEFISH
4              BLUEFISH
5 SHARK DOGFISH SMOOTH
6  SHARK DOGFISH SPINY
```

To ensure that fisheries and environmental data are on the same scale, and to convert any abundance data to presence/absence, the fisheries csvs produced by `standardize_data` are then converted into rasters of effort/presence/absence with the function `create_rast`. While the source datasets are generated for the entire dataset, one rasterStack is produced for each target species and data source. `create_rast` produces rasters with the same resolution as the environmental data where values range from 0 to 2, where 0 means that no effort was present in the grid cell, 1 means that there was fishing effort but the target species was not caught, and 2 indicates that the target species was caught in that grid cell.

Users must provide the standardized csv files generated by `standardize_data`, a link to a netcdf file that the desired grid and timesteps can be extracted from, the start date of the timeseries, and a vector of alternate names for the target species to account for any differences in species names across data sources. The user must also indicate if the data come from fisheries independent (surveys) or dependent sources (observer or logbook programs). If data come from fisheries dependent sources, then a threshold is applied before the raster is created, following McHenry et al 2019; rasters are only created if the species is present at least 30 times throughout the dataset.

The resulting rasterStack will have a range from 0-2 with a number of layers equal to the length of the timeseries. Below is an example layer from the produced rasterStack for Atlantic Cod



from the NEFSC survey data.

The wrapper function `saveRast` is provided for `create_rast`. This wrapper function provides logging functionality to keep track of progress, which is especially helpful if running the code in parallel, and also adds a skip functionality to skip making the raster if it already exists for that species and data source. It also saves the resulting rasterStack in the `input_rasters` directory within the species-specific folder in the working directory.

Here is an example of using `saveRast` in parallel using the package `furrr` with `future_pmap`:

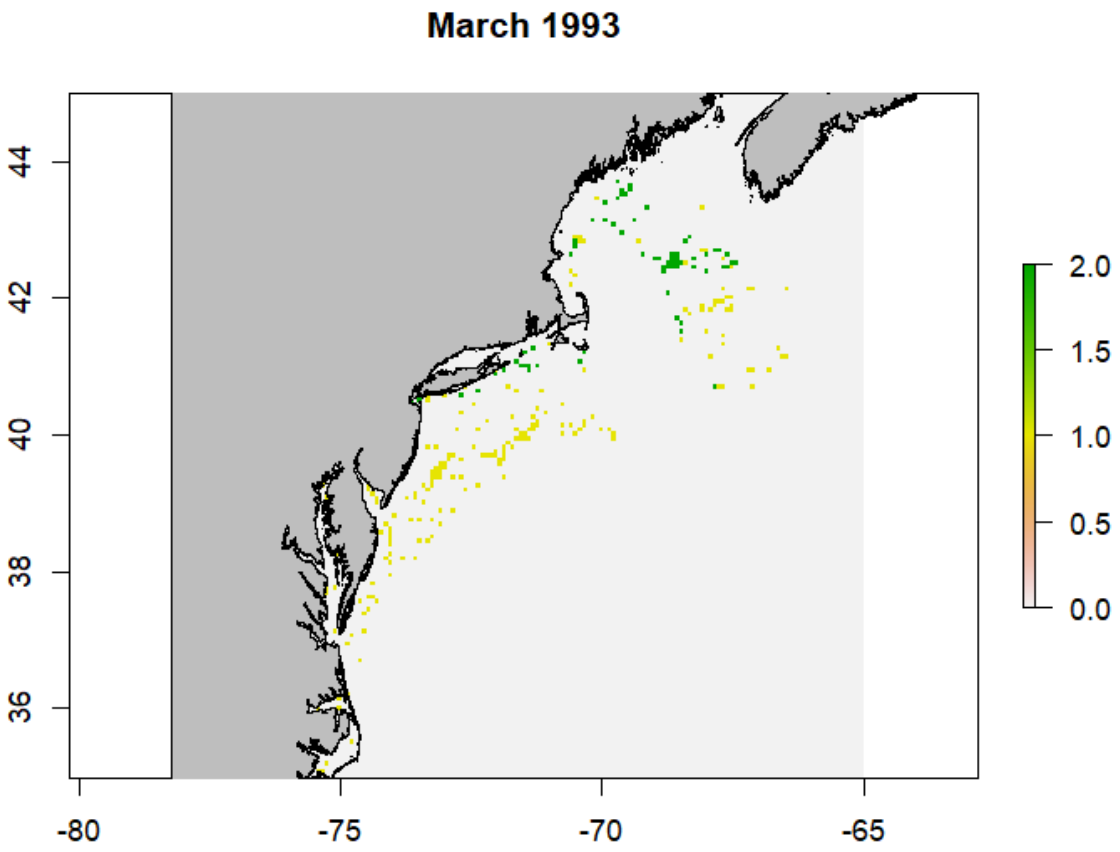
Where `argsList` is a dataframe with a row for each desired call of `saveRast`, and a column associated with most arguments for `saveRast` and `create_rast`. This is generated using `tidyr::expand_grid` to create a dataframe of every combination of target species and data sources.

`create_rast` creates a seperate rasterStack for each species, which is saved to the species-

specific `input_rasters` folder bv the `saveRast` wrapper function. This allows the user to examine the raster for each data source independently to ensure that it meets expectations. Before these can be matched with the environmental data, the rasters must be combined into a single rasterStack, while maintaining the 0-2 range to document presences, absences, and lack of effort. This is done with the `merge_rasts` and associated `combineSave` wrapper function. The `combineSave` wrapper function, in addition to providing logging and skip functionality, also automatically creates a list of raster objects in the `input_rasters` folder for the `merge_rasts` function.

The wrapper function `combineSave` is similar to `saveRast` where is provides logging and skip functionality. This function also handles creating the list of rasterStacks necessary to provide to `create_rast`. It is also written to be run in parallel:

The resulting combined rasterStack will retain the range from 0-2 with a number of layers equal to the length of the timeseries. It works by retaining the maximum value in a grid cell across a given layer (which represent different timestamps) of the provided rasterStacks. Below is an example layer from the merged rasterStack for Atlantic Cod from the NEFSC survey data.



8.2.2 Environmental Data

The example below illustrates how to grab and normalize monthly data from the Modular Ocean Model (MOM6) from the Northeast Shelf. The `pull_hind` and `pull_forecast` functions are designed to work with any MOM6 output. Other environmental data can be used as well, and normalized using the same functions, as long as the raw data is in the expected format.

The first step in getting the environmental data is pulling and spatially subsetting the raw MOM6 data. The functions provided default to pulling the entire time series of the specified model release. Functionality to subset the data by time may be added at a later date, as the length of the timeseries impacts how long the functions take the spatially subset the data.

The function `pull_hind` is designed to pull data from the hindcast simulations:

The function 'pull_forecast' pulls data from the specified forecast similations:

Note that `pull_forecast` has the additional arguments 'init' and 'ens', which represent the initalization date and the ensemble member desired. As of October 2025, the seasonal and decadal forecasts from MOM6 include 10 ensembles with slightly different initial forcing parameters, and as a result can be averaged across ensembles if desired, whereas the long-term forecasts have ensembles that represent different projected CO2 and climate conditions.

For either function, the JSON file link can be found on the CEFI portal in the 'Data Access' tab. The arguments `reqVars`, `release`, `gt`, `of`, `init`, and `ens` help specify which variables to pull, at what output frequency, and what grid to pull (raw vs regrid; the latter is recommended). These need to match their corresponding columns in the JSON file **exactly**. See the documentation for these functions for more details.

Since spatially subsetting the data can take some time (upwards of 30 minutes per variable), if you are pulling a lot of environmental variables, it is recommended to run this command in parallel. Generally, this is not a memory-intensive activity; it just takes some time depending on the length of the timeseries (AKA the number of layers in the resulting rasterBrick). Below is an example of running the `pull_hind` function in parallel with `doParallel`:

Where `var.list` is a two column data frame with the full names of the desired variable (`Long.Name`) and an abbreviated variable name (`Short.Name`) which will eventually become the column name in the final dataframe.

Once the raw hindcast or forecast data is pulled, the workflow moving forward is the same: mean and standard deviations are calculated monthly to calculate z-scores for the complete timeseries. It is recommended to normalize the data in some way prior to building the models to help with interpretation since all the environmental data will likely have different units. We chose to normalize using a z-score to be consistent with exposure calculation methods.

You could also perform each of these steps in parallel following the example above. However, it is not really necessary since these steps are faster than the initial data pull. That being said, it is important to note that the `avg_env`, `sd_env`, and `norm_env` functions expect lists of rasterStacks (which would be the outcome of running `pull_hind` or `pull_forecast` in parallel as illustrated above).

Unlike other functions in this package, no wrapper function is provided. It is not shown here but the resulting lists should be stored in the `./Data/MOM6/` folder so that the data pull only needs to be performed once; it is also recommended to save the raw, mean, and standard deviations, even if they aren't used in the models. This will allow you to 1) easily examine the raw data and 2) calculate z-scores relative to different time series if you wish to do so (for example, normalize forecast data to the present-day mean and standard deviations).

If you have static environmental data such as bathymetry or distance to shore, these should also be normalized to their respective means and standard deviations. You should create an R object that is a named list of your static variables and put it in the `./Data/MOM6/` folder. Names will correspond to the columns in the data frame if you choose to include these data in

your final data frame. Cropping these rasters to the same extent as the MOM6 data is helpful for managing file size. We do not provide functions for this process as these data can come from a variety of sources.

8.2.3 Creating Data Frames

At this point, you should have two sets of raster files: one containing the fisheries effort/presence/absence data and another containing your environmental data. If you want to include static environmental covariates such as bathymetry, you will have a third set of raster files. While rasters are easy to plot, most of our ensemble model members do not take rasters as inputs, so these need to be converted to data frames.

The function `merge_spp_env` is provided to merge species and environmental rasters together, as well as add static environmental variables if provided.

The resulting data frame will have the following structure:

```
'data.frame':   100 obs. of  26 variables:
 $ x         : num  -66.8 -66.7 -66.7 -66.6 -66.5 ...
 $ y         : num  45 45 45 45 45 ...
 $ variable  : chr  "X01.1993" "X01.1993" "X01.1993" "X01.1993" ...
 $ value     : num  0 0 0 0 0 0 0 0 0 0 ...
 $ month     : num  1 1 1 1 1 1 1 1 1 1 ...
 $ year      : num  1993 1993 1993 1993 1993 ...
 $ bottomT   : num  0.182 0.179 0.178 0.172 0.17 ...
 $ bottomO2  : num  -0.14 -0.141 -0.147 -0.145 -0.148 ...
 $ bottomS   : num  0.00701 0.00852 0.01051 0.01072 0.01158 ...
 $ bottomArg : num  -0.000851 -0.000806 -0.000774 -0.000747 -0.000733 ...
 $ surfaceT  : num  0.123 0.126 0.129 0.127 0.125 ...
 $ surfaceS  : num  -0.02366 -0.01261 -0.00579 -0.00424 -0.00458 ...
 $ surfacepH : num  -0.1078 -0.1075 -0.1067 -0.1008 -0.0945 ...
 $ MLD       : num  -0.0925 -0.1174 -0.1207 -0.1069 -0.1362 ...
 $ diazPP    : num  0.132 0.137 0.143 0.145 0.145 ...
 $ smallPP   : num  0.038 0.0368 0.0471 0.0567 0.0567 ...
 $ mediumPP  : num  -0.0751 -0.0712 -0.0546 -0.0295 -0.0275 ...
 $ largePP   : num  -0.026092 -0.024324 -0.01777 -0.007 -0.000244 ...
 $ smallZoo  : num  -0.0764 -0.0848 -0.1159 -0.116 -0.172 ...
 $ mediumZoo : num  -0.0322 -0.0385 -0.0549 -0.0562 -0.0859 ...
 $ largeZoo  : num  0.0372 0.0369 0.0403 0.0377 0.0361 ...
 $ intNPP    : num  -0.043317 -0.03148 -0.016401 0.000677 0.000198 ...
 $ POC       : num  0.00419 -0.00173 -0.00279 -0.001 -0.00336 ...
 $ bathy     : num  -79 -51.6 -62.6 -57.6 -69.2 ...
 $ rugosity  : num  0.512 1.224 0.597 0.132 0.236 ...
```

```
 $ dist2coast: num  4.665 0.926 3.021 9.073 10.255 ...
```

The 'x', 'y', 'variable', 'value', 'month', 'year' columns will always be present. 'variable' is equal to the name of the rasterStack layers, which should be in MM.YYYY format to pull month and year. 'value' is the value in the fisheries raster grid cell and should be equal to 0, 1, or 2.

Additional functions are provided to clean/simplify the data frame: `match_guilds`, `clean_data`, and `remove_corr`. Of these, only `clean_data` is required, `remove_corr` is recommended, and `match_guilds` is optional.

These could be run in any order, with the exception of if the user is utlizing the `match_guilds` function. This should be run prior to `remove_corr` if using.

Here, we will run `match_guilds` first. This function uses the habitat and feeding guilds keys to subset the columns in the data frame to only those relevant to the target species. For example, you may have environmental covariates from the ocean surface and sea floor, like surface and bottom temperatures. Species more associated with the sea floor, such as groundfish or flatfish, are more likely to be impacted by bottom temperatures than surface temperatures they may rarely interact with.

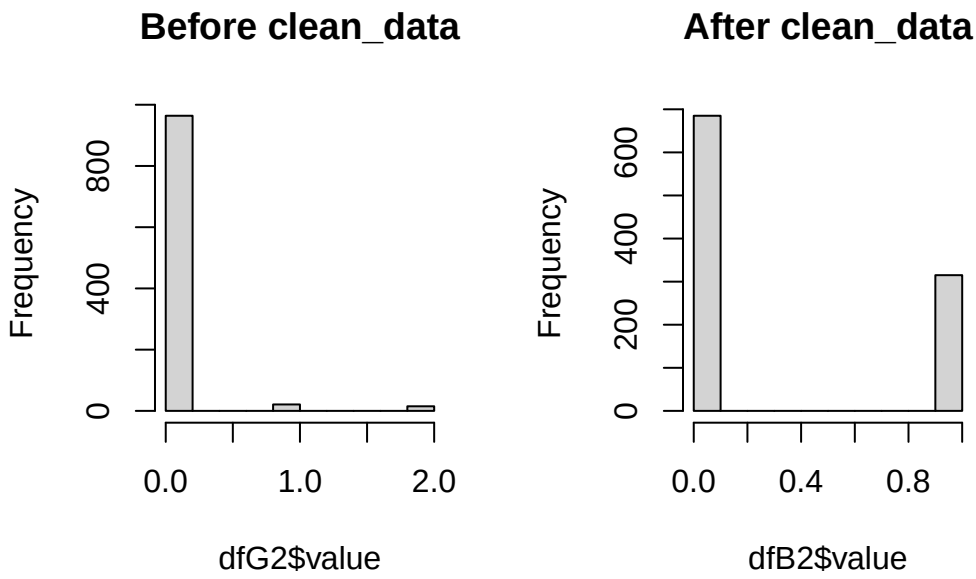
For example, based on the habitat and feeding keys provided, the data frame for Summer flounder will be subset to the following variables:

```
'data.frame':   1000 obs. of  15 variables:
 $ x         : num  -66.8 -66.7 -66.7 -66.6 -66.5 ...
 $ y         : num  45 45 45 45 45 ...
 $ value     : num  0 0 0 0 0 0 0 0 0 0 ...
 $ month     : num  1 1 1 1 1 1 1 1 1 1 ...
 $ year      : num  1993 1993 1993 1993 1993 ...
 $ bottomT   : num  0.182 0.179 0.178 0.172 0.17 ...
 $ bottomO2  : num  -0.14 -0.141 -0.147 -0.145 -0.148 ...
 $ bottomS   : num  0.00701 0.00852 0.01051 0.01072 0.01158 ...
 $ bottomArg : num  -0.000851 -0.000806 -0.000774 -0.000747 -0.000733 ...
 $ smallZoo  : num  -0.0764 -0.0848 -0.1159 -0.116 -0.172 ...
 $ mediumZoo : num  -0.0322 -0.0385 -0.0549 -0.0562 -0.0859 ...
 $ largeZoo  : num  0.0372 0.0369 0.0403 0.0377 0.0361 ...
 $ bathy     : num  -79 -51.6 -62.6 -57.6 -69.2 ...
 $ rugosity  : num  0.512 1.224 0.597 0.132 0.236 ...
 $ dist2coast: num  4.665 0.926 3.021 9.073 10.255 ...
```

Next we will run `clean_data` to turn the data into "true" binary data. The fisheries rasters, and resulting data frames, had values ranging from 0 to 2, representing no effort (0), effort but no catch (1), and catch (2). For presence/absence modeling, the only accepted values are

0 and 1, and these models do not need to know where there was an absence of effort (while is generally good for users to know to understand the scope of the fishing effort). The function `clean_data` will remove all existing 0 values, and remap the presence and absence data to 1s and 0s.

The histograms below illustrate how the data have been transformed.



Note that these datasets are random subsets of the complete dataset so the number of presences/absences is not equal in each. These histograms are only to illustrate the shift in range from 0-2 to 0-1.

Finally, we will run `remove_corr` to remove any correlated environmental covariates before building the model. Position and time (month & year) are retained.

If covariates are removed due to correlation with other covariates, the corresponding column names will be printed in the console or the log file.

The wrapper function `makeDF` includes all of the functions related to making and cleaning the data frame and provides skip and log file functionalities. This function also makes sure that the combined fisheries raster is loaded and the layers have the appropriate names. Like other wrapper functions, this function also saves the results from each call in the appropriate species-specific folder.

This function is generally not very computationally expensive (takes <10 minutes), unless you are matching a lot of environmental covariates. If you need to perform this operation for a long list of species, the wrapper function is designed to be run in parallel.

8.3 Building & Predicting Species Distribution Models

Now that the data frames have been constructed and cleaned, you can build the species distribution models. A series of functions are provided to build, perform cross-validation on, evaluate, and extract variable importance from the models for each of the component models for the ensemble: Generalized Additive Model (GAM), MAXENT, Random Forest (RF), Boosted Regression Trees (BRT), sdmTMB.

The first step is building the base model:

`model` is one of the five component models: `gam`, `maxent`, `rf`, `brt`, `sdmtmb` or the ensemble (`ens`). This function first creates the base model and simplifies it as necessary.

The `sdm_cv` function performs a cross-validation of the model with k = 5.

The predicted values are extracted from the cross-validation output to help build the ensemble and calculate the evaluation metric using the `sdm_preds` function.

The model is then evaluated using the predicted values. This function can calculate a Root Mean Squared Error (RMSE) or Area under the Curve (AUC).

Variable importance for each of the ensemble model components can be extracted from the models:

It is important to note that variable importance from the different models should not be compared without normalizing the values between 0 and 1 as they are all generated using different methods.

The wrapper function `makeMods` combines the above 5 functions, and includes skip and logging functionality. Like previous wrapper functions, it is easy to launch this in parallel with the `furrr` functionality.

Where args is the combination of target species and models to be run, along with the flag to enable the skip functionality.

Here, it is important to note that the different models have different memory demands. The GAM, RF, and MAXENT models are the least computationally expensive, followed by BRT. The sdmTMB is the most expensive. Run times will depend on the dataset size, computational resources, and your R environment. Therefore, it is recommended that you generate sdmTMB models in sequence, but all other models can be generated in parallel.

Below are example run times and run conditions for the final model runs for the Northeast CVA2.0 with 36 target species. This includes all of the steps outlined above: 1) building & simplying the model, 2) performing cross validations, and extracting 3) predictions, 4) AUC, and 5) variable importance. All models were run in a remote R terminal, in an environment linked to Intel's oneAPI, which includes a BLAS library (sdmTMB recommends linking R to an optimized BLAS library to improve runtimes; see the sdmTMB installation guide), with 96 GB of available memory. Final datasets for each species had 100,000+ observations and were

not downsampled. The exception to this was the random forest models which perform better with even distributions of presences and absences. All presences were retained and absences were randomly selected across regions in time and space to ensure an even distribution of absences.

Table 8.1: Model Run Times

	GAM	MAXENT	RF	BRT	sdmTMB
Number of Cores	10	12	6	10	1
Total Run Time (hrs)	15	60	2	26	82
Average Model Run Time (hrs)*	4.2	20*	0.3	7.2	2.5

- *Note 1: These are average run times calculated by dividing the total run time by the number of models run per core (36/number of cores). Some individual models were built faster or slower than the times listed due to the number of presences and the complexity of the environmental relationships*
- *Note 2: The addition of the oneAPI BLAS library increased MAXENT model run times in comparison to previous runs in environments not linked to oneAPI. These previous MAXENT run times in a standard R environment were more similar to GAM run times. However, sdmTMB run times decreased by ~2x (based on small scale tests), but needed to be run in sequence due to memory constraints (yes, even with 96GB of memory, there were memory contrains due to the size of the data set and memory needs of the model), whereas MAXENT models were memory-light enough to be run in parallel.*

The function `make_predictions` will predict the model to a provided timeseries. It has the option to mask off the predictions based on a provided bathymetry raster (`bathyR`).

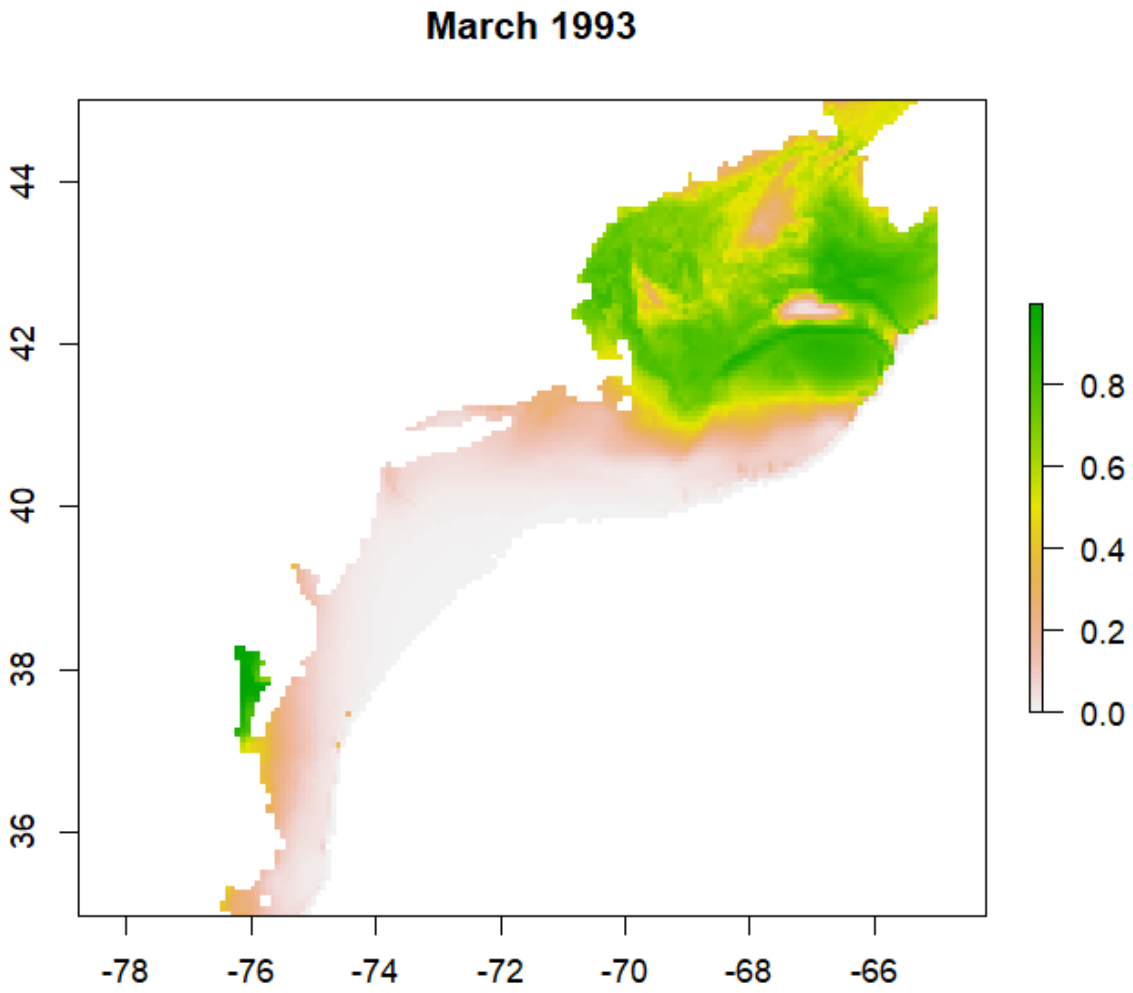
The output will be a rasterStack with the same resolution and number of layers as the time series of environmental data provided.

The wrapper function `predictMods` loads the necessary data frames and models, and saves the output to the `output_rasters` directory within the working directory. Similar to other wrapper functions, it also includes skip and logging functionality, and it is written to be easily run in parallel.

It is important to note that for whatever reason, the MAXENT model cannot be predicted in parallel, so it must be run in sequence. However, predicting the models is much quicker and less memory-intensive than making the models themselves.

`make_predictions` will produce a list of rasters, one raster for each time in the time series, that can be converted to a rasterStack with the `stack` function. Predictions will range between 0 and 1, representing the probability of a species occuring in that grid cell at that time. If `mask = T`, predictions will be limited to a certain depth range. For example

if `bathy_max = 1000` then predictions will not be made for areas where the bathymetry is greater than 1000 m as in the example from a GAM model for Atlantic cod below:



8.3.1 Alternative for sdmTMB Predictions

Regardless of working environment, predictions of sdmTMB models with `predictMods` can take significant amounts of time. The functions `makePredDF` and `predictSDM` offer an alternative workflow similar to `predictMods`. While `predictMods` performs the predictions for each

timestep, this workflow generates a large dataframe of all environmental raster data across all timesteps, performs the prediction once for all timesteps, then creates the individual rasters for each timestep.

This workflow can be easily modified to run in a loop across multiple species. It still takes significant amounts of memory (~50GB for the NE Shelf CVA2.0) so running in parallel is not recommended. Luckily, this code takes approximately 25 minutes to run for a single species.

8.4 Building & Predicting Ensemble

The ensemble is built using the framework from the EFHSDM package. Here it is included in the `make_sdm` function:

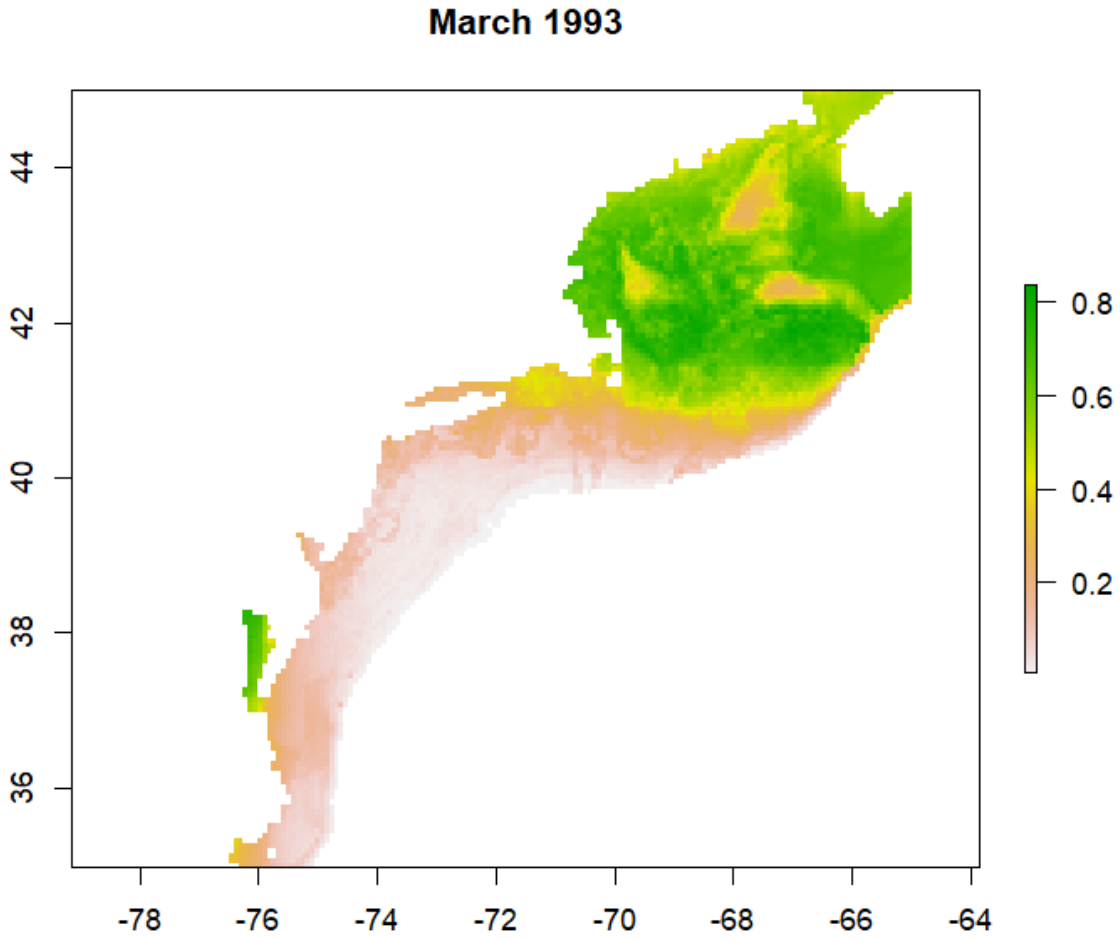
Where the `weights` is a vector of model weights produced either by the EFHSDM function `MakeEnsemble` if using RMSE or another method and `pds` is a list of prediction outputs from the cross-validation of the models. The weights vector and predictions list must be in the same order to correctly assign weights to predictions.

To predict the ensemble model, the `make_predictions` function is used:

Where `abds` is a list of predicted rasters from each of the component models. Similar to building the model, the order of the rasters must match the order of the models in the `weights` vector so that the weights can be assigned appropriately. To predict the ensemble, which is just a weighted average of the component model predictions, only a list of predictor rasters and the model weights are required, so many of the other arguments are set to NULL.

The wrapper function `makeEns` is provided to combine both the creation and prediction of the ensemble. Creating and predicting the ensemble is fast and quick since it is just performing a weighted average of the existing data, so no skip functionality is provided, but logging functionality remains. This can be run in parallel if desired.

The resulting list of rasters, which is again a list of raster layers, will be a weighted average of all provided model predictions. So, like the original predictions the values will range between 0 and



1.

9 Calculating Exposure

9.1 Overview

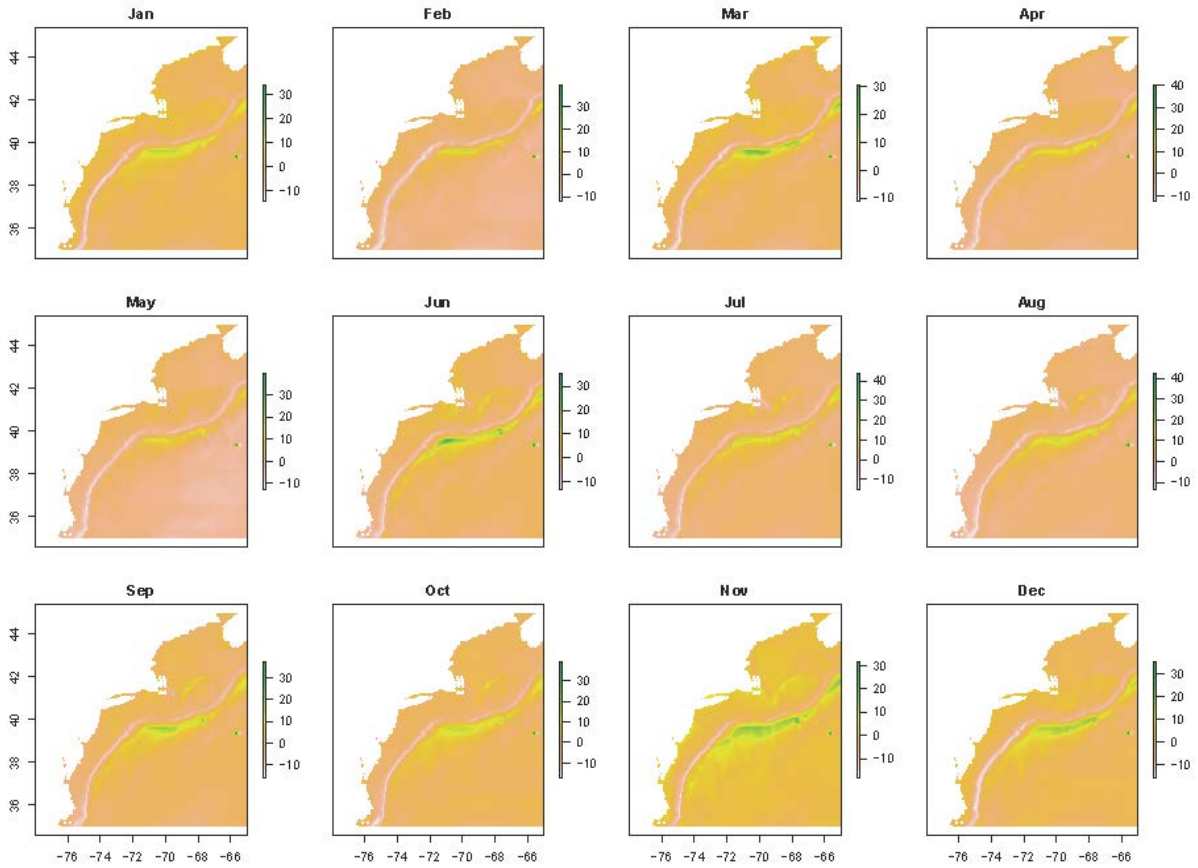
Exposure is calculated from MOM6 forecast data using the same methods to calculate and rank raw exposure as the CVA1.0 on the MOM6 environmental data. We then use the predicted species distributions and the relative importance of covariates in the ensemble models to calculate variable-specific and total exposure for each species.

9.2 Calculating and Ranking Raw Exposure

As in the CVA1.0, raw exposure for each environmental variable is calculated using a z-score, which is the difference in mean conditions in a future and present timeseries divided by present day standard deviations.

The function `calcExposure` will take the output from `pull_hind` and/or `pull_forecast` (a list of rasterStacks) and calculate monthly raw exposure for each environmental variable in the list:

Where `presentRasts` and `futureRasts` are lists of rasterStacks with the same length. These are the raw timeseries over the desired time periods. Remember that `pull_hind` and `pull_forecast` pull the complete timeseries for the desired MOM6 run, so you will need to subset their outputs accordingly if desired. `calcExposure` will produce a list of rasterStacks with the same length as the provided lists. Each rasterStack will have 12 layers, one for each month.



The next step is to rank the raw exposure. Since the CVA is designed to help determine which species are at risk to the greatest exposure, these ranks are based on this goal, rather than the distribution of the data themselves. We use the same ranks as the CVA1.0:

Table 9.1: Raw Exposure Ranks

Score	Raw Exposure	Rank
Low	< 0.5	1
Medium	0.5 - 1.5	2
High	1.5 - 2	3
Very High	> 2	4

The function `rankExposure` performs the ranking. It also provides the option to multiply raw exposure values for certain by -1 (aka flip). This option was included because the direction of change and the meaning of that change may differ across environmental variables. For example, an increase in surface temperatures would have a positive exposure, whereas a decrease in pH would have a negative exposure; both changes are considered bad for the environment, but

the signs differ due to how these variables are quantified. The `flip` and `noFlipList` options provide the option to flip the signs of raw exposure for all variables *except* those listed in `noFlipList`.

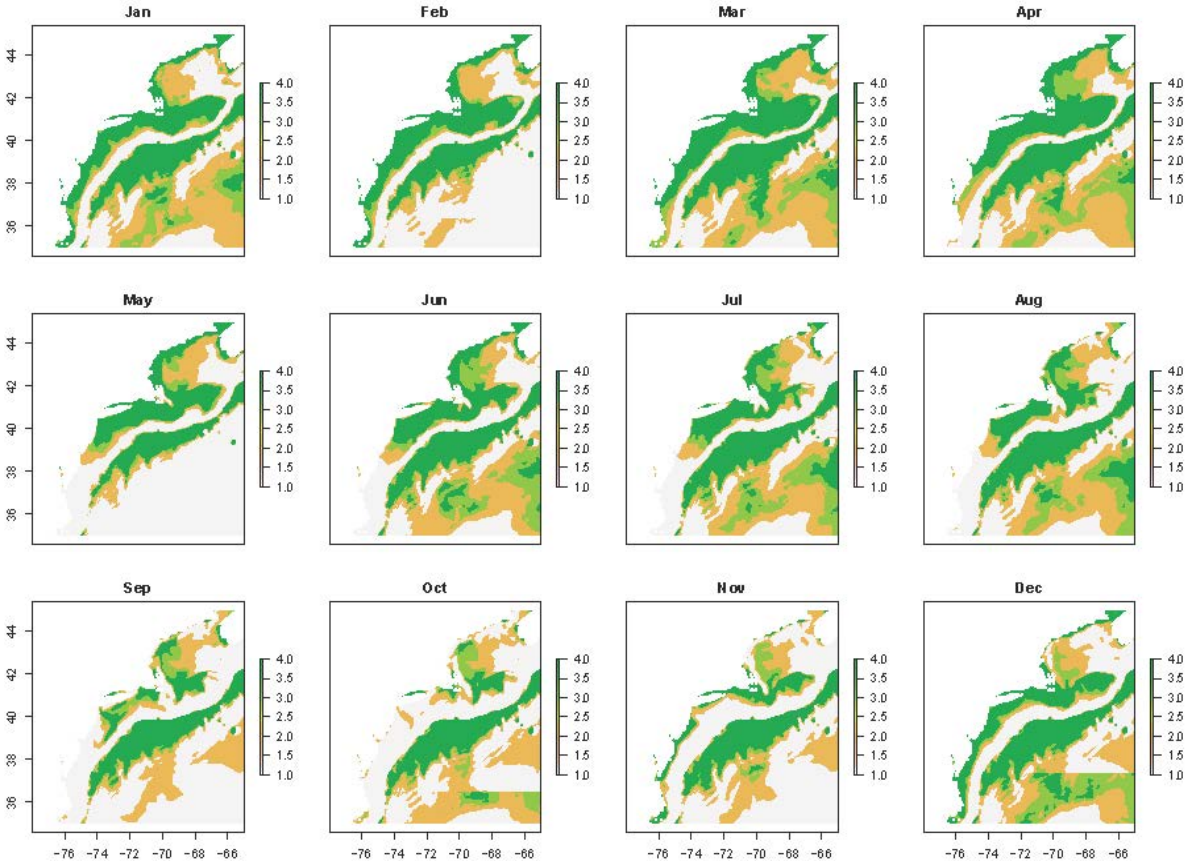


Figure 9.1: Bottom Temperature Ranked Exposure comparing 2019-2019 and 2025-2035

9.3 Creating Species-Specific Exposure Time Series and Maps for Each Variable

Once raw exposure is ranked, species-specific exposure for each environmental variable can be calculated using the functions `makeExposureTimeseries` and `makeExposureMaps`. These functions are nearly identical and only require the ranked exposure values and the mean ensemble species distribution model (SDM) predictions. Ranked exposure values are averaged across space within each month in `makeExposureTimeseries` and averaged across months in each model grid cell in `makeExposureMaps`, using the SDM results as weights to account for

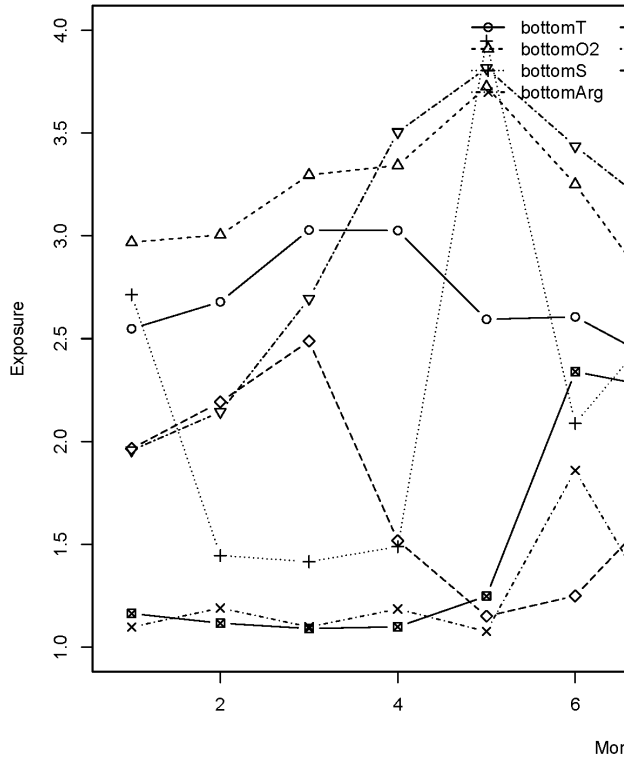
species distributions and habitat use. The results are a time series of average ranked species-specific exposure within each month and a raster of average species-specific exposure across time for each environmental variable.

The ensemble SDM results produced by `make_predictions` should be converted into a monthly average prior to using `makeExposureTimeseries` and `makeExposureMaps`. It is also recommended that you remove low SDM values after averaging. In the NE Shelf CVA2.0, we removed all SDM values less than 0.1 (representing areas with less than a 10% likelihood of finding the species). Not removing these low values resulted in higher than expected exposure results in areas where the species is rare.

The output from `makeExposureTimeseries` is a matrix with 12 columns (1 for each month) and a number of rows equal to the number of environmental variables in `expRanked`. `makeExposureMaps` produces a rasterStack, with each layer representing one of the environmental variables. Values should range from 1 to 4.

The wrapper function `makeVariableAverages` is provided and can be run in parallel using the `furrr` package, similar to the SDM wrapper functions. Like those wrapper functions, your directory must be set up according to the Installation and Setup Guide provided for them to function. `makeVariableAverages` also generates the monthly average SDM results, and removes low values for you.

Like the SDM wrapper functions, `args` is a dataframe where each row is a set of arguments to be passed to the function. Here, the arguments include the species name (`spp`), ensemble name to help locate the appropriate ensemble model (`ensName`), and start and end years for the present and future timeseries (`pStart`/`pEnd`/`fStart`/`fEnd`) to load in the correct exposure values, as well as save the results from `makeExposureTimeseries` and `makeExposureMaps` appropriately. These are not memory or time-intensive calculations, so it is not necessary to run them in parallel - it is just helpful if you have a lot of species to work through.



Below are example variable-specific exposures for Atlantic cod:

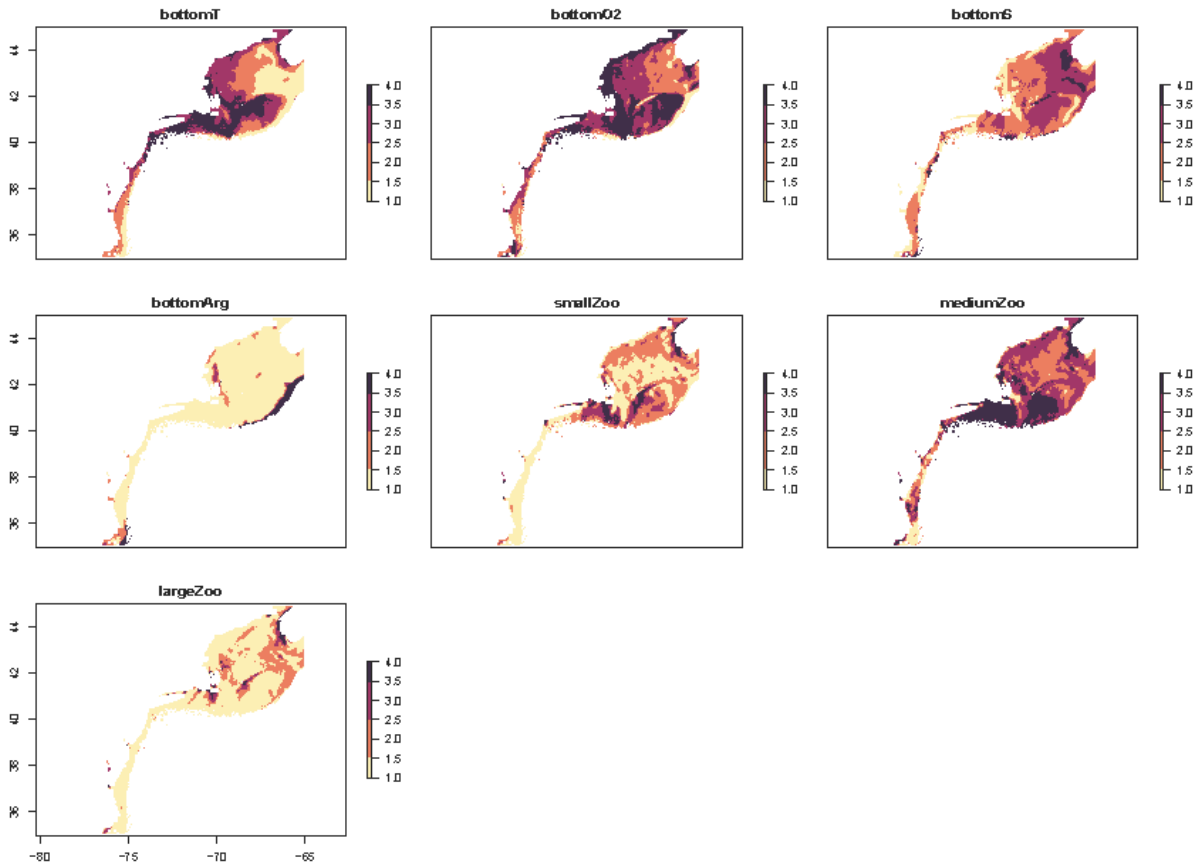


Figure 9.2: Variable-specific exposures across space

9.4 Calculating Total Exposure for Each Species

Once you have the species-specific exposure calculated for each variable, they must be combined to estimate total species-specific exposure. These will be the final spatially and temporally explicit values that are combined with species sensitivity to estimate vulnerability.

To combine the species-specific exposures for each variable, we use the same logic rule as the CVA1.0:

Table 9.2: Logic Rule used to Calculate Total Species Exposure

Overall Score	Numeric Score	Logic Rule
Low	1	All other scores
Medium	2	2 or more attributes >= 2.5
High	3	2 or more attributes >= 3.0

Overall Score	Numeric Score	Logic Rule
Very High	4	3 or more attributes >= 3.5

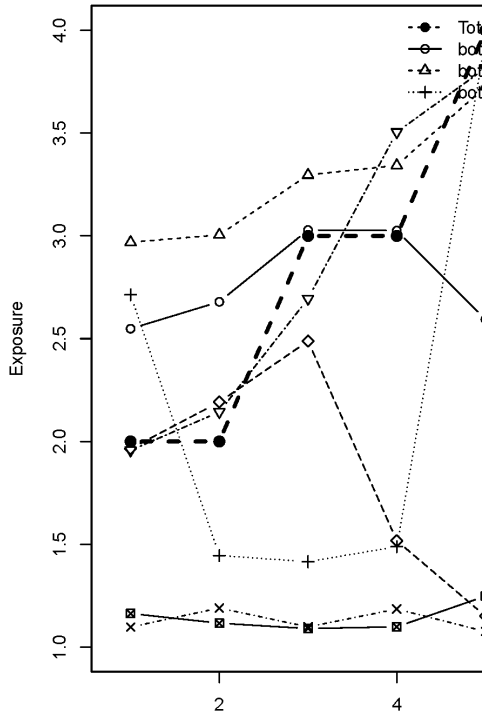
We calculate total exposure in two ways: 1) Applying the logic rule to all ecologically relevant environmental variables as defined by the feeding and habitat keys and 2) Only applying the logic rule to ecological variables that have relative weights in the final ensemble SDM greater than a set value (for the NE Shelf CVA2.0, we chose 10% or 0.1). This was done to account for possible unknown changes in variable importance under future climate change. The latter method (only counting the most important variables), assumes that these values will continue to be very important under future climate change scenarios. It is unclear if this will be the case, especially for variables such as pH where it is unclear how they will impact species. Calculating total exposure using all the ecologically relevant variables allows us and stakeholders to consider all variables regardless of their model importance.

The functions `combineTimeseries` and `combineMaps` provide options for both methods with the option `countAll`. Setting `countAll` to TRUE will automatically count all provided variables in `matExp` for `combineTimeseries` or `mapExp` for `combineMaps`, which are the outputs from `makeExposureTimeseries` and `makeExposureMaps`, respectively. If set to FALSE, you will need to provide the relative weights and the threshold to use to count variables in the logic rule. It is important that, if you have not yet done so, that you subset `matExp` and `mapExp` to only the ecologically relevant variables before running the `combineTimeseries` and `combineMaps` functions. This can be done using the relative weights of the variables.

The function `combineWeights` estimates the relative variable importance across the models. Since every component model of the ensemble SDM estimates variable importance differently, all importance data is normalized to their respective sums such that they total 1. A weighted average is then performed across models using the weights of each component model in the ensemble. The result is a vector of values representing the relative importance of each of the ecologically relevant variables to the model.

Where `vars` is a vector of character strings including the ecologically relevant variables, `ensWeights` is the weights of each component model of the ensemble model generated by `make_sdm`, `impFlist` is a vector of character strings including the file paths to the model variable importance files made by `sdm_importance`, and `staticNames` are the names of any static environmental variables to be excluded.

The names of the vector resulting from `combineWeights` can be used to subset the outputs from `makeExposureTimeseries` and `makeExposureMaps`, since these output contain all possible environmental variables from MOM6, before running `combineTimeseries` and `combineMaps`.



Below are examples of total exposure using both methods for Atlantic cod:

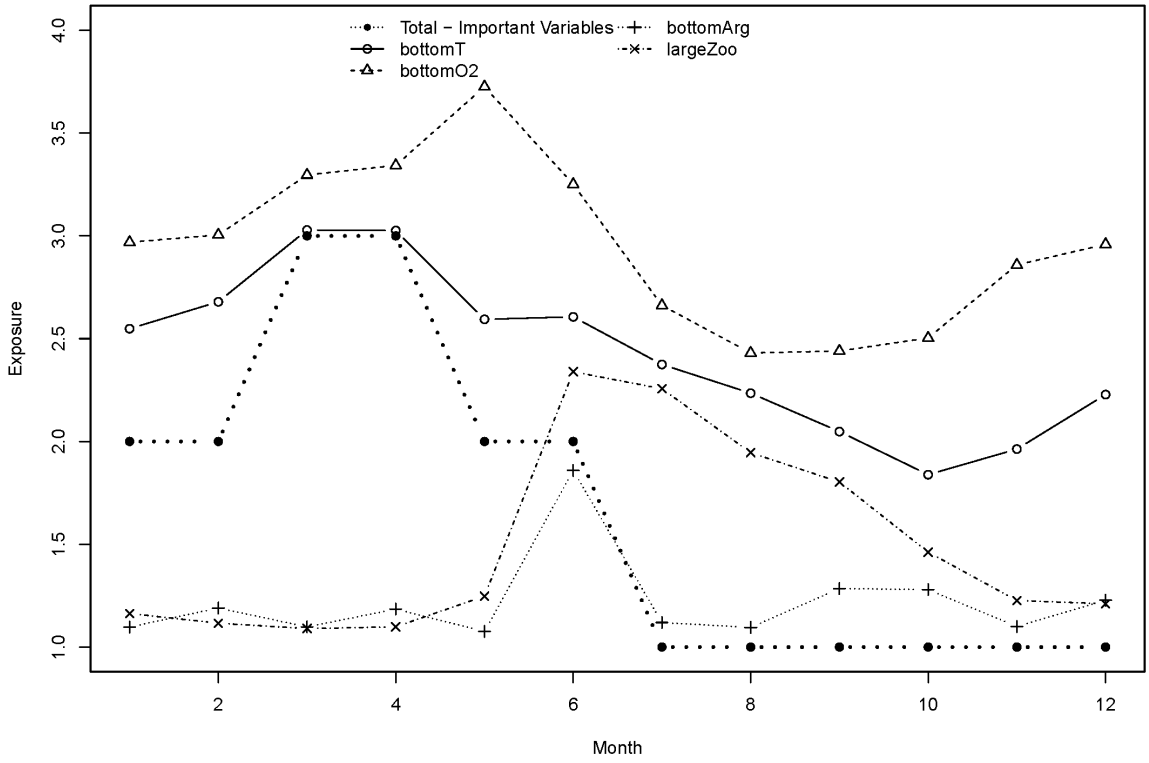


Figure 9.3: Total Exposure using only important variables across time

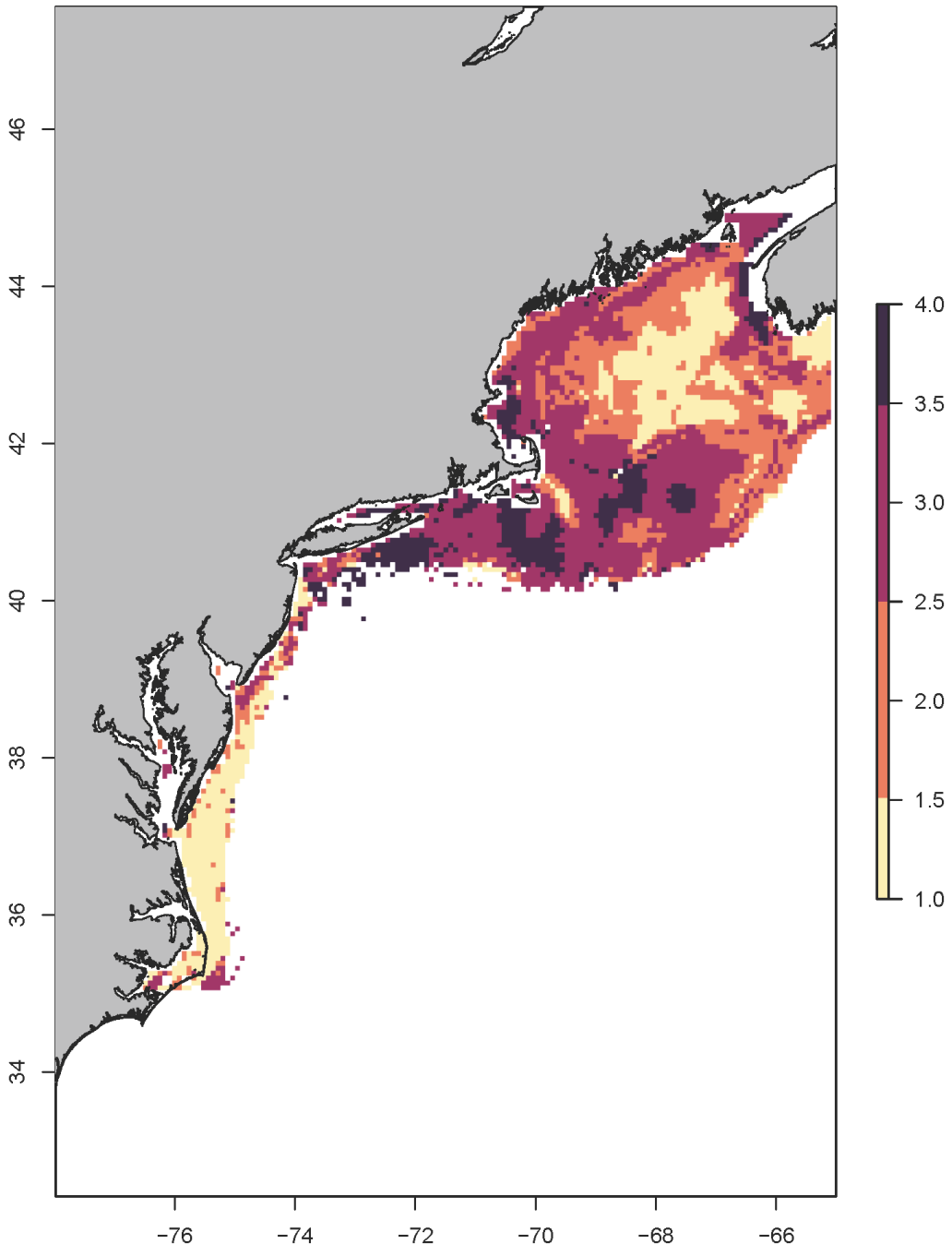


Figure 9.4: Total Exposure using all ecologically-relevant variables across space

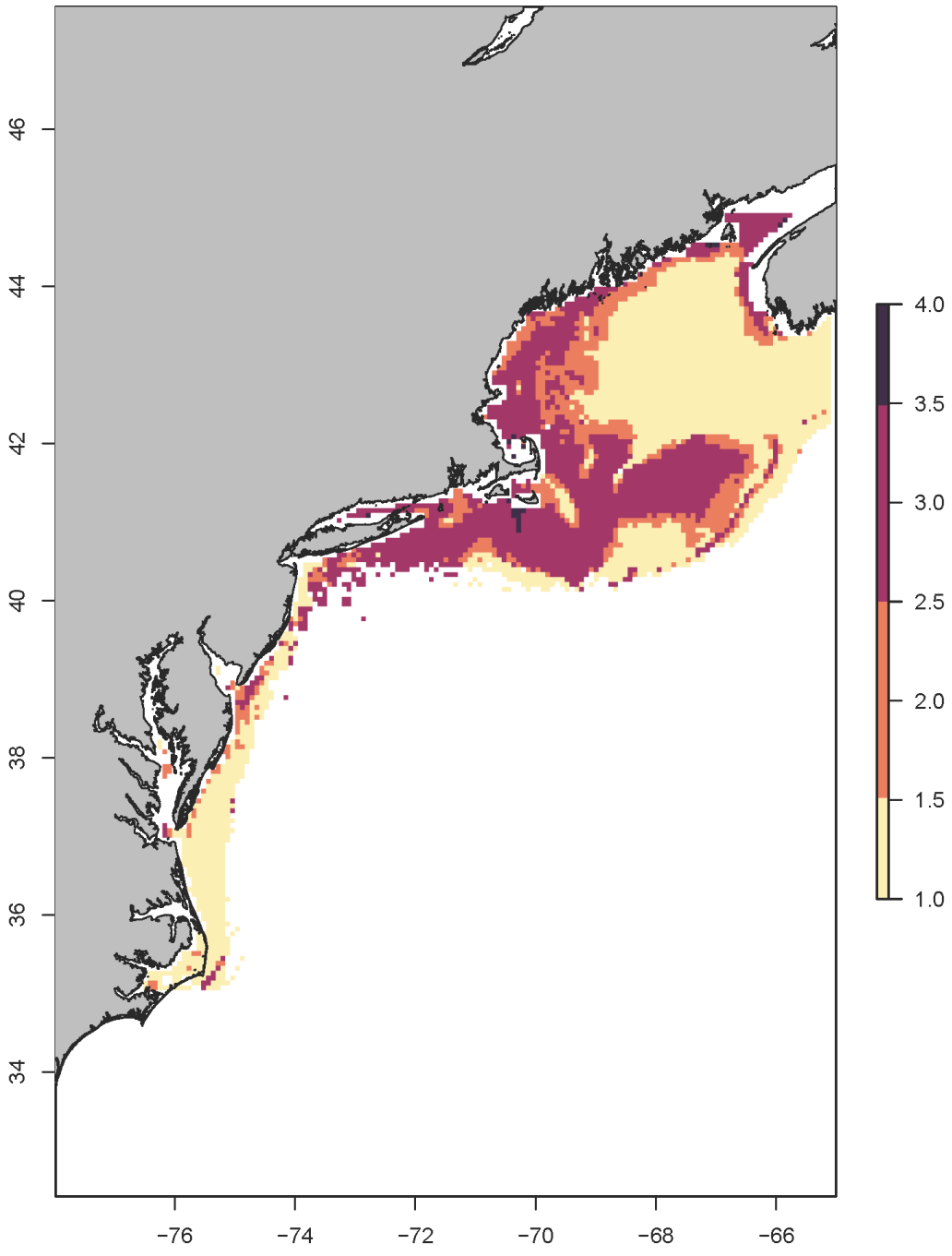


Figure 9.5: Total Exposure using only important variables across space

10 Sensitivity Calculation

10.1 Analysis

Sensitivity scores were collected using the Fish Stock Climate Vulnerability Assessment Portal, and exported as a csv. Scores were calculated for each attribute and these scores were used to calculate a final exposure score for the species. Unlike exposure, and similar to the original analysis, a single value is produced for each species.

10.1.1 Attribute Scores

Attribute scores were calculated using a weighted average following Equation 1 in Morrison et al 2015. Briefly, the number of tallies in each of the four bins is multiplied by a weight, summed, and divided by the total number of tallies (which would be 5 scorers per species x 5 tallies per scorer = 25). The weights that correspond to each bin are:

Table 10.1: Bin Weights

Bin	Weight
Low	1
Medium	2
High	3
Very High	4

The function `attribute.score` performs the weighted average. This function is based on code provided by the Tyler Loughran and the Atlantic Highly Migratory Species Management Division, Office of Sustainable Fisheries. The workflow is outlined below. Individual attribute scores are also accompanied by a data quality score, which are provided by expert scorers for each species and attribute to help identify data gaps. This is calculated using the `data.quality.score` function.

10.1.2 Total Sensitivity Scores

Following the original CVA methods, total sensitivity is calculated with a logic rule. This is the same logic rule used to calculate total exposure from individual variable exposure scores:

Table 10.2: Logic Rule used to Calculate Total Species Sensitivity

Overall Score	Numeric Score	Logic Rule
Low	1	All other scores
Medium	2	2 or more attributes >= 2.5
High	3	2 or more attributes >= 3.0
Very High	4	3 or more attributes >= 3.5

The total sensitivity score is calculated using the `logic.rule` function. Certainty was calculated by resampling the tallies with replacement across the four bins, and recalculating the weighted average attribute score, then the total sensitivity. Following the original CVA, certainty is represented as the proportion or percent of bootstrapped total sensitivity scores that match the expert-derived total scores. The certainty associated with the total sensitivity score is calculated with the `bootstrap.certainty` function.

10.1.3 Workflow

10.1.3.1 Calculating Scores

A single function, `calculate.sensitivity`, combines both the attribute score calculation done by `attribute.score` and the total sensitivity score calculation done by `logic.rule` in a single function. This function is modeled on provided code from the Southeast Fisheries Science Center.

`calculate.sensitivity` returns a vector, containing the individual attribute scores and the total score. When used with `lapply` as in the example above, a list of vectors will be produced, with one vector per species in the provided `species.data.list`.

10.1.3.2 Calculating Data Quality

The function `data.quality.score` calculates the mean data quality score for each attribute and species. It produces a vector similar to `calculate.sensitivity`, with an entry for each attribute, and works with the `lapply` family of functions.

10.1.3.3 Calculating Certainty

Certainty was quantified using the bootstrapping methods described above. For the default number of iterations (10,000), this function takes just under a minute (~45 seconds) per species on a standard workstation.

Both `attribute.score` and `logic.rule`, and subsequently `calculate.sensitivity` have the ability to also perform the bootstrap calculation.

When `bootstrap = TRUE`, a data.frame, rather than a vector, is produced. This data.frame will have a number of rows equal to the number of `samples`, with each row representing one of the iterations. Columns will contain the attribute scores and total sensitivity score for that iteration. When used with `lapply` as in the example above, a list of data.frames will be produced, with one data.frame per species in the provided `species.data.list`.

The function `bootstrap.certainty` calculates the final certainty value (AKA the proportion of bootstrap iterations that match the expert scores). This function will calculate the proportion of bootstrapped total sensitivity scores correspond to the total sensitivity score (1-4). It adds a column called 'Certainty' to the vector generated by `calculate.sensitivity(bootstrap = FALSE)`.

The entire workflow can be run using `lapply` and `mapply` functionality on the entire raw dataset from the FSCVA portal using only a few lines of code:

The resulting sensitivity data.frame shows the scores for each attribute, plus the total sensitivity and corresponding certainty:

```
                X Adult.Mobility Complexity.in.Reproductive.Strategy
1  Acadian redfish           2.72                                1.84
2          Alewife           1.60                                3.25
3  American conger           1.45                                2.15
4     American eel           1.15                                2.05
5 American lobster           1.90                                1.40
6  American plaice           2.75                                1.45
  Dispersal.of.Early.Life.Stages
1                           2.04
2                           2.80
3                           1.35
4                           1.10
5                           1.80
6                           1.45
  Early.Life.History.Survival.and.Settlement.Requirements Habitat.Specificity
1                                                    2.20                1.72
2                                                    3.28                2.56
3                                                    2.25                2.55
```

```
4                                                 2.45                2.05
5                                                 2.35                1.85
6                                                 2.15                1.80
  Other.Stressors Population.Growth.Rate Prey.Specificity
1            1.92                  3.52             1.96
2            2.68                  2.24             1.68
3            2.24                  2.00             1.65
4            2.45                  2.70             1.10
5            2.70                  3.45             1.15
6            1.25                  2.45             2.10
  Sensitivity.to.Ocean.Acidification Sensitivity.to.Temperature Spawning.Cycle
1                               1.12                       1.80           2.60
2                               1.56                       2.16           3.16
3                               1.25                       1.70           2.90
4                               1.15                       1.40           2.50
5                               3.85                       2.70           2.90
6                               1.90                       1.65           2.40
  Stock.Size.Status Total.Sensitivity Certainty
1              1.84                 2    0.9998
2              2.52                 3    1.0000
3              2.40                 2    0.6946
4              2.55                 2    0.7234
5              2.20                 3    1.0000
6              2.95                 2    0.9882
```

Not shown here, but the data quality data frame has a similar layout, with a row for each species and a column for each attribute. There is no total sensitivity or certainty column.

Part III

Species Narratives

Species Narratives summarize the vulnerability results for each species, and provide additional context on known climate change impacts on the species and known life history characteristics. PDF versions of the narratives will be available for download.

Each narrative contains the following information:

Summary

A summary section providing the final average Vulnerability, Exposure, and Sensitivity scores. If scores were calculated across species stocks and scores differed across stocks, a range of scores will be provided. Data quality scores associated with both Exposure and Sensitivity scores will also be provided.

This section also contains a summary table. Click on the table to enlarge it.

Changes from Previous Assessments

This section contains a table highlighting the scores from this assessment, and compares them to previous assessments. If a citation for the previous assessment is not provided, it comes from (**hare2016vulnerability--10?**).

Stocks

If a species was scored within different stock groups, a map of the different stocks is provided here and referenced throughout the narrative.

Score Descriptions

These sections go into the scores in-depth, describing average scores, spatial patterns, and drivers. Each section has a map of scores. Climate Exposure has two sets of maps - one illustrating total exposure and the second illustrating factor exposures.

Data Quality Metrics

This section describes the Life History Data Quality and Model Confidence Scores. Both were provided by the experts and range from 0 - 3. Following the original analysis, a score above 2 was deemed “acceptable”. The rubrics for these scores are provided below

Life History Data Quality Rubric

Data Quality Score	Description
3	**Adequate Data.** The score is based on data which have been observed, modeled or empirically measured for the species in question and comes a from reputable source.
2	**Limited Data.** The score is based on data which has a higher degree of uncertainty. The data used to score the attribute may be based on related or similar species, come from outside the study area, or the reliability of the source may be limited.
1	**Expert Judgment.** The attribute score reflects the expert judgment of the reviewer and is based on their general knowledge of the species, or other related species, and their relative role in the ecosystem.
0	**No Data.** No information to base an attribute score on. Very little is known about the species or related species and there is no basis for forming an expert opinion (please use judiciously).

This is the same rubric used in (**hare2016vulnerability--10?**).

Model Confidence Rubric

Model Confidence Score	Description
3	**High Confidence.** The model highly accurately reflects the species' realized niche. It captures fine-scale nuances, matches expert field observations, and handles environmental transitions realistically.

Model Confidence Score	Description
2	**Moderate Confidence.** The model aligns well with known distributions and environmental drivers. Most key habitats are represented, though there may be minor inaccuracies at the range edges or in data-poor regions.
1	**Low Confidence.** The model captures broad trends but has significant "noise." Environmental drivers likely don't reflect actual species needs.
0	**No Confidence.** The model fails to represent known species range. It predicts species in unsuitable areas or misses well-known core areas entirely.

This rubric was developed to mimic the Life History Data Quality rubric.

The Area under the Receiver-Operator Curve (AUC) is also provided for the ensemble model with the Model Confidence Scores. This statistic quantifies how well models classify binary data. The closer the value is to one, the better the model did at predicting observed presences and absences. This metric was not provided to the experts while scoring Model Confidence.

Additional Metrics

This section describes the Distribution Shifts and Directional Effect results. Directional Effect is consistent with the previous analysis - experts were asked to score how they left the species would be impacted (postive/negative/neutral) by climate change.

Distribution Shifts were previously assessed using a subset of Sensitivity Attributes. In this analysis, they are quantified with the ensemble species distribution models when possible. The provided map illustrates average species ranges using 5 year averages from the first and last 5 years of the hindcast data and the last 5 years of the most recent decadal forecast. The provided timeseries illustrate how core and total (quantified by 50 and 95% KDE respectively) habitat areas, latitude range, and longitude ranges change throughout the MOM6 hindcast (black points) and decadal forecast (yellow points) datasets.

In the online version of the narratives, an animated gif is also provided to illustrate how species distributions and the edges of these distributions shift through time in both the hindcast and forecasts.

Climate Effects & Life History Synopsis

The final two sections of the narratives include summaries of known climate impacts on a species and a summary of their life histories.

References

Hare, Jonathan A, Wendy E Morrison, Mark W Nelson, Megan M Stachura, Eric J Teeters, Roger B Griffis, Michael A Alexander, et al. 2016. "A Vulnerability Assessment of Fish and Invertebrates to Climate Change on the Northeast US Continental Shelf." *PloS One* 11 (2): e0146756.

11 American lobster (*Homarus americanus*)

11.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = High (Gulf of Maine/Georges Bank stock) - Very High (Southern New England stock)**Data Quality**Model Confidence Score =Life History Data Quality = 97%

```
[1] "C:/Users/katherine.gallagher/Documents/READ-EDAB-CVA2.0Manual/test_images/American lobst
```

```
[1] "C:/Users/katherine.gallagher/Documents/READ-EDAB-CVA2.0Manual/test_images/American lobst
```

11.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	High (Gulf of Maine/Georges Bank) - Very High (Southern New England)	Moderate (All Stocks)	Increase
Directional Effect	Negative	Neutral	Decrease

11.3 Score Drivers

11.3.1 Vulnerability:

add text here

11.3.2 Climate Exposure:

add text here

11.3.3 Biological Sensitivity:

The **Gulf of Maine/Georges Bank** stock of American lobster received a **High** sensitivity score and the **Southern New England** stock scored **Very High.** Certainty was high for both stocks, with 99% certainty for the Gulf of Maine/Georges Bank stock and 92% certainty for the Southern New England stock.

Population growth rate and other stressors drove the scores for both stocks. Relationships to calcified organisms contributed to the High score for the Gulf of Maine/Georges Bank stock, and received a score of 3.5 for the Southern New England stock. Stock size had the highest score for the Southern New England Stock.

Lobsters are long-lived, slow growing species that can reach ages of 50+ years. Disease and the introduction of new predators such as black sea bass and spiny dogfish impact lobsters across their range (**asmfc2025--11?**; **goldstein2025--11?**). American lobsters are omnivores but diets are dominated by mollusks (**elner1987--11?**), which are likely to be impacted by ocean acidification.

Sensitivity has increased since the last assessment, where lobsters received a Moderate score (**hare2016vulnerability--11?**). Population growth rate and other stressors drove scores in both assessments.

11.4 Data Quality Metrics

11.4.1 Life History Data Quality:

The percent of data quality scores greater than 2 was 97% for the Gulf of Maine/Georges Bank stock and 95% for the Southern New England stock, indicating excellent data quality. For both stocks, Stock Size/Status was the only data gap (mean attribute data quality score < 2).

11.4.2 Model Confidence:

add text here

11.5 Additional Metrics

11.5.1 Distribution Shifts:

add text here

11.5.2 Directional Effect:

Both stocks are likely to be **negatively** impacted by climate change (certainties range from 81 – 100% based on bootstrap analysis of expert scores). Increased thermal stress and disease will continue to decrease the productivity of the stocks. In addition, new demersal predators such as black sea bass and spiny dogfish will increase predation pressures of juvenile and adult lobster. The directional effect was estimated to be neutral in the previous Assessment (**hare2016vulnerability--11?**).

11.6 Climate Effects on Abundance and Distribution

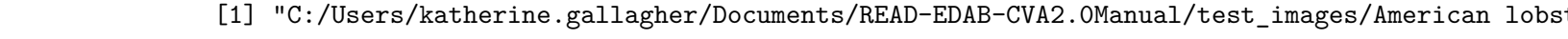
The Northwest Atlantic has been warming at an exceptionally fast rate (**saba2016--11?**), which has contributed to declines in fecundity and population size in Southern New England (**goldstein2022--11?**) while the fishery in the central Gulf of Maine has increased dramatically (**lebris2018--11?**). Temperatures have increased in the lobster's inshore range, and evidence suggests that the productivity of the stock is severely compromised with increased thermal stress and disease (**asmfc2025--11?**). The assemblage of predators of American lobster has shifted with warming conditions in the Gulf of Maine, with benthic and demersal predators, such as black sea bass (*Centropristis striata*) and spiny dogfish (*Squalus acanthias*) becoming increasingly present throughout the lobster's range, taking the place of traditional predators, such as Atlantic cod (*Gadus morhua*). The effects of invasive and range-expanding species may also lead to greater predation risk for juvenile lobsters (**goldstein2025--11?**).

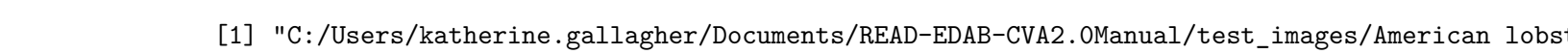
11.7 Life History Synopsis

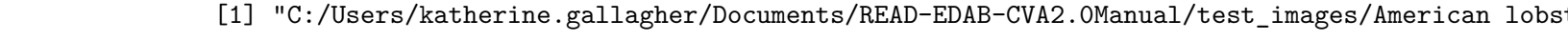
American lobsters inhabit the coastal and oceanic waters of the Atlantic Ocean from Labrador to North Carolina. They are widely distributed within most of the coastal waters in the Northeast U.S. with some major offshore concentrations along the outer edge of the Continental Shelf and upper slope between Georges Bank and Delaware Bay (**mackenzie1985--11?**). Spawning takes place during a 4 month period from late May through much of September, when females leave their solitary shelters about 7 days before molting to share the shelter of a dominant male (**factor1995--11?**) (**mackenzie1985--11?**). Lobster larvae are planktonic for 6-8 weeks (**factor1995--11?**) and travel from surface waters offshore to inshore areas of Southern

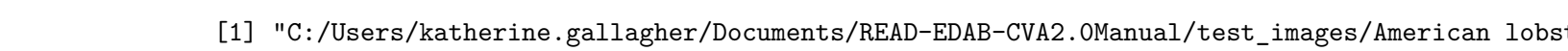
New England and the Gulf of Maine. Larval distribution is affected by the distribution of spawning females and by surface current velocity and direction, temperature, salinity, light intensity, and hydrostatic pressure (**katz1994--11?**) (**harding1988--11?**) (**hudson1993--11?**) (**mackenzie1985--11?**). Larval survival is likely affected by pollution of the marine environment by human activity, especially coastal waters (**connor1972--11?**). There is an inverse relationship between temperature and the time from hatching to the postlarval stage (when they settle). Larvae take as few as 11 days at 22°C and as many as 54 days at 10°C to reach the postlarval stage. It is speculated that thermoclines are a negative cue for larvae/post larvae settlement (**factor1995--11?**) (**templeman1936--11?**). Juvenile lobsters utilize shelters (rocks and vegetation) in the benthic habitat, and may also burrow in mud (**wahle1991--11?**). Juveniles are omnivorous but also scavenge for food. Much of their diet consists of bottom invertebrates, crabs, urchins, mussels, polychaetes, periwinkles, and sea stars, but they may also consume fish and plants (**lawton1987--11?**). Adult lobsters are also considered omnivores and scavengers . However, their diet is dominated by mollusks (especially mussels), echinoderms, and polychaetes (**elner1987--11?**). Juveniles and adults prefer bottom water temperatures of 8-14°C (**mackenzie1985--11?**). Inshore, adult lobsters appear to have a limited home range of about 2.2 km, however offshore lobsters tend to migrate greater distances (up to 345 km in 71 days) (**mackenzie1985--11?**). American lobster are managed by the Atlantic States Marine Fisheries Commission as three stock units based on life history differences: Gulf of Maine, Georges Bank, and southern New England (**asmfc2025--11?**).

11.8 Figures









11.9 References

Atlantic States Marine Fisheries Commission (ASMFC). 2025. "ASMFC Stock Assessment Overview: American Lobster." Atlantic States Marine Fisheries Commission. https://asmfc.org/wp-content/uploads/2025/11/AmericanLobsterStockAssmtOverview_Oct2025.pdf.

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12 Atlantic cod (*Gadus morhua*)

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12.1 Summary

ScoresVulnerability = Moderate - HighClimate Exposure = ModerateBiological Sensitivity = High - Very High

Data Quality Model Confidence Score = 2.1 +/- 0.32Life History Data Quality = 82-83%

Sensitivity

Sensitivity Attributes	Range Score	Range Data Quality	Range Tally	Eastern Gulf of Maine Score	Eastern Gulf of Maine Data Quality	Eastern Gulf of Maine Tally	Western Gulf of Maine Score	Western Gulf of Maine Data Quality	Western Gulf of Maine Tally	Georges Bank Score	Georges Bank Data Quality	Georges Bank Tally	Southern New England Score	Southern New England Data Quality	Southern New England Tally
Adult Mobility	1.91	2		1.8	2		1.84	2		1.92	2		2.08	2	
Complexity in Reproductive Strategy	3.46	2.5		3.48	2.5		3.48	2.5		3.32	2.5		3.56	2.5	
Dispersal of Early Life Stages	2.12	2.22		2.16	2.2		2.16	2.2		2.04	2.3		2.12	2.2	
Egg and Larval Survival and Settlement Requirements	3.41	2.4		3.36	2.4		3.44	2.4		3.28	2.4		3.56	2.4	
Historical Variability in Temperature	3.14	2.6		3.08	2.6		3.2	2.6		3.16	2.6		3.12	2.6	
Juvenile Habitat Requirements	2.42	2.7		2.44	2.7		2.44	2.7		2.2	2.7		2.6	2.7	
Other Stressors	2.37	2.2		2.2	2.2		2.44	2.2		2.24	2.2		2.6	2.2	
Population Growth Rate	2.7	1.85		2.76	1.9		2.84	1.9		2.48	1.8		2.72	1.8	
Prey Specificity	2.22	2.2		2.24	2.2		2.2	2.2		2.2	2.2		2.24	2.2	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.63	2.15		1.6	2.3		1.64	2.1		1.6	2.1		1.68	2.1	
Spawning Cycle	3.2	2.5		3.24	2.5		3	2.5		3.24	2.5		3.32	2.5	
Stock Size/Status	3.93	2.6		3.96	2.6		3.92	2.6		3.92	2.6		3.92	2.6	
Total Sensitivity | Certainty	**High (3) | 0.87**			**High (3) | 0.82**			**High (3) | 0.66**			**High (3) | 0.92**			**Very High (4) | 0.64**		

Exposure

Exposure Factors	Range Mean	Range SD	Range Tally	Eastern Gulf of Maine Mean	Eastern Gulf of Maine SD	Eastern Gulf of Maine Tally	Western Gulf of Maine Mean	Western Gulf of Maine SD	Western Gulf of Maine Tally	Georges Bank Mean	Georges Bank SD	Georges Bank Tally	Southern New England Mean	Southern New England SD	Southern New England Tally
Bottom Temperature	*2.37*	*1.10*		*2.01*	*0.99*		*2.89*	*0.94*		*1.93*	*1.02*		*3.2*	*0.88*	
Bottom Oxygen	2.8	1.01		2.79	0.74		3.46	0.77		2.21	0.95		3.07	0.96	
Bottom Salinity	*2.58*	*1.04*		*2.42*	*0.83*		*1.82*	*0.88*		*3.34*	*0.91*		*2.82*	*0.44*	
Bottom Aragonite Solubility	1.22	0.61		1.03	0.11		1.02	0.11		1.52	0.85		1	0.00	
Small zooplankton integrated biomass	1.27	0.28		1.3	0.26		1.31	0.25		1.1	0.14		1.52	0.38	
Medium zooplankton integrated biomass	1.35	0.28		1.28	0.19		1.47	0.29		1.25	0.22		1.43	0.40	
Large zooplankton integrated biomass	*1.3*	*0.36*		*1.36*	*0.43*		*1.29*	*0.31*		*1.26*	*0.33*		*1.09*	*0.15*	
Total Exposure	**Moderate (1.97)**	**0.93**		**Moderate (1.67)**	**0.83**		**Moderate (2.25)**	**0.89**		**Moderate (1.76)**	**0.94**		**Moderate (2.45)**	**0.77**	
Total Exposure - Important Variables	***Low (1.29)***	***0.63***		***Low (1.01)***	***0.10***		***Low (1.22)***	***0.54***		***Low (1.42)***	***0.77***		***Moderate (1.87)***	***0.72***	

Vulnerability

Overall Vulnerability Rank	Range Mean	Range SD	Range Tally	Eastern Gulf of Maine Mean	Eastern Gulf of Maine SD	Eastern Gulf of Maine Tally	Western Gulf of Maine Mean	Western Gulf of Maine SD	Western Gulf of Maine Tally	Georges Bank Mean	Georges Bank SD	Georges Bank Tally	Southern New England Mean	Southern New England SD	Southern New England Tally
Total Vulnerability	**Moderate (6.17)**	**3.09**		**Moderate (5.01)**	**2.50**		**Moderate (6.75)**	**2.68**		**Moderate (5.28)**	**2.82**		**High (9.8)**	**3.09**	
Total Vulnerability - Important Variables	***Moderate (4.09)***	***2.23***		***Low (3.02)***	***0.31***		***Moderate (3.65)***	***1.61***		***Moderate (4.25)***	***2.30***		***High (7.48)***	***2.88***	

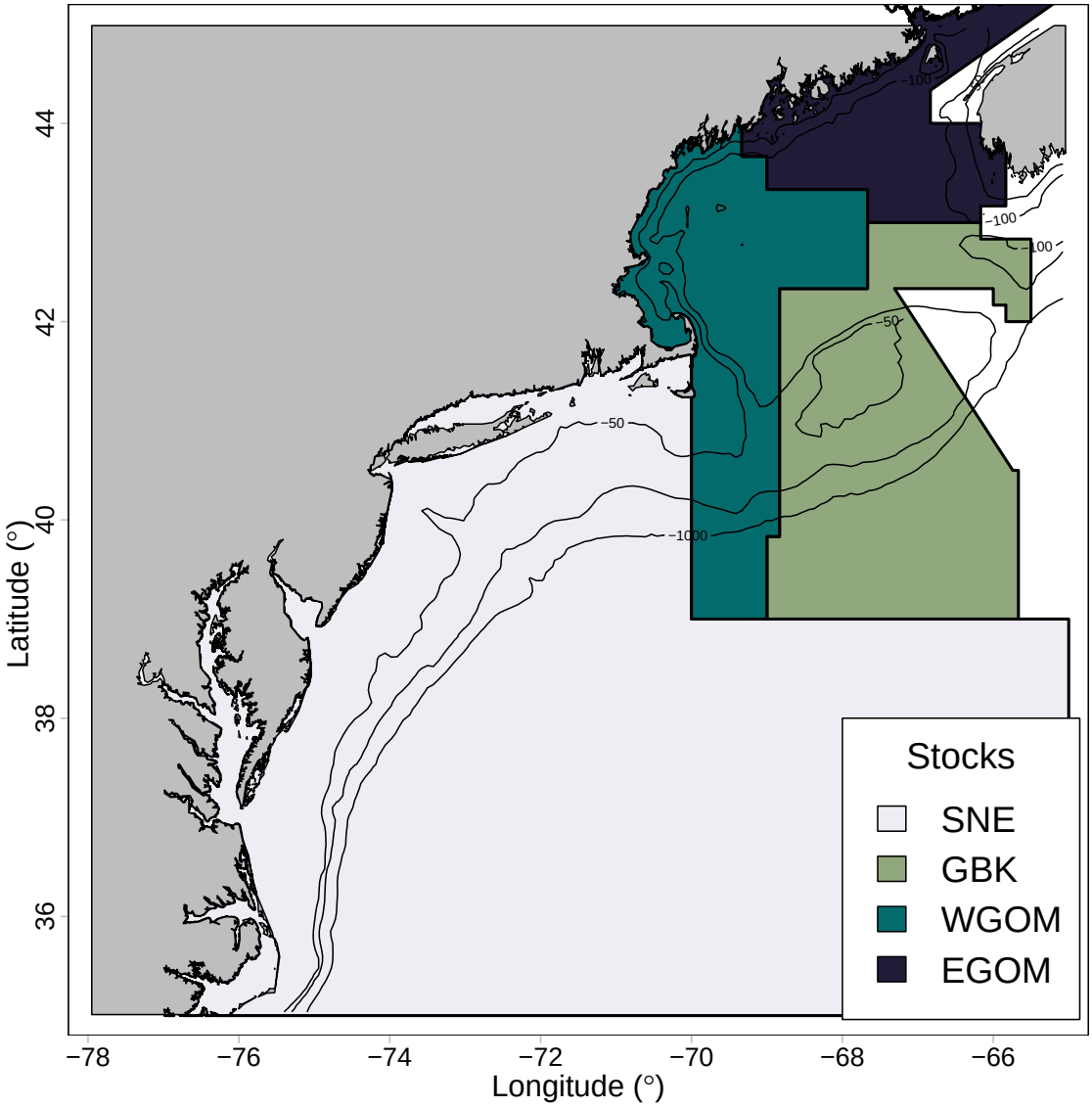
Low Moderate High Very High

12.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability	Moderate - High	Moderate	None - Increase
Exposure	Moderate	High	Decrease
Sensitivity	High – Very High	Moderate	Increase
Directional Effect	Negative	Negative	None

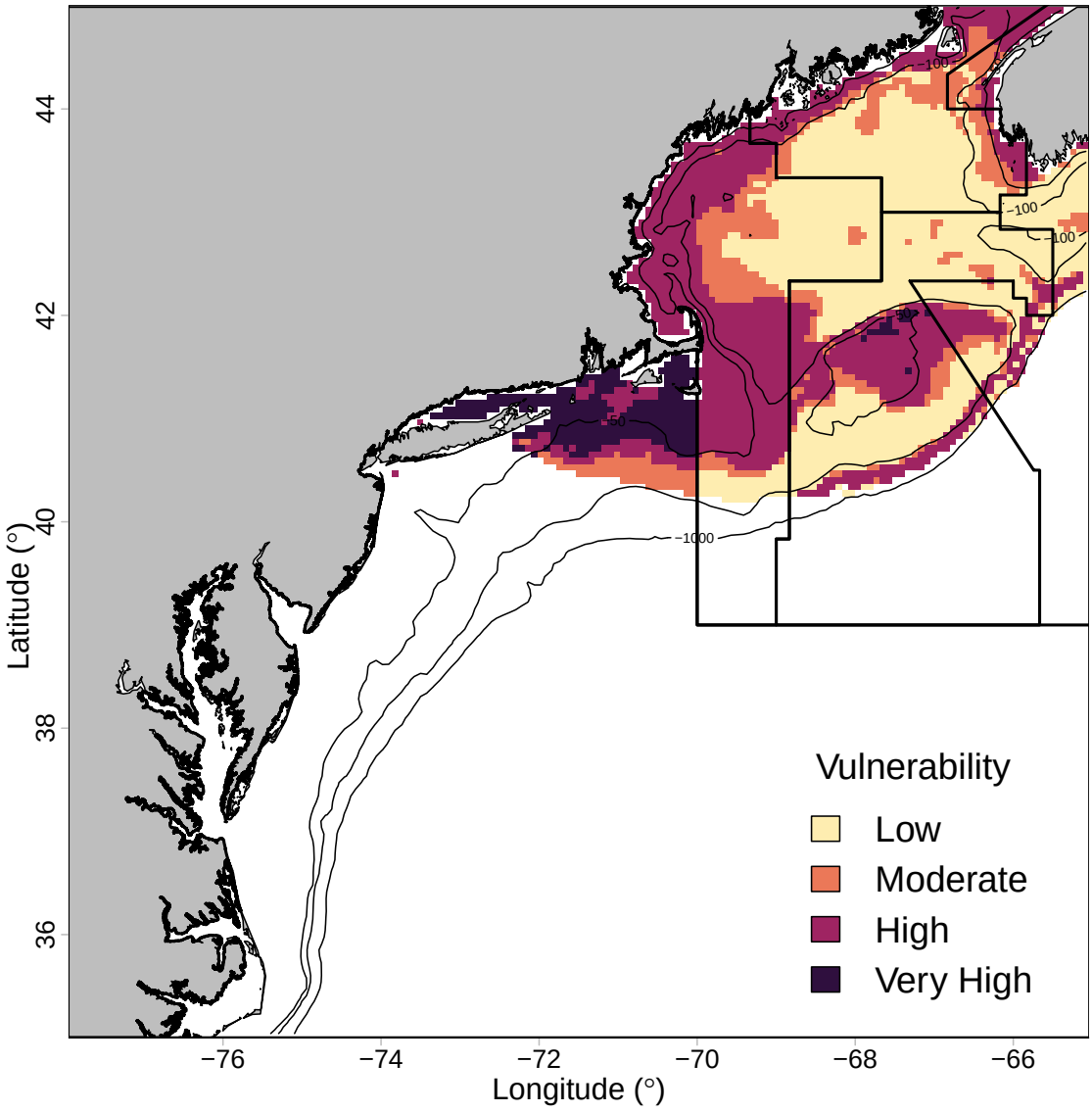
12.3 Stocks

Atlantic cod has four stocks on the Northeast Shelf: Eastern Gulf of Maine (EGOM), Western Gulf of Maine (WGOM), Georges Bank (GBK), and Southern New England (SNE).


Latitude (°)
44
42
40
38
36
−78
−76
−74
−72
−70
−68
−66
Longitude (°)
−100
−100
−100
−50
−50
−1000
Stocks
SNE
GBK
WGOM
EGOM

12.4 Score Descriptions:

12.4.1 Vulnerability:



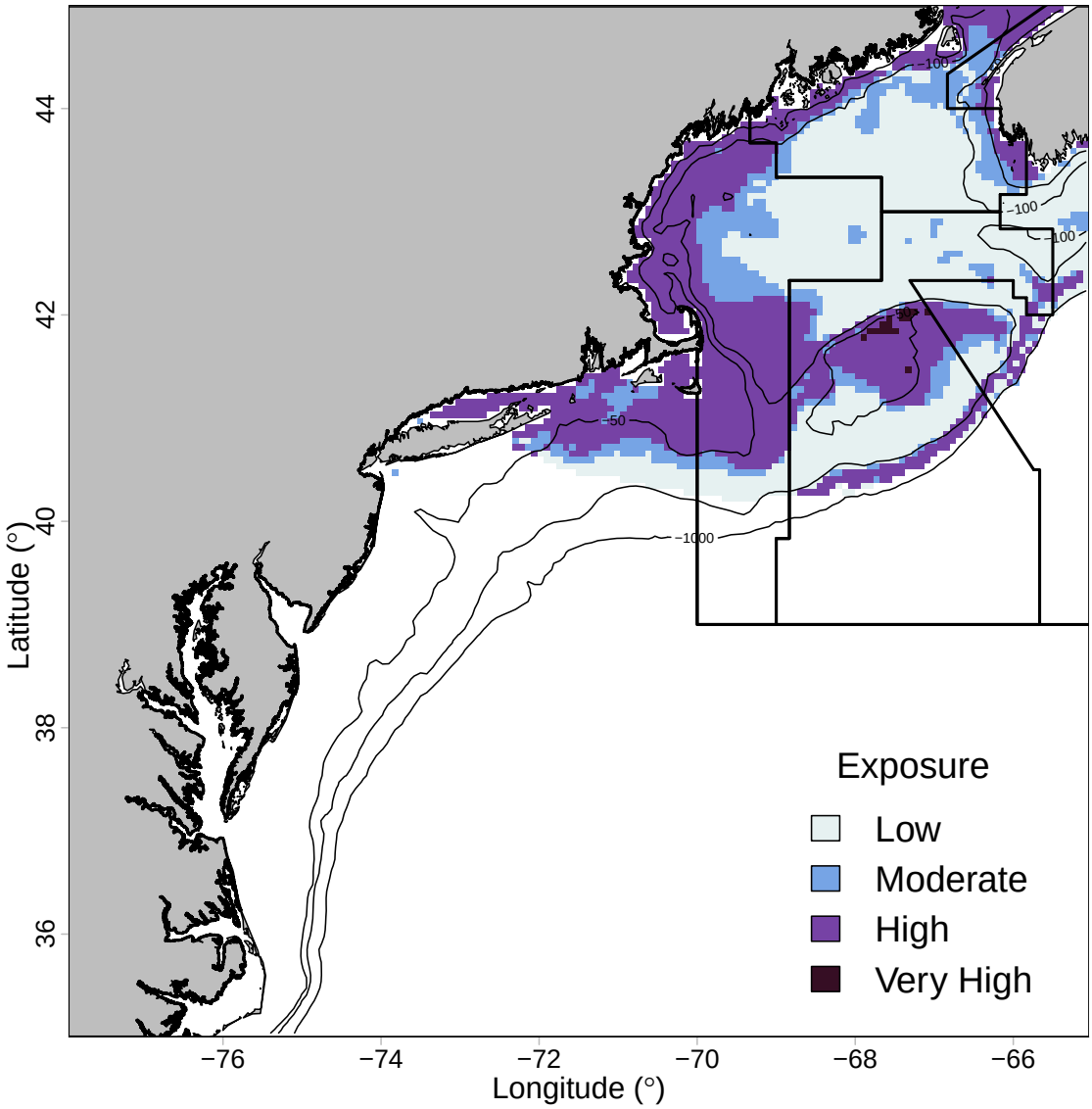
Mean vulnerability scores within the **Eastern Gulf of Maine, Western Gulf of Maine, and Georges Bank** stocks were **Moderate**, ranging from 5.01 to 6.75 (+/- 2.50 - 2.82). Mean scores in the **Southern New England** stock were **High** (9.8 +/- 3.09). This was driven by the higher Sensitivity score within the stock.

Very High Vulnerability scores were found south of Cape Cod, due to high exposure and high sensitivity within the Southern New England stocks. High Vulnerability scores were found along the coast of the Gulf of Maine and over Georges Bank due to high exposure, whereas exposure was Low within the Gulf of Maine basin.

If Exposure scores based only on factors important to the species distribution model were used, Vulnerability scores dropped to Low in the Eastern Gulf of Maine stock. The results in the remaining stocks did not change. In this case, Vulnerability was still high south of Cape Cod, but decreased throughout the Gulf of Maine, only remaining high along Georges Bank.

Vulnerability scores in the previous Assessment were Moderate for Atlantic cod, similar to our results (**hare2016vulnerability--12?**).

12.4.2 Climate Exposure:

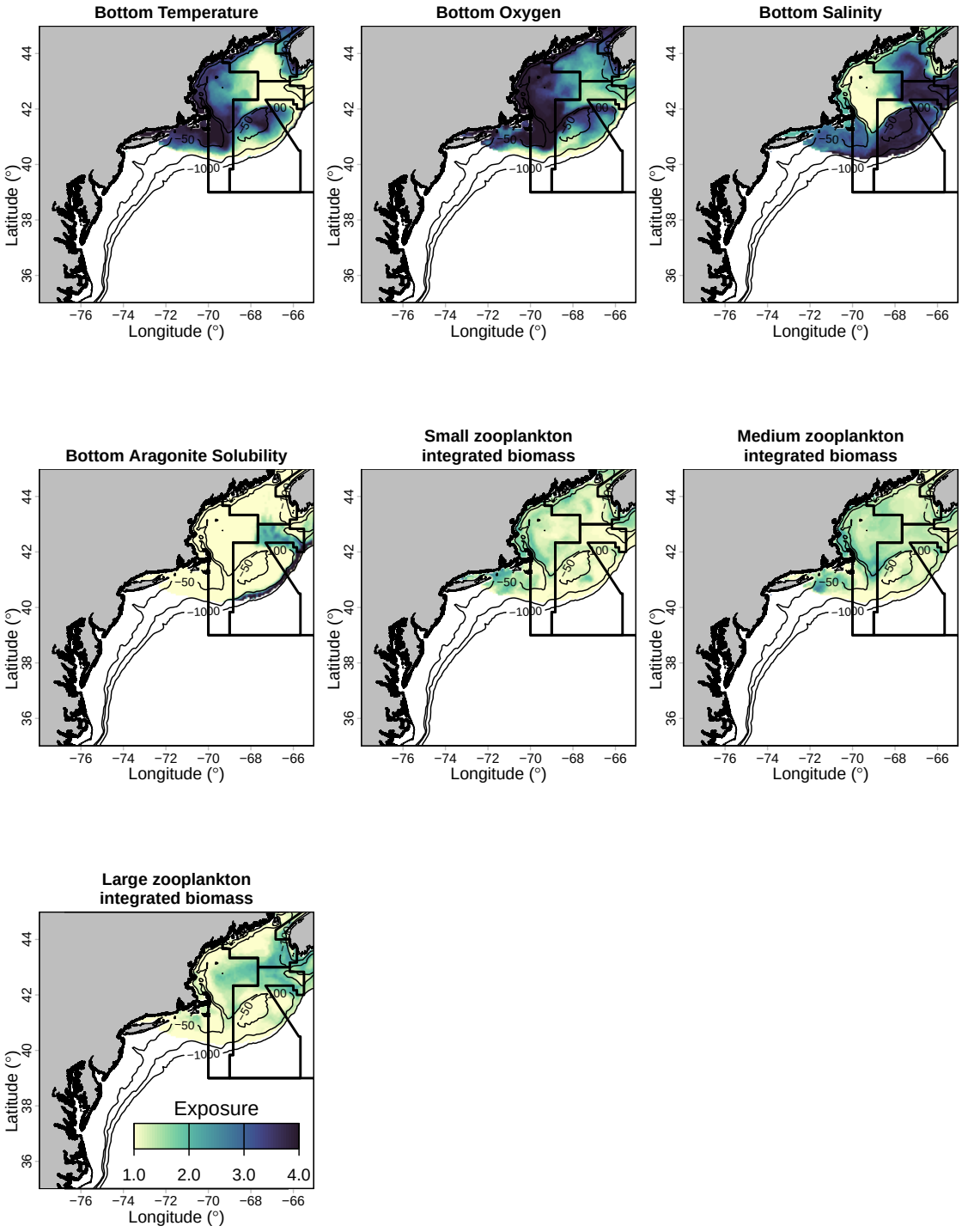


Mean (+/- standard deviation) exposure scores within all four cod stocks were **Moderate** when all exposure factors were considered. Bottom temperature, bottom salinity, and large zooplankton biomass were the most important dynamic variables in the Atlantic Cod species distribution model. Cod are piscivores, feeding on other small fishes that likely prey on zooplankton (**bowman1975food--12?**). Spawning is closely linked to temperature (**lough2004essential--12?**).

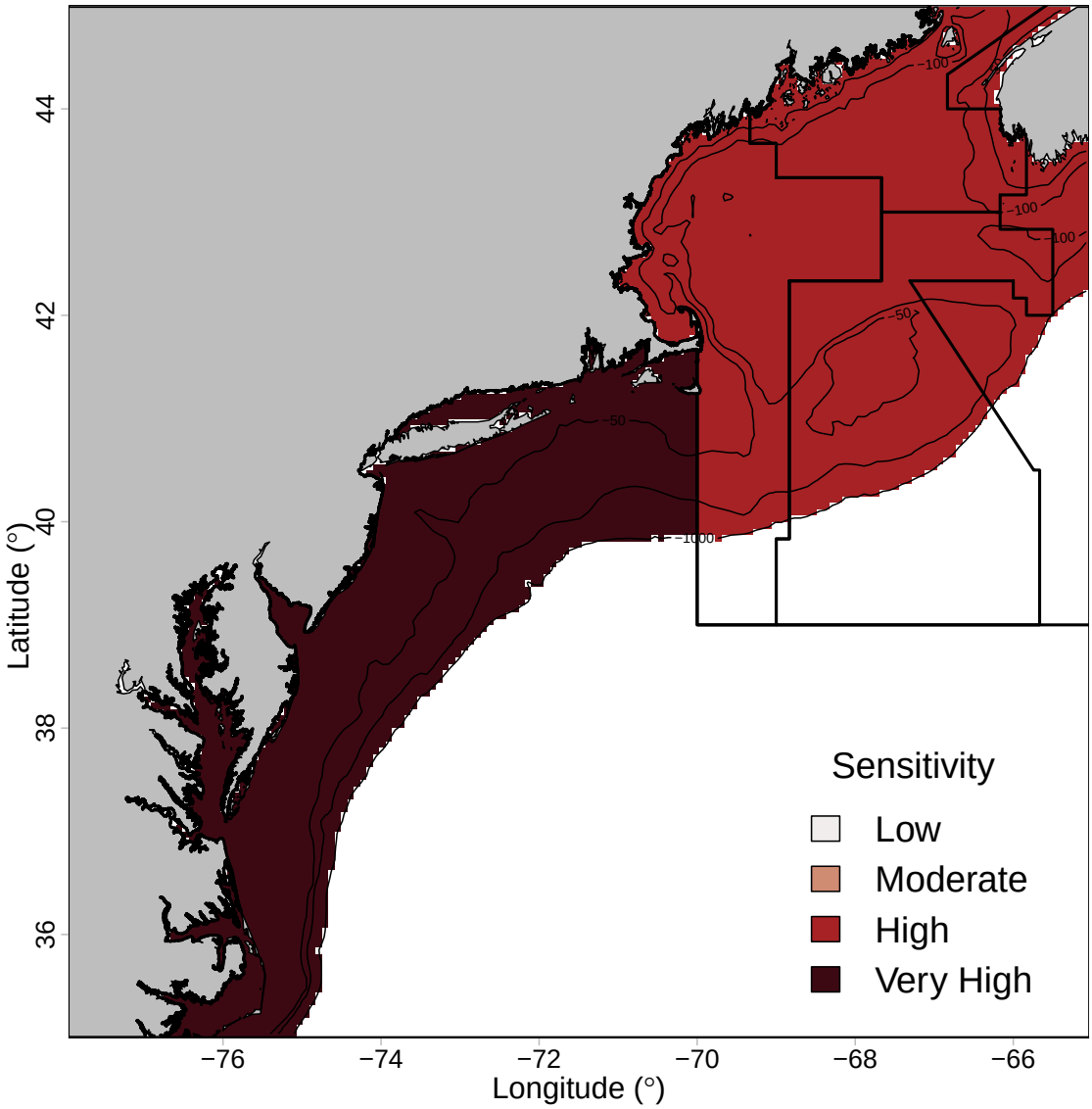
When only important factors were used to calculate exposure, exposure in the Eastern and Western Gulf of Maine stocks, as well as the Georges Bank stock, decreased to Low. The mean exposure score only remained Moderate in the Southern New England stock.

Across the species range, exposure scores were greatest along the coastline, southeast of Cape Cod, and over Georges Bank. These patterns were driven by high exposure to changes in bottom temperature, salinity, and oxygen. High bottom temperature exposure south of Cape Code and Long Island drove the higher scores observed in the Western Gulf of Maine and Southern New England Stocks.

In the previous assessment, cod received a High exposure score, driven by high exposure to increased sea surface temperatures and ocean acidification (**hare2016vulnerability--12?**).


Bottom Temperature
Bottom Oxygen
Bottom Salinity
Bottom Aragonite Solubility
Small zooplankton integrated biomass
Medium zooplankton integrated biomass
Large zooplankton integrated biomass
Latitude (°)
Longitude (°)
36
38
40
42
44
−76
−74
−72
−70
−68
−66
−50
−100
−1000
Exposure
1.0
2.0
3.0
4.0

12.4.3 Biological Sensitivity:



Of the four cod stocks, three (**Eastern Gulf of Maine, Western Gulf of Maine, and Georges Bank**) received **High** sensitivity scores. The **Southern New England** stock received a **Very High** score. Certainty was highest for the Georges Bank (92%) stock, followed by the Eastern Gulf of Maine (82%), Western Gulf of Maine (66%), and Southern New England (64%).

Stock size, complexity in reproductive strategy, and egg and larval survival and settlement requirements drove the scores across all four stocks. Spawning cycle and historical variability in temperature also scored above 3 for all stocks.

Cod are currently overfished in the Gulf of Maine (**nefsc2021update--12?**). Spawning is coupled with copepod production (**brander1992distribution--12?**) and may be impacted by warming temperatures (**behan2023profile--12?**; **drinkwater2005response--12?**). Warming temperatures may also reduce adult fitness, and thus reduce egg quality (**klein2016effects--12?**). Recruitment success is linked to retentive circulation patterns in the Gulf of Maine and on Georges Bank (**cong1996retrospective--12?**; **lough1989movement--12?**), which can be altered by large scale climate patterns and overall regional warming (**pershing2015slow--12?**).

Atlantic cod sensitivity has increased since the last assessment. Atlantic cod were previously scored as Moderate (**hare2016vulnerability--12?**). Stock size/status and egg and larval survival and settlement requirements (formally early life history requirements) drove the scores in both assessments.

12.5 Data Quality Metrics

12.5.1 Life History Data Quality:

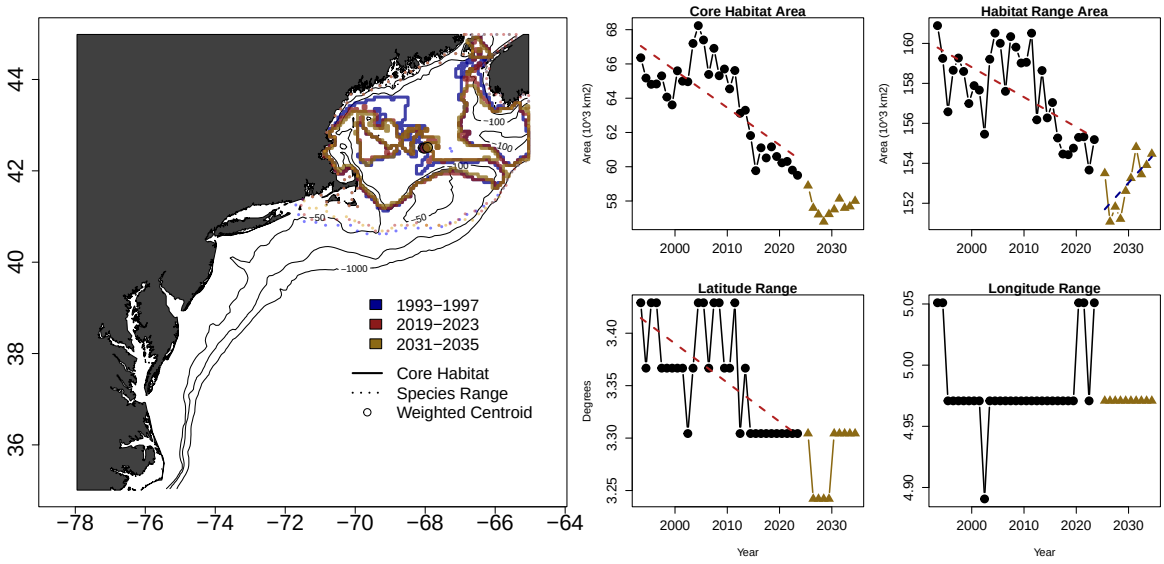
82% of the data quality scores were greater than or equal to 2 for the Georges Bank, Western Gulf of Maine, and Southern New England stocks. This value was 83% for the Eastern Gulf of Maine stock, indicating strong data quality. For all stocks, only Population Growth Rate was considered a data gap (mean attribute data quality score < 2).

12.5.2 Model Confidence:

Experts had moderate confidence in the Atlantic cod model (mean confidence = 2.1 +/- 0.32). Experts noted that cod are often associated with rocky bottoms, which are undersampled by traditional trawl surveys used to build the models. The ensemble model AUC was 0.84, indicating good performance.

12.6 Additional Metrics

12.6.1 Distribution Shifts:



Note: For an animated version of this plot, please see the online version of this document.

While cod populations have declined over the past 30 years, distributions have not shifted significantly. The weighted centroid of the distribution shifted 3.8 km ESE between 1993-1997 and 2019-2023. Significant declines in latitudinal range led to significant losses in core and total habitat area. This was driven by the southern edge of the species' range moving further north.

In the next ten years, the distribution centroid is expected to shift a further 7.3 km ENE in comparison to its current location. Core habitat and total habitat areas are expected to decline, in comparison to 2019-2023 averages, but models predict a significant increase in total habitat area from 2025-2035. This is likely driven by the predicted cooling of bottom temperatures along the Northeast Shelf.

Time Period	Centroid Shift (km & bearing)	Core Habitat Area (change) (km2)	Habitat Total Area (change) (km2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	65,441	159,006	3.43	4.97
2019-2023	3.8 km ESE	60,182 (-5,258)	154,872 (-4,133)	3.3 (-0.12)	4.97 (0)

Time Period	Centroid Shift (km & bearing)	Core Habitat Area (change) (km2)	Habitat Total Area (change) (km2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
2031-2035	7.3 km ENE	57,770 (-2,412)	153,955 (-917)	3.3 (0)	4.97 (0)

12.6.2 Directional Effect:

All four stocks of Atlantic Cod will likely be negatively impacted by future climate change (100% certainty in expert scores). This is similar to the previous Assessment (**hare2016vulnerability--12?**). Continued warming will likely drive declines in recruitment and shifts in spawning phenology.

12.7 Climate Effects on Abundance and Distribution

Recent warming in the Gulf of Maine relating to a northward shift in the Gulf Stream and changes in the Atlantic Multidecadal Oscillation and Pacific Decadal Oscillation led to reduced Atlantic cod recruitment and increased mortality (**pershing2015slow--12?**). Rising temperatures may affect Atlantic cod spawning locations and timing (**behan2023profile--12?**). (**drinkwater2005response--12?**) suggested Atlantic cod spawning grounds will experience a shift northward, with spawning timing occurring later in the season as a response to warming ocean temperatures. Climate change may also increase in natural mortality in adult Atlantic cod due to declines in fish condition (**behan2023profile--12?**). (**drinkwater2005response--12?**) predicted large-scale declines in Atlantic cod in the Gulf of Maine if bottom temperatures increase above 12°C, either through changes in distribution to colder waters or natural mortality. Reduced fitness in Atlantic cod adults from climate impacts may result in lower quality eggs (**klein2016effects--12?**); however, there may be some capacity to adapt to slower rates of warming (**langan2020abundance--12?**).

12.8 Life History Synopsis

Cod in the Northwest Atlantic range from Greenland to Cape Hatteras, NC, with a majority of biomass in the Gulf of Maine and on Georges Bank (**behan2023profile--12?**). Spawning timing varies by location - later in the year with increasing latitude - although peaks generally occur in the winter and spring (**lough2004essential--12?**) in water temperatures between 5-7°C. In spring, Atlantic cod spawning matched the variation in peak abundance of *Calanus finmarchicus*. In fall, cod spawned in some, but not all, areas where *Calanus* were

abundant. This is consistent with the hypothesis that cod spawning is coupled to copepod production (**brander1992distribution--12?**). Atlantic cod eggs are pelagic and hatch after 8-60 days, varying with temperature (**hardy1978development--12?**). Recruitment success increases when dispersion of larvae by currents is reduced (**cong1996retrospective--12?**). On Georges Bank, a clockwise gyre circulation pattern tends to retain and isolate eggs and larvae (**lough1989movement--12?**). Transformation from larvae to the juvenile stage occurs at sizes greater than 20 mm, where they then move to the bottom (**lough2004essential--12?**). Early juveniles are dependent on gravel substrates or grass beds in nearshore nurseries to favor survival through predator avoidance and increased food availability (**lough1989movement--12?**). In the Gulf of Maine, juveniles prey on mollusks and other invertebrates, while adults consume a greater proportion of fish including herring, redfish, mackerel, and other cod (**bowman1975food--12?**). Adult cod have few predators, other than large sharks and seals (**neuenhoff2019continued--12?**). Recent modeling studies illustrate growing populations of grey seals, harp seals, and dogfish are a major barrier to cod population recovery (**behan2023profile--12?**). Adults in the Newfoundland stock migrate, leaving overwintering areas in deep oceanic waters and follow tongues of deep, relatively warm, oceanic waters ("highways") across the shelf to summer feeding areas nearshore. They then move northward along the Newfoundland coast in late summer, and eventually return to wintering areas (**lough2004essential--12?**). Previous research suggested that in the middle part of their range, Atlantic cod only undertake minor seasonal movements in reaction to changing temperatures. However, more recent research suggests that there are inshore/offshore migrations that correspond to spawning and feeding activity in the Gulf of Maine (**zemeckis2017seasonal--12?**). As of 2025, Atlantic cod are divided into 4 stock units (Western & Eastern Gulf Maine, Georges Bank, and Southern New England) based on spawning locations, larval transport pathways, and genetic markers (**nefmc2025multispecies--12?**). All stocks are managed under the New England Fishery Management Council's Northeast Multispecies Fishery Management Plan (**kleinmacphee2012atlantic--12?**). The stock is overfished and overfishing is occurring in the Gulf of Maine (**nefsc2021update--12?**).

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13 Atlantic croaker (*Micropogonias undulates*)

13.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = Moderate

Data Quality Model Confidence Score =Life History Data Quality = 73%

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[1] "C:/Users/katherine.gallagher/Documents/READ-EDAB-CVA2.0Manual/test_images/Atlantic croa

13.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	Moderate	Low	Increase
Directional Effect	Neutral	Positive	Increase

13.3 Score Drivers

13.3.1 Vulnerability:

add text here

13.3.2 Climate Exposure:

add text here

13.3.3 Biological Sensitivity:

Atlantic croaker received a sensitivity score of **Moderate**. Certainty was high (90%). Three attributes received scores greater than 2.5: 1) Egg and Larval Survival and Settlement Requirements (2.9), 2) Other Stressors (2.7), and 3) Stock Size (2.5).

Croaker eggs and larvae rely on currents and tidal activity for transport to suitable nursery habitats (**sempsrott2017biological--13?**). Larvae are selective feeders, only feeding on a subset of plankton species (**govoni1986selective--13?**). Juveniles are also dependent on native salt marsh habitats, which themselves are highly vulnerable (**guyant2022northeast--13?**) and are susceptible to eutrophication and hypoxic events. Croaker had a low sensitivity score in the previous assessment (**hare2016vulnerability--13?**). Egg and Larval Survival and Settlement Requirements drove the previous sensitivity score.

13.4 Data Quality Metrics

13.4.1 Life History Data Quality:

73% of sensitivity attribute data quality scores were greater than or equal to 2, indicating good data quality. Four attributes were considered data gaps (mean attribute data quality score < 2): Complexity in Reproductive Strategy, Egg and Larval Survival and Settlement Requirements, Population Growth Rate, and Stock Size/Status.

13.4.2 Model Confidence:

add text here

13.5 Additional Metrics

13.5.1 Distribution Shifts:

add text here

13.5.2 Directional Effect:

The effect of climate change on croaker is estimated to be **neutral** with 89% certainty in expert scores. Warmer temperatures may decrease young-of-year mortality rates. However, juvenile habitats are highly vulnerable to climate change. In the previous assessment, the effects of climate change were estimated to have a positive impact on croaker (**hare2016vulnerability--13?**).

13.6 Climate Effects on Abundance and Distribution

As a sub-tropical species, Atlantic croaker may experience both positive and negative effects on their abundance and distribution as a result of climate change. Interannual variability in croaker abundance may be climate related, with colder winters causing increased mortality in overwintering young-of-the-year (**lankford2001low--13?**). Warm winters, however, may result in increased abundances, or "outbursts" of juveniles in the spring that contribute to expansion of the population's distribution (**hare2007mechanistic--13?**). (**diamond2013simulating--13?**) modeled Atlantic croaker abundance in the Mid-Atlantic with global climate change projections and found increased croaker abundance with lower inter-annual variability under global climate change. Positive responses in sensitivity analyses stemmed from warmer temperatures, and negative responses were due to rising variability in salinity and increases in offshore transport and sea level rise (**diamond2013simulating--13?**). Common juvenile habitat types (submerged aquatic vegetation, salt marshes, and intertidal habitats) are highly vulnerable to climate change (**guyant2022northeast--13?**).

13.7 Life History Synopsis

Atlantic croaker are a demersal species found from the Gulf of Maine to Argentina. Along the United States Atlantic coast, they are most abundant from Chesapeake Bay to northern Florida (**asmfc2020addendum--13?**). Atlantic croaker spawn from July - December in inshore shelf and estuarine waters between Delaware Bay and Cape Hatteras, peaking in August - October near Chesapeake Bay (**able1998first--13?**). Their name comes from the croaking noises the fish make during spawning by vibrating their abdominal muscles against their swim bladder (**asmfc2020addendum--13?**). After 30-90 days post-hatch Atlantic croaker larvae are driven inshore by winds (**thorrold1997response--13?**) where they migrate to estuarine habitats (**sempsrott2017biological--13?**). Initial fall recruitment of croaker depends on fall continental shelf winds to provide transport into Chesapeake Bay (**norcross1991estuarine--13?**). Recruitment to specific estuaries is influenced by tidal activity and currents, such that eggs and larvae are dependent on transport to suitable habitats for survival (**sempsrott2017biological--13?**). Atlantic croaker larvae are selective feeders, as they eat fewer taxa of suitable size than are present in ambient

plankton (**govoni1986selective--13?**). Juvenile croaker are highly dependent on marine, estuarine, and riverine mud and sand substrates, as well as native salt marsh habitats (**guyant2022northeast--13?**). The major prey of young-of-the-year Atlantic croaker are polychaetes, copepods, and mysids. Detritus is also a major component of the juvenile diet (**ffwcc2010croaker--13?**) (**sink2011species--13?**). Adult Atlantic croaker perform seasonal migrations, moving northward and inshore to estuaries and bays in the spring - fall and southward and offshore to shelf waters in the winter (**asmfc2020addendum--13?**). They reach maturity by 2 years of age and can live up to 17 years, although lifespans are commonly below 10 years (**asmfc2020addendum--13?**). Adult Atlantic croaker are opportunistic feeders and are known to consume polychaetes, anchovies, mysid shrimp, amphipods, fishes and crabs, and detritus in Chesapeake Bay (**nye2011annual--13?**). Atlantic croaker are managed under Amendment 1 to the Atlantic States Marine Fisheries Commission's Interstate Fishery Management Plan for Atlantic Croaker (2005) and Addenda I, II, and III (2011, 2014, 2020). (**asmfc2020addendum--13?**). The most recent benchmark stock assessment was completed in 2017, where the current stock status could not be determined because the assessment results were sensitive to certain modeling assumptions, but any further depletion in the stock was unlikely (**asmfc2017croaker--13?**).

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14 Atlantic halibut (*Hippoglossus hippoglossus*)

14.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = High **Data Quality**Model Confidence Score =Life History Data Quality = 87%

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14.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	High	High	None
Directional Effect	Negative	Negative	None

14.3 Score Drivers

14.3.1 Vulnerability:

add text here

14.3.2 Climate Exposure:

add text here

14.3.3 Biological Sensitivity:

Atlantic halibut scored **High** with a certainty of 100%. The score was driven by Stock Size (3.8) and Population Growth Rate (3.6) attributes. While it did not contribute to the score, it is worth noting that the only other attribute that scored higher than 2.5 was Spawning Cycle (2.6). Halibut mature slowly and are long lived species, living upwards of 50 years and maturing between 5-13 years (**Armsworthy2014--14?**). While the most recent stock assessment could not determine the stock status analytically, previous assessments have suggested that halibut are overfished (**Northeast_Fisheries_Science_Center_US2015-cv--14?**). Halibut also received a score of high sensitivity in the previous Climate Vulnerability Assessment (**hare2016vulnerability--14?**). This score was also driven by Stock Size/Status and Population Growth Rate.

14.4 Data Quality Metrics

14.4.1 Life History Data Quality:

87% of data quality scores for the sensitivity attributes were greater than or equal to 2, indicating strong data quality. Stock Size/Status was the only identified data gap (mean attribute data quality score < 2).

14.4.2 Model Confidence:

add text here

14.5 Additional Metrics

14.5.1 Distribution Shifts:

add text here

14.5.2 Directional Effect:

Halibut will likely be **negatively** impacted by climate change (100% certainty in expert scores). While halibut distributions are expanding due to warmer temperatures, they are shifting northward and off the Northeast shelf into Canadian waters. This is similar to the result from the 2016 Assessment (**hare2016vulnerability--14?**).

14.6 Climate Effects on Abundance and Distribution

(**Nye2009--14?**) observed a northward shift in Atlantic halibut distribution and an increase in the depth of occurrence on the Northeast U.S. continental shelf. The species' thermal habitat has widened, likely contributing to its range expansion (**Shackell2022--14?**), and biomass and landings in Canadian waters have thus increased greatly since the early 2000s (**Trzcinski2016--14?**). (**Shackell2022--14?**) calculated growth rates from tagging studies and found faster growth in Atlantic halibut in warmer, southern waters. From 1990 to 2018, thermal habitat rose by 11.6% (+/- 7.35%) and the number of degree days where Atlantic halibut are experiencing growth increased by 13.5 °C days (+/- 7.86 °C days) (**Czich2023--14?**). Continued warming is likely to increase abundance in the northern end of the species' range; however elevated CO^2^ levels and rising temperatures may lead to oxidative stress (**Almroth2019--14?**).

14.7 Life History Synopsis

Atlantic halibut are found from the Labrador Shelf in the North to Virginia, although they are rare south of Long Island (**Cargnelli1999--14?**). Adult halibut spawn from late fall to late spring in deep continental shelf waters, with presumed spawning peaks occurring at 2 - 9 day intervals between October and January (**Collette2002--14?**; **Armsworthy2014--14?**). Eggs hatch in 13-20 days depending on the temperature (**Blaxter1983--14?**); larvae have been found in the plankton as much as a year after spawning (**Collette2002--14?**). Depending on current speeds and larval duration, eggs and larvae may drift anywhere from 300-500 km. Tagging studies of spawning individuals suggest that larvae can be transported into the Gulf of St Lawrence, which may provide nursery habitats (**Armsworthy2014--14?**). Juveniles are most common in shallow water from 20-60 m in nursery areas on the shelf (**Cargnelli1999--14?**). Juvenile halibut up to 30 cm in length consume invertebrates, mostly annelid worms and crustaceans; they then feed on fish and invertebrates from 30-80 cm in length, and then almost exclusively on fish as they grow larger (**Scott1988--14?**) (**Bowman2000--14?**). Adults normally inhabit benthic waters from 2-15°C, but occupy the entire water column to feed and spawn (**Cargnelli1999--14?**) (**Collette2002--14?**) (**Armsworthy2014--14?**). In the Gulf of Maine, tagged individuals have been observed conducting both shore and long distance migrations, as well as being resident year-round (**Shackell2022--14?**). Age at maturity ranges from 7 - 13 years (females) and 5 - 12 years (males), and their maximum age is between 38-50 years (**Armsworthy2014--14?**). Atlantic halibut is currently managed by the New England Fishery Management Council under the Northeast Multispecies Fishery Management Plan. The most recent management track assessment was completed in 2024, where stock status could not be determined analytically due to a lack of biological reference points (**NEFSC2024_halibut--14?**).

14.8 Figures

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15 Atlantic herring (*Clupea harengus*)

15.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = Moderate

Data Quality Model Confidence Score =Life History Data Quality = 95%

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15.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	Moderate	Low	Increase
Directional Effect	Negative	Negative	None

15.3 Score Drivers

15.3.1 Vulnerability:

add text here

15.3.2 Climate Exposure:

add text here

15.3.3 Biological Sensitivity:

Atlantic herring had a **Moderate** sensitivity score, with a certainty of 90%. This score was driven by two attributes: Stock Size (4) and Spawning Cycle (2.8).

Spawning success and recruitment is highly correlated with temperatures. The optimal spawning temperature window has shortened over the past 30 years (**Kanwit2009--15?**; **Molina2025--15?**). Mismatches between larvae and prey abundance lead to higher mortality (**Randall2022--15?**). As of the 2024 assessment, herring are overfished but not experiencing overfishing (**NEFSC2024_herring--15?**).

In the previous assessment, herring received a low sensitivity score. This score was driven by Early Life History Requirements, citing the need for demersal spawning habitat (**hare2016vulnerability--15?**).

15.4 Data Quality Metrics

15.4.1 Life History Data Quality:

95% of data quality scores were greater than or equal to 2, indicating excellent data quality. No data gaps (mean attribute data quality score < 2) were identified for this species.

15.4.2 Model Confidence:

add text here

15.5 Additional Metrics

15.5.1 Distribution Shifts:

add text here

15.5.2 Directional Effect:

Herring are very likely to be **negatively** impacted by future climate change along the Northeast Shelf (100% certainty in expert scores). Warming temperatures and increasing bottom salinities are decreasing the amount of favorable habitat for herring (**Suca2021--15?**). Mismatches between larvae and prey abundances have resulted in slower growth and higher larval mortality (**Randall2022--15?**). The 2016 Climate Vulnerability Assessment for the Northeast Shelf also suggested that herring would be negatively impacted, but with more uncertainty (**hare2016vulnerability--15?**).

15.6 Climate Effects on Abundance and Distribution

A number of Atlantic herring life history parameters are linked to drivers of climate change. Climate-related mismatches between Atlantic herring larval feeding and prey abundance have resulted in slower growth and higher mortality (**Randall2022--15?**). (**Groger2010--15?**) found that lags in the North Atlantic Oscillation (NAO) and Atlantic Multi-decadal Oscillation (AMO) were the primary drivers of Atlantic herring recruitment. Winter NAO has also indirectly influenced recruitment in multiple herring stocks (**Axenrot2003--15?**; **Oskarsson2010--15?**; **Lynam2005--15?**) and may be used to predict recruitment with lead times up to 5 years (**Groger2010--15?**). Atlantic herring recruitment is also influenced by larval temperatures, the Gulf Stream Index, and food availability from water mass movement (**Molina2025--15?**). Favorable habitat for Atlantic herring in the Northwest Atlantic is currently decreasing with rising temperatures and bottom salinity (**Suca2021--15?**), with herring populations in warmer waters responding more negatively to increasing temperatures than those in colder waters (**Beaudry2024--15?**). Herring larvae in the Baltic Sea are facing more heat stress and cardiac dysfunction with an increase in the number of marine heat waves (**Groger2024--15?**). Migration and distribution of Atlantic herring is also linked with thermal oceanic fronts between colder, less saline continental shelf water and warmer, more salient slope water (where plankton are also usually more abundant) (**Kelly1986--15?**).

15.7 Life History Synopsis

Atlantic herring in the Northwest Atlantic reside from southern Labrador in Canada to Cape Hatteras, NC (**Stevenson2005--15?**). They spawn primarily on Jeffreys Ledge and Georges Bank between July and November, with peaks in August - September in the Gulf of Maine. However, recent research has suggested that the optimal spawning temperature window has shortened over the past 30 years (**Kanwit2009--15?**; **Molina2025--15?**). Early life history development is inversely related to water temperature, with slower growth occurring at higher temperatures (**Stevenson2005--15?**). Eggs hatch after 6-15 days, and larval development time ranges from 4-8 months. Larvae absorb their yolk sac within 2.5 - 6 days, at which point

the larvae become pelagic and can be transported away from or retained near spawning habitats, depending on prevailing current systems (**Lough1982--15?**). Predation of eggs is linked to high mortality. Predators include haddock, cod, mackerel, winter flounder, moon snails, hermit crabs and starfish. Strong haddock recruitment has been linked to low abundances of larval herring (**Boyar1973--15?**) (**Burbank2023--15?**). Larval herring prey on copepods, with size depending on the size of the larvae (**Sherman1971_larval--15?**). Juveniles form large schools in coastal waters (i.e. estuaries and embayments) throughout the Gulf of Maine in the spring and summer, and overwinter in deep bays or offshore (**Stevenson2005--15?**). Juveniles and adults are planktivorous and opportunistic feeders. As they grow and the size of their jaws increases, they consume larger organisms. Their diet therefore varies with season, their age and size, and location. (**Stevenson2005--15?**) (**Sherman1971_juvenile--15?**). A lack of availability of preferred prey in adults (due to strong mixing in the water column) can lead to suboptimal feeding and slow growth resulting in increased mortality from starvation and predation (**Stevenson2005--15?**). Adult herring make extensive seasonal migrations between summer spawning grounds on Georges Bank and in the Gulf of Maine and overwintering areas in southern New England and the Mid-Atlantic region (**Stevenson2005--15?**). There are two major spawning stocks in the Northwest Atlantic: the Gulf of Maine and Georges Bank/Nantucket Shoals. Stocks are separated based on larval distribution patterns, spawning times/locations, and growth rates (**Iles1982--15?**) (**Boyar1973--15?**) (**Anthony1980--15?**). Both stocks are currently assessed in the U.S. as a single coastal stock complex managed by the New England Fishery Management Council. As of the most recent assessment (**NEFSC2024_herring--15?**), the stock is overfished but not experiencing overfishing based on the biological reference points for spawning stock biomass (SSB) and fishing mortality (F).

15.8 Figures

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16 Atlantic mackerel (*Scomber scombrus*)

16.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = Low

Data Quality Model Confidence Score =Life History Data Quality = 95%

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16.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	Low	Moderate	Decrease
Directional Effect	Negative	Negative	None

16.3 Score Drivers

16.3.1 Vulnerability:

add text here

16.3.2 Climate Exposure:

add text here

16.3.3 Biological Sensitivity:

Atlantic mackerel received a **Low** sensitivity score with a certainty of 100%. The only score greater than 2.5 was Stock Size (3.2). Stocks are not currently overfished or experiencing overfishing according to the most recent stock assessment. The stock is rebuilding to target levels (**nefsc2025_mack--16?**).

In the previous Climate Vulnerability Assessment, mackerel were given a moderate sensitivity score. This previous score was driven by Stock Size, Egg and Larval Survival and Settlement Requirements (Early Life History Requirements), and Spawning Cycle. Stock sizes were not classified at the time of the last assessment, and it was unclear how spawning cycle and early life stages would be impacted by warming waters (**hare2016vulnerability--16?**). Recent work has suggested that spawning habitats will likely shift west and north under future warming conditions (**bruge2016--16?**).

16.4 Data Quality Metrics

16.4.1 Life History Data Quality:

95% of sensitivity attribute data quality scores were greater than or equal to 2, indicating excellent data quality. Only Relationship to Calcified Organisms was considered a data gap (mean attribute data quality score < 2).

16.4.2 Model Confidence:

add text here

16.5 Additional Metrics

16.5.1 Distribution Shifts:

add text here

16.5.2 Directional Effect:

Atlantic mackerel are likely to be **negatively** impacted by climate change. Expert certainty was 83%. Regional warming has resulted in northeasterly shifts in mackerel distributions. Spawning habitat could increase in more northern regions such as the Gulf of St Lawrence as mackerel seek out preferred temperatures (**overholtz2011--16?**). This result is similar to the results from the 2016 Climate Vulnerability Assessment for the region (**hare2016vulnerability--16?**).

16.6 Climate Effects on Abundance and Distribution

Atlantic mackerel are sensitive to temperature changes, particularly temperatures greater than 5°C (**overholtz2011--16?**). From 1968 to 2008, the distribution of mackerel in the Northwest Atlantic has shifted roughly 250km north and east in addition to changes in depth distribution to more shallow on-shelf waters. These changes in position and depth are known to be correlated with the gradual warming of the Northwest Atlantic (**overholtz2011--16?**). Atlantic mackerel egg distribution has also shifted northeastward as a consequence of warming, depleting egg production in important spawning locations in southern New England (**richardson2020--16?**). (**bruge2016--16?**) projected changes in Northeast Atlantic mackerel spawning distribution under future climate scenarios, and found that the spawning center of gravity is expected to shift westward and northward. (**mbaye2019--16?**) investigated effects of environmental conditions on Atlantic mackerel spawning habitat in the Gulf of St. Lawrence. Generalized additive model (GAM) results showed that spawning habitat in the Gulf of St. Lawrence could expand and extend to western and southern Newfoundland.

16.7 Life History Synopsis

Atlantic mackerel in the Northwest Atlantic inhabit cold and temperate continental shelf waters from the Gulf of St Lawrence to Cape Lookout, North Carolina (**studholme1999--16?**). Spawning occurs from April to August in the Northwest Atlantic and progresses from south to north with warming temperatures and following migration patterns (**sette1943--16?**). There are two spawning contingents: the southern contingent spawns from April to July in the Mid-Atlantic Bight and Gulf of Maine, and the northern contingent spawns in the southern Gulf of St. Lawrence in June and July (**berrien1982--16?**) (**morse1987--16?**). Temperature is a cue for spawning. Spawning does not occur until water has warmed to 8°C with most egg production occurring between 9-14°C (**sette1943--16?**). More recent work has shown that mackerel spawning and migration timing is negatively correlated with sea surface temperature (**jansen2011--16?**). Early life stage development happens quickly, with eggs hatching in 4-7.5 hours, depending on temperature (**berrien1982--16?**). Larvae ($<$ 13 mm) occur primarily in offshore waters from Chesapeake Bay to southern Gulf of St. Lawrence (**sette1943--16?**).

Larval and juvenile depth varies diurnally and with age (**collette2002__mack--16?**). Larvae feed primarily on zooplankton and copepods, and larvae > 6.4 mm are known to experience cannibalistic behavior (**castonguay2008--16?**). Juveniles rely on water column habitats, both on the shelf, and in estuaries (**guyant2022--16?**). Juveniles and adults are opportunistic feeders that prey on a wide range of crustaceans, copepods, planktonic organisms, and fishes (**sette1950--16?**). Squid, seabirds and larger piscivores, including Atlantic cod, prey on juvenile mackerel (**studholme1999--16?**). Adults over winter in moderately deep water, 70-200 m, along the continental shelf. After migrating inshore in April-May, they form the two spawning aggregations (**berrien1982--16?**). Predators of adult mackerel include larger fish, sharks, skates, seabirds, and marine mammals (primarily whales and seals) (**studholme1999--16?**). Atlantic mackerel are managed by the Mid-Atlantic Fishery Management Council under the Mackerel, Squid, and Butterfish Fishery Management Plan. As of the 2025 assessment, the stock is not overfished and overfishing is not occurring, but is rebuilding to target levels (**nefsc2025__mack--16?**).

16.8 Figures

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17 Atlantic menhaden (*Brevoortia tyrannus*)

17.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = Low

Data Quality Model Confidence Score =Life History Data Quality = 88%

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17.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	Low	Low	None
Directional Effect	Positive	Positive	None

17.3 Score Drivers

17.3.1 Vulnerability:

add text here

17.3.2 Climate Exposure:

add text here

17.3.3 Biological Sensitivity:

Atlantic menhaden received a **Low** sensitivity score, with a certainty of 93% from bootstrap analysis. The greatest attribute score was Prey Specificity (2.5). Juveniles and adults both consume zooplankton, but are generalist feeders that are not likely to be impacted by mismatches (**rogers1989--17?**).

The previous Climate Vulnerability Assessment also gave menhaden a low sensitivity score (**hare2016vulnerability--17?**). The highest attribute in that assessment was Early Life History Requirements because late larval and juvenile life stages depend on estuaries (**able2005re--17?**). None of the estuarine habitats that these life stages rely on were classified as vulnerable by the habitat climate vulnerability assessment (**guyant2022northeast--17?**).

17.4 Data Quality Metrics

17.4.1 Life History Data Quality:

88% of data quality scores were greater than or equal to 2, indicating strong data quality. Complexity in Reproductive Strategy and Spawning Cycle were identified as data gaps (mean attribute data quality score < 2).

17.4.2 Model Confidence:

add text here

17.5 Additional Metrics

17.5.1 Distribution Shifts:

add text here

17.5.2 Directional Effect:

Atlantic menhaden are estimated to be **positively** impacted by climate change. Expert certainty was 58% from bootstrap analysis. Menhaden distributions have been expanding northward (**morano2025--17?**) likely due to warming and increased spawning activity. These results are similar to those from the 2016 Climate Vulnerability Assessment, albeit with decreased certainty (**hare2016vulnerability--17?**).

17.6 Climate Effects on Abundance and Distribution

Atlantic menhaden are estuary dependent species, and their habitats may be impacted by future development or trawling and dredging activities (**rogers1989--17?**). Atlantic menhaden distribution is expanding northward, likely to due warmer water temperatures, although other local ecological factors may also influence distribution (**morano2025--17?**). Recruitment of Atlantic menhaden larvae to nursery habitats appears to be driven primarily by larval supply and the oceanographic properties driving transport, as well as nursery habitat quality, rather than specific settlement cues (**quinlan1999--17?**). Climate change could impact these transport mechanisms or the quality of nursery habitats. In a study modeling menhaden recruitment trends in Chesapeake Bay and Southern New England, a broad-scale climate driver, the Atlantic Multidecadal Oscillation (AMO) was one of the best predictors of recruitment (**buchheister2016--17?**). (**midway2020--17?**) used hierarchical models to determine significant effects of age 1 Atlantic menhaden growth caused by the AMO (negative effect), easterly (positive effect) and northerly (negative effect) wind velocity. (**turner2017--17?**) investigated declines in size-at-age for age 3-5 Atlantic menhaden and attributed these size changes to coastal and ocean warming. This study also predicted future reductions in size-at-age as ocean temperatures rise.

17.7 Life History Synopsis

Atlantic menhaden inhabit coastal and estuarine waters from Nova Scotia to Florida (**rogers1989--17?**). They are a key forage species in the North and South Atlantic, and are consumed by a wide range of predators, including striped bass, bluefish, and weakfish in the North Atlantic (**asmfc2025_menhaden--17?**). Menhanden spawning peaks in December - February (**sedar2025--17?**). Spawning was originally thought to be concentrated off the coast of the Carolinas following the fall migration. However, more recent analysis of larval distributions suggests that spawning occurs throughout the range, as far north as Long Island (**nelson1977--17?**; **simpson2017--17?**). Menhaden spawning periods coincide with presence of phytoplankton and zooplankton, the dominant food sources of larval menhaden (**rogers1989--17?**). Eggs are buoyant and reliant on transportation by wind and tidal currents towards nursery estuary habitats, where they hatch in a few days (**sedar2025--17?**). After around 30 days in the larval stage, juveniles move to upper portions of estuaries including river shoals and the heads of small tidal creeks, where they congregate over vegetation in open water (**rogers1989--17?**). Young juvenile menhaden consume individual zooplankton; this reliance on zooplankton decreases as menhaden move from open waters to marshes, although adult menhaden also consume zooplankton (**rogers1989--17?**). Adults prefer temperatures near 18°C and generally remain within the top 100m of the water column (**sedar2025--17?**). After the winter spawning season, adults migrate inshore and north following spawning grounds in the Carolinas to the Gulf of Maine (**rogers1989--17?**). Atlantic menhaden reach maturity around 2 years of age and a maximum age is listed at 10 years, although individuals

older than 6 years of age are rarely caught in the fishery (**sedar2025--17?**). Menhaden in the Northwest Atlantic are managed as a single stock unit by the Atlantic States Marine Fisheries Commission. The most recent stock assessment update in 2025 used ecological reference points to determine stock status. As of 2025, the stock is not overfished and overfishing is not occurring (**asmfc2025_menhaden--17?**).

17.8 Figures

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18 Atlantic sea scallop (*Placopecten magellanicus*)

18.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = High

Data Quality Model Confidence Score =Life History Data Quality = 95%

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18.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	High	High	None
Directional Effect	Negative	Negative	None

18.3 Score Drivers

18.3.1 Vulnerability:

add text here

18.3.2 Climate Exposure:

add text here

18.3.3 Biological Sensitivity:

Both the **Georges Bank** and **Mid Atlantic Bight** stock units of Atlantic Sea Scallop received a **High** sensitivity score. Certainties for both stocks were 100%. Relationship to Calcified Organisms (4) and Adult Mobility (3.8) drove the scores in both stocks. Scores for each attribute were the same for both stocks.

Scallops are subject to stress due to increasing ocean acidification due to their calcium carbonate shells (**zang2022modeling--18?**). Adults are primarily sessile, but can make small-scale movements to avoid predators.

Atlantic sea scallops also received a high sensitivity score in the previous assessment. This score was also driven by Sensitivity to Ocean Acidification and Adult Mobility (**hare2016vulnerability--18?**).

18.4 Data Quality Metrics

18.4.1 Life History Data Quality:

95% of sensitivity attribute data quality scores were greater than or equal to 2 for both stocks of Atlantic Sea Scallop, indicating excellent data quality. No data gaps (mean attribute data quality score < 2) were found in either stock.

18.4.2 Model Confidence:

add text here

18.5 Additional Metrics

18.5.1 Distribution Shifts:

add text here

18.5.2 Directional Effect:

Both stocks of Atlantic sea scallop are estimated to be **negatively** impacted by future climate change. Certainty ranged from 88 – 99%, with higher certainty for the Mid Atlantic Bight stock. Rising temperatures and ocean acidification induce stress in sea scallops. In addition, anthropogenic activities such as dredging and offshore wind construction degrade habitats and impact behaviors. These results are similar to those from the 2016 Climate Vulnerability Assessment (**hare2016vulnerability--18?**).

18.6 Climate Effects on Abundance and Distribution

The Atlantic sea scallop population in the Northwest Atlantic is subject to climate-related environmental stress from rising temperatures and ocean acidification (**zang2022modeling--18?**). Scallop habitat can be degraded by trawling/dredging activity (**maguire2002effects--18?**). Scallop feeding behaviors are temporarily impacted by noise pollution associated with offshore wind construction (**jezequel2022pile--18?**). Changes in temperature and food availability are likely the predominant drivers for reduced growth in sea scallops (**cranford1998situ--18?**). (**rheuban2018projected--18?**) modeled the Atlantic sea scallop fishery under a range of climate and management scenarios and found the possibility of reduced biomass up to 50% by the end of this century. Results of a scope for growth model with increased warming food deficiency indicated a range contraction of suitable sea scallop habitat in the Mid-Atlantic Bight (**zang2022modeling--18?**). A second scope for growth model by (**zang2023effects--18?**) investigated the relationship between sea scallop size and temperature. Under future climate warming, sea scallops exhibited a range contraction and decline in maximum shell height. An integrated assessment model simulating sea scallop population dynamics with oceanographic and socioeconomic influences indicates that sea scallop harvests could experience a substantial decline by 2050 under increased CO^2 emissions (with the assumption that ocean acidification decreases recruitment and slows growth and warming increases growth) (**cooley2015integrated--18?**).

18.7 Life History Synopsis

Atlantic sea scallops are one of the most economically important fisheries in the Northwest Atlantic, with average annual ex-vessel revenues of over $500 million (**zang2022modeling--18?**). Their habitat ranges from the Gulf of St. Lawrence to Cape Hatteras, North Carolina (**stewart1994environmental--18?**) with primary fishing areas in Georges Bank, the Great South Channel, and the Mid-Atlantic Bight (**hart2006long--18?**). After a period of severe overfishing in the 1970s to 1990s, regulations including effort reductions, gear restrictions, and closed areas were introduced to rebuild the stock (**hart2006long--18?**). Spawning occurs in late summer or early fall, although smaller spawning events at other times of the year are

possible (**culliney1974larval--18?**). Gametes from other scallops, agitation, or tides may trigger scallop spawning (**parsons1992intra--18?**). Larvae reside in the water column for 5-7 weeks (**thouzeau1991spatial--18?**) and circulate with residual currents ranging from 6-25 cm/s (**culliney1974larval--18?**). Larval exchange is influenced by duration and depth of planktonic drift, gyre strength, and weak cross-isobath flow (**thouzeau1991spatial--18?**). The transition to the benthos of the freeswimming larvae occurs at a size of about 0.25 mm shell height and is accompanied by drastic changes in diet, morphology, and locomotory ability (**bourne1965comparison--18?**). Mean fall temperatures impact the time larvae take to settle (**tremblay1994drift--18?**). Sea scallop larvae are potentially preyed on by filter feeders and larval and adult planktivores (**tremblay1986horizontal--18?**). Juvenile Atlantic sea scallops are mainly found on gravel, small rocks, shells, and among branching animals and plants that permit attachment of juveniles (**thouzeau1991spatial--18?**) and feed on suspended particles, primarily phytoplankton and microzooplankton (**shumway1987food--18?**). Adult sea scallops are filter feeders on phytoplankton and other suspended organic particulates from the water (**shumway1987food--18?**). Adult sea scallops prefer temperatures of 5-18°C (**hart2004essential--18?**). Predation on juvenile and adult sea scallops decreases with shell height; common predators include Atlantic cod, wolffish, sculpins, winter and yellowtail flounder, as well as boring polychaetes, sea stars, crabs, and lobsters (**hart2004essential--18?**). The New England Fishery Management Council manages Atlantic sea scallops in federal waters under the Atlantic Sea Scallop Fishery Management Plan. As of the 2020 Management Track Assessment, the stock is not overfished and overfishing is not occurring (**nefsc2020sea--18?**).

18.8 Figures

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19 Atlantic surfclam (*Spisula solidissima*)

19.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = High

Data Quality Model Confidence Score =Life History Data Quality = 95%

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19.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	High	High	None
Directional Effect	Negative	Negative	None

19.3 Score Drivers

19.3.1 Vulnerability:

add text here

19.3.2 Climate Exposure:

add text here

19.3.3 Biological Sensitivity:

Atlantic surfclams received a **High** sensitivity score, with 100% certainty. Two attributes had scores greater than 3: Adult Mobility (4) and Relationship to Calcified Organisms (4). Like other bivalves, they have a calcium carbonate shell and are therefore susceptible to ocean acidification (**Cargnelli1999_surfclam--19?**). In addition, they are primarily sessile, other than burrowing into sediments to avoid predators (**Cargnelli1999_surfclam--19?**).

Surfclams received the same sensitivity score in the 2016 Climate Vulnerability Assessment. Adult Mobility and Relationship to Calcified Organisms also drove that score (**hare2016vulnerability--19?**).

19.4 Data Quality Metrics

19.4.1 Life History Data Quality:

95% of data quality scores were greater than or equal to 2, indicating excellent data quality. No data gaps (mean attribute data quality score < 2) were identified.

19.4.2 Model Confidence:

add text here

19.5 Additional Metrics

19.5.1 Distribution Shifts:

add text here

19.5.2 Directional Effect:

Atlantic surfclams are expected to be **negatively** impacted by climate change. Certainty in expert scores was 80%. Warming bottom waters have shifted surfclam populations from nearshore to more northern and offshore waters (**Timbs2019--19?**; **Stromp2023--19?**; **Hofmann2018--19?**). This northern shift may lead to future competition with ocean quahogs (**Spencer2026--19?**; **Stromp2023--19?**). Ocean acidification will also likely impact the surfclam. These results are similar to those from the 2016 Climate Vulnerability Assessment. Certainty was higher in the previous assessment (**hare2016vulnerability--19?**).

19.6 Climate Effects on Abundance and Distribution

Warming bottom water temperatures in the Northwest Atlantic have produced a range shift in Atlantic surfclams from southern, inshore waters of the Mid-Atlantic Bight into deeper waters north and offshore (**Timbs2019--19?**; **Stromp2023--19?**). The southern range boundary of the species likely began to migrate northward and offshore in the 1970s (**Hofmann2018--19?**). A mass mortality event in the inshore waters of Delmarva occurred in 2000, where surfclams were in a "transient event margin", a physiological range where scope for growth is negative but temperatures remain non-lethal, leading to decreased energetics and mortality (**Weinberg2005--19?**) (**Woodin2013--19?**). Thermal stress in surfclams occurs at 21°C and mortality above 25°C (**Munroe2013--19?**) (**Powell2017--19?**). Model projections through the end of the 21st century show a positive trend in Atlantic surfclam biomass as their range continues to shift, with biomass expanding in deeper waters off of New Jersey, Long Island, and Southern New England and declines in Georges Bank and Delmarva (**Spencer2026--19?**). Warming waters may result in declines in maximum size in the southern portion of the species' range (**Powell2025--19?**). This range shift may lead to potential competition between Atlantic surfclams and ocean quahogs (*Arctica islandica*) and changes to the spatial extent of the fishery (**Spencer2026--19?**) (**Stromp2023--19?**).

19.7 Life History Synopsis

Atlantic surfclams are a benthic biomass dominant species on the Northeast U.S. continental shelf from the Gulf of St. Lawrence to Cape Hatteras, North Carolina (**Hofmann2018--19?**). The fishery operates in the Mid-Atlantic Bight and Georges Bank and produces over $30 million in exvessel annual revenue (**Murray2016--19?**). Spawning occurs in summer to early fall, with earlier spawning occurring in southern habitats (**Cargnelli1999_surfclam--19?**). Temperature is a determinant for gonadal development and spawning (**Loosanoff1963--19?**). Spawning occurs when temperatures exceed 15°C, and may be delayed due to cold temperatures (**Cargnelli1999_surfclam--19?**). Fertilized eggs develop into planktonic larvae approximately 9 hours after fertilization at 21.7°C and 40 hours at 14°C (**Ropes1980--19?**). Larvae mature into the pediveliger stage, a transitional "swimming-crawling" larval stage with development of a foot for burrowing, around 18 days after fertilization (**Loosanoff1963--19?**). Larvae are dispersed by currents (generally southward along the Northeast shelf) and settlement coincides with a relaxation of upwelling events (**Fay1983--19?**). Variable reproductive success has been linked to variability in current dynamics in the Mid-Atlantic Bight (**Zhang2016--19?**). Metamorphosis to juveniles, which consists of complete absorption of the velum and settlement to the substrate, occurs anywhere from 19 to 35 days after fertilization depending on temperature (**Fay1983--19?**). Juveniles settle in sandy habitats in the intertidal and up to depths of 178m (**Cargnelli1999_surfclam--19?**). Juveniles and adults are planktivorous siphon feeders, feeding on phytoplankton 4µm diameter. Food varies with the season,

geographic location, and depth of shell bed (**Cargnelli1999_surfclam--19?**). Atlantic surfclams have many predators, including the naticid snails *Euspira heros* and *Neverita duplicata*, the sea star *Asterias forbesi*, lady crabs (*Ovalipes ocellatus*), Jonah crabs (*Cancer borealis*) and horseshoe crabs (*Limulus polyphemus*). Fish predators include haddock (*Melanogrammus aeglefinus*) and Atlantic cod (*Gadus morhua*) (**Ropes1980--19?**). Age at maturity ranges anywhere from 3 months to 4 years, however size is likely more important to maturation rates (**Cargnelli1999_surfclam--19?**). Atlantic surfclams have a maximum age of 31 years (**Cargnelli1999_surfclam--19?**). The stock is managed by the Mid-Atlantic Fishery Management Council under the Atlantic surfclam and ocean quahog Fishery Management Plan. As of the 2024 Management Track Assessment, the stock is not overfished and overfishing is not occurring. However, spawning stock biomass continues to decline and is currently at 95% of the target biomass (**NEFSC2024_surfclam--19?**).

19.8 Figures

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20 Atlantic wolffish (*Anarhichas lupus*)

20.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = High

Data Quality Model Confidence Score =Life History Data Quality = 97%

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20.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	High	High	None
Directional Effect	Negative	Negative	None

20.3 Score Drivers

20.3.1 Vulnerability:

add text here

20.3.2 Climate Exposure:

add text here

20.3.3 Biological Sensitivity:

Atlantic Wolffish received a **High** sensitivity score. This score had a certainty of 99% from bootstrap analysis. Three attributes had scores greater than 3: Stock Size (3.5), Population Growth Rate (3.3), and Historical Variability in Temperature (3).

Wolffish are a long lived (maximum documented age is 33 years) and late maturing species (most reach maturity by age 6) (**Rountree2002--20?**; **McBride2022--20?**). Furthermore, wolffish prefer colder waters and previous studies have illustrated that they will shift to seek out preferred temperature (**Kulka2011--20?**; **Rountree2002--20?**). Wolffish are not currently overfished and overfishing is not occurring (**NEFSC2022_wolffish--20?**).

In the previous Climate Vulnerability Assessment, Atlantic wolffish also received a High sensitivity score (**hare2016vulnerability--20?**). This was driven by Stock Status, Population Growth Rate, and Spawning Cycle, similar to our results. Previous research suggested that spawning only occurred in the autumn in the Northwest Atlantic, which likely drove the high Spawning Cycle score in the previous Assessment. However, more recent work has shown that timing may vary according to local conditions (**Rountree2002--20?**; **McBride2022--20?**).

20.4 Data Quality Metrics

20.4.1 Life History Data Quality:

97% of data quality scores were greater than or equal to 2, indicating excellent data quality. No data gaps (mean attribute data quality score < 2) were identified.

20.4.2 Model Confidence:

add text here

20.5 Additional Metrics

20.5.1 Distribution Shifts:

add text here

20.5.2 Directional Effect:

Atlantic wolffish are expected to be **negatively** impacted by future climate change. Certainty scores from experts were 100%. Atlantic wolffish are known temperature seekers and prefer cold waters (**Kulka2011--20?**). Low dissolved oxygen concentrations may also impact wolffish (**Bianucci2016--20?**). These results are similar to those from the 2016 Climate Vulnerability Assessment for the region (**hare2016vulnerability--20?**).

20.6 Climate Effects on Abundance and Distribution

Atlantic wolffish is a cold water species and has been termed a "temperature seeker" (**Kulka2011--20?**). Early life history stages and fish near the southern stock boundary may be the most vulnerable to surface water temperatures exceeding 9-11°C (**Rountree2002--20?**). Juvenile Atlantic wolffish exposed to temperatures of 15°C exhibited reduced growth and higher metabolic rates compared to those at 10°C (**Hinchcliffe2025--20?**). In the North Sea, Atlantic wolffish have shifted their distribution and decreased abundance at the southern end of their range and in shallow waters (**Bluemel2022--20?**). (**Bianucci2016--20?**) observed a connection between low dissolved oxygen and contraction of the Atlantic wolffish population on the Scotian Shelf. Model results of future dissolved oxygen conditions also extend to areas with high Atlantic wolffish biomass (**Bianucci2016--20?**). Similar results were also observed in (**Brennan2016--20?**) which predicted declines in dissolved oxygen concentrations and increases in temperature outside of known adult wolffish tolerances.

20.7 Life History Synopsis

Atlantic wolffish inhabit deep, rocky waters of the Northwest Atlantic from Labrador to the Great South Channel and Georges Bank (**Rountree2002--20?**). In the Northwest Atlantic, wolffish are believed to spawn primarily in autumn (**Rountree2002--20?**). Spawning is likely temperature dependent (**AWBRT2009--20?**). Wolffish will migrate to colder temperatures before the spawning season, and egg laying and guarding behavior could be disrupted if bottom temperatures increase above 10°C (**Collette2002--20?**). However, laboratory experiments suggest that timing may be highly plastic and dependent on local conditions, including photoperiod (**McBride2022--20?**). Spatial and temporal variation in spawning may be contributing to the stock's regional reproductive isolation (**AWBRT2009--20?**). Eggs are deposited in rocky dens and incubation time is highly dependent on water temperature, varying between 3 - 9 months (**AWBRT2009--20?**). After hatching, prolarvae stay close to the bottom. Larvae consume a broad range of amphipod, crustacean, and fish larvae (**Rountree2002--20?**). The prolarval stage can last from a few hours to 6 days (**Collette2002--20?**). The larvae become pelagic shortly after and stay within the water column from 20 days to 2 months, with colder temperatures prolonging the larval period, before migrating back to the bottom

(**Rountree2002--20?**). Juvenile and adult wolffish occupy the bottom habitats of the continental shelf and slope, using large stones and rocks for shelter and nesting and soft substrates for feeding (**AWBRT2009--20?**) (**Collette2002--20?**). Juvenile wolffish primarily consume echinoderms (**AWBRT2009--20?**). Adults prey upon hard-shelled invertebrates such as mollusks, crustaceans and echinoderms. Stomach contents have included sea urchins, whelks, cockles, sea clams, brittle stars, crabs, scallops and other shellfish in addition to an occasional redfish (**AWBRT2009--20?**) (**Rountree2002--20?**). While previous studies have suggested that wolffish migrate inshore/offshore seasonally for spawning or maturity, it is unclear if this is a true migration or if behavioral changes are affecting catchability as eggs are found both inshore and offshore. More recent tagging studies have illustrated that movements in the Northwest Atlantic are limited, but the species has been shown to migrate in the Northeast Atlantic (**Rountree2002--20?**) (**Simpson2015--20?**) (**Gunnarsson2019--20?**). Spiny dogfish, sea raven and Atlantic cod are the dominant predators for juveniles. For adults, the occurrence of predation is low and is not likely to be having a significant effect at the population level (**AWBRT2009--20?**). The maximum reported age for wolffish in the Gulf of Maine is 33 years (**Fairchild2015--20?**). Most individuals reach maturity by age 6, but this may vary regionally due to temperature (**Rountree2002--20?**) (**McBride2022--20?**). Atlantic wolffish are managed by the New England Fishery Management Council under the Northeast Multispecies (Groundfish) Fishery Management Plan. As of the 2022 Management Track Assessment, the stock is not overfished and overfishing is not occurring (**NEFSC2022_wolffish--20?**).

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21 Black sea bass (*Centropristis striata*)

21.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = Low

Data Quality Model Confidence Score =Life History Data Quality = 95%

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21.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	Low	Moderate	Decrease
Directional Effect	Positive	Positive	None

21.3 Score Drivers

21.3.1 Vulnerability:

add text here

21.3.2 Climate Exposure:

add text here

21.3.3 Biological Sensitivity:

Black Sea Bass received a **Low** sensitivity score. Certainty was 64%. The only score greater than 2.5 was Complexity in Reproductive Strategy (2.7). This is likely because black sea bass as protogynous hermaphrodites (they transition from female to male as they age). Black Sea Bass also spawn in aggregations that consist of a single dominant male and many females.

In the 2016 Climate Vulnerability Assessment, Black Sea Bass received a Moderate sensitivity score. This was driven by Complexity in Reproductive Strategy (which received the same score in both assessments) and Early Life History Requirements (**hare2016vulnerability--21?**). Juvenile Habitat Requirements was likely important because Black Sea Bass rely on seagrasses and shellfish beds as nursery habitats (**guyant2022northeast--21?**). This attribute received a score of 2.4 in our assessment.

21.4 Data Quality Metrics

21.4.1 Life History Data Quality:

95% of data quality scores were greater than or equal to 2, indicating excellent data quality. No data gaps (mean attribute data quality score < 2) were identified.

21.4.2 Model Confidence:

add text here

21.5 Additional Metrics

21.5.1 Distribution Shifts:

add text here

21.5.2 Directional Effect:

Black sea bass are expected to be **positively** impacted by climate change. Expert certainty was 100%. Black sea bass have been expanding their distributions northward into the Gulf of Maine (**mcmahan2020geographic--21?**), likely due to increasing temperature supporting recruitment. This is the same result as the 2016 assessment (**hare2016vulnerability--21?**).

21.6 Climate Effects on Abundance and Distribution

Black sea bass in the Mid-Atlantic Bight are currently expanding their distribution northward into the Gulf of Maine (**mcmahan2020geographic--21?**). Black sea bass in their newly-expanded range have a less diverse diet, lower condition, and a younger age-at-maturity than their counterparts at the historic northern range of the stock (**mcmahan2020geographic--21?**). A laboratory study on Mid-Atlantic black sea bass observed a decline in bodily processes that require oxygen demand above 24°C, suggesting black sea bass will continue to shift their range poleward with warming conditions (**slesinger2020effect--21?**). (**tabandera2024black--21?**) found that winter bottom temperature was positively correlated with recruitment deviations in black sea bass. Black sea bass in the southern Mid-Atlantic Bight also appear to occupy a smaller portion of their habitat in years with an increase in cold shelf water on the shelf slope. Northward distributions of black sea bass are resulting in new species interactions and habitat overlaps. In Narragansett Bay, black sea bass juveniles now share nursery habitats with juvenile tautog and cunner, however dietary overlap was size-dependent and ontogenic diet shifts eventually led to niche separation (**bell2015disentangling--21?**; **taylor2023habitat--21?**).

21.7 Life History Synopsis

Black sea bass are found along the U.S. Atlantic coast from Cape Cod to the Gulf of America in structured habitats such as reefs, shipwrecks, or oyster beds (**drohan2007essential--21?**). In the Mid-Atlantic Bight, spawning occurs from April to October, beginning first in the southern portion of their range and northern latitudes starting later in the year (**able1998first--21?**). Early larval stages last approximately 72 hrs at 21°C but can last anywhere from 2-6 days depending on temperature (**berlinsky2004effects--21?**). Afterwards, larvae enter a 21-24 day planktonic phase where they can disperse up to 50 miles by currents (**watanabe2021status--21?**). Eggs and pelagic larvae are vulnerable to predation, especially by gelatinous zooplankton (**arai1988interactions--21?**). Larvae settle in nearshore shelf habitats and then migrate to estuarine nurseries to settle and metamorphose into juveniles (occurs at 13-24mm) (**able1995early--21?**). Larvae are abundant at temperatures between 13-21°C and leave estuaries for deeper water when temperatures dip below 14°C (**kendall1972description--21?**). Larvae consume microalgae and zooplankton, including rotifers (**richards1963demersal--21?**). Metamorphosis is likely the biggest factor influencing settlement (**musick1977seasonal--21?**). Structural complexity appears to be an essential component of juvenile black sea bass habitat offshore as well as inshore nurseries throughout the species range. In offshore areas, recently settled fish occur in accumulations of shell on sand substrata, complex microtopographies on exposed clay, on rocky reefs, and on wrecks (**able1995early--21?**). Primary black sea bass settlement habitats are probably located along the near shore continental shelf in accumulations of the shells of bivalves, including Atlantic surfclams (*Spisula solidissima*) (**able1995early--21?**).

Juvenile growth rates impact winter mortality. Smaller juveniles have smaller lipid reserves and higher metabolic rates and therefore face higher predation or starvation risks in the winter (**geissinger2021food--21?**). Juveniles, which are diurnal, visual predators, prey on benthic and epibenthic crustaceans (isopods, amphipods, small crabs, sand shrimp, copepods, mysids) and small fish. As they mature, their diet diversifies to include crabs, krill, isopods, shrimp, and small fish (**kimmel1973food--21?**). Adult black sea bass perform seasonal inshore/offshore migrations between coastal spawning grounds and offshore overwintering habitats. Migrations are triggered by declining temperatures (around 10-12°C) and begin in the North moving southward (**tabandera2024black--21?**). In the Mid-Atlantic Bight, individuals have home ranges as large as 7km^2 (**tabandera2024black--21?**). Observations suggest that spawning aggregations consist of a single dominant male and a harem of females. These spawning aggregations may rely on visual cues rather than chemical or environmental cues (**tabandera2024black--21?**). Black sea bass are protogynous hermaphrodites, with fish changing sex from female to male as they increase in age and size (**mcgovern2002temporal--21?**). Age of sexual transition varies with latitude with females maturing and undergoing sexual transition at greater ages in northern latitudes (**obrien1993maturation--21?**). On average, age at maturity in the Mid-Atlantic is 2-3 years but can extend up to 5 years (**obrien1993maturation--21?**). Maximum age for females is 8 years and 12 years for males (**nefsc2017_62saw--21?**). The black sea bass fishery in the Northwest Atlantic is divided into two stock subunits (north and south) and is managed by the Mid-Atlantic Fishery Management Council along with the Atlantic States Marine Fisheries Commission. As of the 2025 Management Track Assessment, the Mid-Atlantic stock is not overfished or subject to overfishing (**asmfc2025_bsb--21?**).

21.8 Figures

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22 Blue crab (*Callinectes sapidus*)

22.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = Moderate

Data Quality Model Confidence Score =Life History Data Quality = 93%

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22.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	Moderate	High	Decrease
Directional Effect	Neutral	Neutral	None

22.3 Score Drivers

22.3.1 Vulnerability:

add text here

22.3.2 Climate Exposure:

add text here

22.3.3 Biological Sensitivity:

Blue crabs had a **Moderate** sensitivity score, with 79% certainty. Three attributes received scores greater than 2.5: Complexity in Reproductive Strategy (3.4), Other Stressors (2.8), and Egg and Larval Survival and Settlement Requirements (2.7).

Blue crabs rely on estuaries throughout their life cycles, including for spawning. Estuaries are highly vulnerable habitats due to urbanization impacts such as hypoxia, loss of seagrass beds, and changes to freshwater inputs (**guyant2022northeast--22?**). Temperatures have been shown to impact spawning behaviors and larvae require low salinities (10-30) for proper development (**hill1989species--22?**).

Blue crabs received a High sensitivity score in the previous Climate Vulnerability Assessment. This was driven by similar attributes to our scores: Other Stressors and Complexity in Reproductive Strategy (**hare2016vulnerability--22?**).

22.4 Data Quality Metrics

22.4.1 Life History Data Quality:

93% of data quality scores were greater than or equal to 2, indicating excellent data quality. Only Stock Size/Status was considered a data gap (mean attribute data quality score < 2).

22.4.2 Model Confidence:

add text here

22.5 Additional Metrics

22.5.1 Distribution Shifts:

add text here

22.5.2 Directional Effect:

Blue crab are expected to be **neutrally** impacted by climate change. Expert scores had 93% certainty. This is consistent with the previous Assessment (**hare2016vulnerability--22?**). While degradation of estuarine habitats may limit food sources or force crabs into new areas, warmer temperatures are predicted to increase survival rates, particularly over winter (**aguilar2008importance--22?**; **glandon2019winter--22?**).

22.6 Climate Effects on Abundance and Distribution

As an estuary dependent species, blue crab habitats have been degraded through urbanization impacts including hypoxia, reduction in seagrass beds, and changes in timing and volume of freshwater inputs. Low oxygen conditions may force crabs into shallower, more spatially restricted areas of Chesapeake Bay, increasing their vulnerability to predation or capture. Prolonged hypoxia may limit food resources (mainly infaunal bivalves) in affected areas even after normoxic conditions resume (**aguilar2008importance--22?**). (**glandon2019winter--22?**) estimated water temperatures in Chesapeake Bay through 2100 and found that a 50% shorter winter will increase overwinter survival rates in blue crab by 20%. Faster growth, a longer growing season, and lower winter mortality would lead to increased productivity (**glandon2019winter--22?**). Blue crabs may also adapt phenologically to warming temperatures by shifting the timing of biological events such as spawning (**schneider2024reproductive--22?**). Blue crabs have been observed in the Gulf of Maine, indicating a possible range extension from climate-driven rising temperatures (**johnson2015savory--22?**).

22.7 Life History Synopsis

The blue crab's native range is along the western Atlantic coast from Nova Scotia to Argentina, including the Gulf of America (**hill1989species--22?**). The blue crab fishery is the most valuable fishery in the Chesapeake Bay (**hines2007ecology--22?**). In Chesapeake Bay, adults spawn from May to October, however spawning is likely more limited in cooler waters (**umces2011stock--22?**). To spawn, adults aggregate in low salinity areas of upper estuaries to mate. Adult females then travel to higher salinity spawning grounds (**hill1989species--22?**). The timing of the migration varies with water temperature and occurs earlier in the year at lower latitudes (**aguilar2008importance--22?**). Eggs (7-14 days) and larvae (40-70 days) are pelagic (**hill1989species--22?**) and are carried by currents offshore before using physical transport methods to return to bays and eventually estuaries (**millikin1984synopsis--22?**). Larvae require temperatures of 20-30°C and salinities of 10-30 ppt for proper development (**hill1989species--22?**). Settlement into estuarine habitats is associated with Ekman flow from southward wind events resulting in strong subtidal flow into the estuaries. Once in the estuary, settlement and metamorphosis of the megalopae are stimulated by chemical cues associated with appropriate nursery habitat (**epifanio2016behavioral--22?**). Larvae are phytoplanktivorous, consuming dinoflagellates and copepod nauplii (**hill1989species--22?**). They are preyed upon by a large number of estuarine and marine animals , including other commercially important species such as the striped bass (**hill1989species--22?**). Once reaching the juvenile stage, blue crabs gradually migrate into shallower, less saline waters in upper estuaries and rivers, where they grow and mature. Males, which prefer low-salinity waters, generally migrate farther upstream than do females, which tend to stay in the lower rivers and estuaries (**hill1989species--22?**). Post-larval and juvenile crabs are considered general

scavengers, bottom carnivores, detritivores, and omnivores, consuming a wide range of dead and live fish, crabs, organic debris, shrimp, mollusks (including mussels, clams, oysters, and snails), and aquatic plants (**hill1989species--22?**). Adult blue crab prey is very similar to juveniles; they will consume most available prey and freely switch among preferred prey types (**hill1989species--22?**). Juvenile and adult blue crabs are consumed by mammals, sea turtles, birds, and several fishes (**hill1989species--22?**). Blue crabs reach maturity at 12-18 months, with a maximum lifespan around 4 years of age and a maximum size of 9 inches carapace length (CL) (**umces2011stock--22?**) (**millikin1984synopsis--22?**). Adults prefer temperatures of 9-10°C for feeding, growth, and migration (**hines2007ecology--22?**). In the Northwest Atlantic, blue crabs are managed by the Maryland Department of Natural Resources, Virginia Marine Resources Commission, and Potomac River Fisheries Commission. These entities work together with NOAA to conduct stock assessments for blue crab. The Chesapeake Bay blue crab population is not overfished based on current biological reference points, but abundance and recruitment remain low. A benchmark stock assessment is underway to improve understanding of blue crab population dynamics and its fishery (**cbsac2025advisory--22?**).

22.8 Figures

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23 Bluefin tuna (*Thunnus thynnus*)

23.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = Low

Data Quality Model Confidence Score =Life History Data Quality = 98%

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23.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	Low	Moderate (**loughran2025climate--23?**)	Decrease
Directional Effect	Neutral	Neutral (**loughran2025climate--23?**)	None

23.3 Score Drivers

23.3.1 Vulnerability:

add text here

23.3.2 Climate Exposure:

add text here

23.3.3 Biological Sensitivity:

Bluefin Tuna received a **Low** sensitivity score, with a certainty of 67%. The only attribute that received a score greater than 2.5 was Egg and Larval Survival and Settlement Requirements (2.6). Early larval stages are susceptible to starvation and older larvae are susceptible to predation (**shropshire2022--23?**; **mcgowan1989--23?**). Larvae also depend on ocean currents for transport to nursery habitats (**shropshire2022--23?**).

Bluefin Tuna were not included in the 2016 Climate Vulnerability Assessment for the Northeast Shelf, but were included in the recent Highly Migratory Species Climate Vulnerability Assessment (**loughran2025climate--23?**). In this analysis, Bluefin Tuna received a Moderate sensitivity score across their range (**loughran2025climate--23?**). This score was driven by population growth rates, reproductive cycles, and early life history requirements, similar to our analysis.

23.4 Data Quality Metrics

23.4.1 Life History Data Quality:

98% of data quality scores were greater than or equal to 2, indicating excellent data quality. No data gaps (mean attribute data quality score < 2) were identified.

23.4.2 Model Confidence:

add text here

23.5 Additional Metrics

23.5.1 Distribution Shifts:

add text here

23.5.2 Directional Effect:

Bluefin tuna were estimated to be **neutrally** impacted by climate change. Expert certainty was 96%. Bluefin distributions have been projected to move poleward into the northeast Atlantic. Habitat losses are expected to be great (60% loss) in the Gulf of America, but smaller (20%) in the Northwest Atlantic (**braun2023widespread--23?**). Spawning may shift to early spring due to increasing temperatures (**muhling2011--23?**). This is similar to the results from the recent Highly Migratory Species Climate Vulnerability Assessment (**loughran2025climate--23?**).

23.6 Climate Effects on Abundance and Distribution

Projected surface temperatures in the Gulf of America may alter the spatial and temporal extent of Atlantic bluefin tuna spawning. Model results from (**muhling2011--23?**) indicate decreased larval bluefin tuna occurrence in late spring in the Gulf by 39-61% by 2050 and 93-96% by 2100, with early spring spawning becoming more prevalent. (**erauskin2026--23?**) also modeled decreased habitat suitability for spawning in the Gulf of America. They predict a poleward shift in Atlantic bluefin tuna distribution and greater spatial overlap with prey in higher latitudes. Nursery habitats in the Gulf of America are impacted by oil spills. Upwards of 5% of predicted spawning habitat within the U.S. Exclusive Economic Zone (EEZ) was impacted by the Deepwater Horizon oil spill (**hazen2016--23?**).

23.7 Life History Synopsis

Western Atlantic bluefin tuna are a highly migratory species observed from Newfoundland to the Gulf of America; they occupy temperate surface waters but can dive to depths of 1000m (**fromentin2006--23?**). Spawning occurs between spring and summer throughout western North Atlantic spawning grounds, namely the Gulf of America (March-June) and Slope Sea (June-September). Spawning in the Slope Sea occurs one to two months later than the Gulf of America due to differences in sea surface temperatures (**baglin1982--23?**). Bluefin tuna eggs hatch after 2 days and larvae remain pelagic for 3-4 weeks (**shropshire2022--23?**). Larval bluefin tuna are found near spawning grounds and dispersed via wind and the margins of ocean currents. Edges of ocean currents and eddies provide access to warmer sea surface temperatures and nutrient-rich waters that support larval foraging of zooplankton prey while avoiding predation (**shropshire2022--23?**). Favorable habitat for bluefin larvae in the Gulf of America was determined to be moderately warm waters (23.5-28.5°C) in the eddies of the Loop Current, and outside of cooler, higher chlorophyll continental shelf waters (**teo2007--23?**). Along the coast of the southeastern United States, larval stages may benefit from nearshore meanders of the Gulf Stream that can produce upwelling of nutrient rich water along the continental shelf edge (**shropshire2022--23?**). Larvae initially feed on zooplankton and

progress to a piscivorous diet at a relatively small size (**llopiz2015--23?**). Starvation is the largest cumulative source of mortality in early larval stages, and predation on older larvae is a main source of mortality in later stages (**mcgowan1989--23?**). Juvenile habitats are located over the continental shelf in the summer and further offshore in the winter. Juvenile life stages follow similar seasonal movements compared to adult life stages, though are generally more restricted to coastal areas (**nmfs2006--23?**). Juvenile and adult bluefin tuna are opportunistic, generalist predators feeding mainly on zooplanktivorous fishes and crustaceans. Predation of larger fish continues with increasing body size, with juveniles moving from sand lance to mackerel and herring as adults (**logan2011--23?**; **logan2015--23?**). Adult bluefin tuna migrate seasonally from winter to early summer spawning grounds in the Gulf of America, Straits of Florida, and Slope Sea to summer and fall foraging grounds throughout the Gulf Stream, South and Mid-Atlantic Bight, Gulf of Maine, Nova Scotia Shelf, and east of the Flemish Cap (**block2001--23?**). Age at 50% maturity estimates for Atlantic bluefin tuna vary between 5 to 12 years, with a maximum age of over 30 years (**scrs2021--23?**). NOAA Fisheries, with recommendations from the International Commission for the Conservation of Atlantic Tunas (ICCAT), manages the Atlantic bluefin tuna fishery in the United States through the Atlantic Highly Migratory Species Management Division. As of 2023, the stock is being managed ICCAT Recommendation 21-10. Population numbers are unknown but the stock is not subject to overfishing (**iccat2022--23?**).

23.8 Figures

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24 Bluefish (*Pomatomus saltatrix*)

24.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = Low

Data Quality Model Confidence Score =Life History Data Quality = 97%

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24.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	Low	Low	None
Directional Effect	Neutral	Positive	Decrease

24.3 Score Drivers

24.3.1 Vulnerability:

add text here

24.3.2 Climate Exposure:

add text here

24.3.3 Biological Sensitivity:

Bluefish received a **Low** sensitivity score, with a certainty of 69%. The only attribute that received a score greater than 2.5 was Juvenile Habitat Requirements (4). Bluefish rely entirely on salt marshes and submerged aquatic vegetation as juveniles; both habitats are considered highly vulnerable to future climate change (**guyant2022northeast--24?**).

Bluefish also received a Low sensitivity score in the 2016 Assessment. Similar to our results, Early Life History Requirements was the only attribute to score above 2.5 (**hare2016vulnerability--24?**).

24.4 Data Quality Metrics

24.4.1 Life History Data Quality:

97% of data quality scores were greater than or equal to 2, indicating excellent data quality. No data gaps (mean attribute data quality score < 2) were identified.

24.4.2 Model Confidence:

add text here

24.5 Additional Metrics

24.5.1 Distribution Shifts:

add text here

24.5.2 Directional Effect:

Bluefish are expected to be **neutrally** impacted by climate change. Expert certainty was 95%. Distributions are shifting north and east and warming temperatures may expand the spawning season (**Tyrell2022--24?**). However, there is a concern that seasonal mismatches may occur if juveniles arrive in estuaries before their prey, which may impact recruitment (**Taylor2006--24?**). In the original Assessment, bluefish were predicted to be positively impacted by climate change due to expanding distributions and spawning seasons (**hare2016vulnerability--24?**).

24.6 Climate Effects on Abundance and Distribution

There has been a documented shift in bluefish distribution, spawning, habitat use in the Mediterranean (**Sabates2012--24?**; **VillegasHernandez2015--24?**). Bluefish distributions in the Northwest Atlantic seem to be shifting north and east (**Tyrell2022--24?**). Increasing fall sea surface temperatures may extend spawning season further into the calendar year (**Tyrell2022--24?**). If warming alters spawning phenology, there are concerns that juveniles may arrive in estuaries before their prey. Seasonal mismatches have been shown to impact growth, body condition, and reproductive success (**Taylor2006--24?**). Changing distributions of prey species as a result of climate change may alter interactions between bluefish and their prey (**Whitfield2026--24?**).

24.7 Life History Synopsis

Bluefish inhabit temperate and tropical surface waters in the North Atlantic from Maine to the Atlantic coast of Florida (**Shepherd2006--24?**). Bluefish in the North Atlantic are a single, migratory stock with two discrete spawning periods. The first occurs during March-May near the edge of the continental shelf of the South-Atlantic Bight. The second occurs between June and August in the Mid-Atlantic Bight (**Shepherd2006--24?**). Eggs hatch after 46-48 hours at 18 - 22.2°C to pelagic larvae (**Deuel1966--24?**). Larvae spawned in the South Atlantic (spring-spawned cohort) are subject to advection north via the Gulf Stream and inshore via warm core rings. Eggs in the southern part of the Mid-Atlantic Bight may be advected south and offshore (**Shepherd2006--24?**). Transport of larvae and/or juveniles across the shelf- slope region may be aided by wind-driven surface flow (**Hare1996--24?**). (**Sherman1984--24?**) note that wide distribution of bluefish larvae over the shelf reduces their vulnerability to the predator field and environmental perturbation. Larvae prey on copepods (**Shepherd2006--24?**). Peak abundance of larval bluefish in Southern New England and the Mid-Atlantic Bight coincided with zooplankton production (**Sherman1984--24?**). Changes in prey availability and distribution can impact bluefish condition and reproductive success (**Taylor2006--24?**). Juvenile bluefish inhabit estuaries, bays, and coastal oceans with sand, oyster beds, and submerged aquatic vegetation habitats (**McBride1991--24?**). Juveniles enter estuaries when temperatures are greater than 13-15°C and leave when temperatures fall below 10-12°C (**Smith1994--24?**). Juvenile migration into estuarine habitats appears tied to a shift in eating other fish. In estuaries, adults and juveniles eat whatever taxa are locally abundant (fish, crustaceans, shrimp, and polychaetes). In ocean habitats, young-of-the-year bluefish switch to piscivory predation with increasing size, similar to estuarine habitats (**Marks1996--24?**). Bluefish are opportunistic foragers, and the composition of adult bluefish diet depends largely on the available prey, but they consume similar prey items as juvenile bluefish (**Shepherd2006--24?**). Bluefish have long been recognized as a notable predator of forage fish (**Tyrell2022--24?**). Adult bluefish conduct seasonal migrations following warmer waters, overwintering in the south and moving north in the spring and

summer (**Shepherd2006--24?**). Migrations may be cued by photoperiod as well, and have been observed to shift northward in warm years (**Nye2009--24?**). Shortfin mako sharks are identified as a major predator of bluefish and may consume more individuals than removed by the fishery (**Wood2009--24?**). Most bluefish are mature by age 2, with approximately half maturing by age 1 (**NEFSC2015_bluefish--24?**). The maximum age for bluefish in the North Atlantic is 12 years (**Salerno2001--24?**). In the Northwest Atlantic, bluefish are managed by the Mid-Atlantic Fishery Management Council and Atlantic States Marine Fisheries Commission. The last stock assessment occurred in 2025, where bluefish were found to not be overfished nor is overfishing occurring (**NEFSC2025_bluefish--24?**).

24.8 Figures

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25 Blueline tilefish (*Caulolatilus microps*)

25.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = Moderate

Data Quality Model Confidence Score =Life History Data Quality = 63%

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25.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	Moderate	High (**Burton2023-sv--25?**)	Decrease
Directional Effect	Neutral	Neutral (**Burton2023-sv--25?**)	None

25.3 Score Drivers

25.3.1 Vulnerability:

add text here

25.3.2 Climate Exposure:

add text here

25.3.3 Biological Sensitivity:

Blueline tilefish received a **Moderate** sensitivity score. Certainty was 90%. Three attributes had scores greater than 2.5: Adult Mobility (2.8), Population Growth Rate (2.8), and Historical Variability in Temperature (2.5).

Adult blueline tilefish are primarily demersal. They are thought to be sedentary and remain close to their dens, similar to goldenline tilefish (**schmidtke2017life--25?**). Maximum ages upwards of 50 years have been recorded (**schmidtke2017life--25?**; **spanik2023validating--25?**). They reach maturity anywhere from a few months to 3.7 years off the coast of Virginia to upwards of 5 years off the southeast US (**ross1982feeding--25?**; **harris2004age--25?**; **low2023characterizing--25?**).

Blueline tilefish were not included in the 2016 Climate Vulnerability Assessment for the Northeast US, but were included in the recent Climate Vulnerability Assessment for the Southeast US (**burton2023climate--25?**). Blueline tilefish received a High sensitivity score in that assessment. This score was driven by Sensitivity to Temperature and Population Growth Rate attributes. The experts in that assessment suggested that long-lived, slow-growing nature of blueline tilefish are believed to make them susceptible to overfishing (**burton2023climate--25?**).

25.4 Data Quality Metrics

25.4.1 Life History Data Quality:

Only 63% of data quality scores were greater than or equal to 2, indicating poor data quality. Most attributes had data gaps (mean attribute data quality score < 2), including Adult Mobility, Complexity in Reproductive Strategy, Egg and Larval Survival and Settlement Requirements, Historical Variability in Temperature, Other Stressors, and Stock Size/Status.

25.4.2 Model Confidence:

add text here

25.5 Additional Metrics

25.5.1 Distribution Shifts:

add text here

25.5.2 Directional Effect:

The impact of climate change on blueline tilefish is expected to be **neutral**, with 92% certainty in expert scores. It is possible that blueline tilefish biomass may increase with increasing temperatures, as landings in the Mid-Atlantic have been correlated to increasing temperatures (**pinsky2018preparing--25?**; **rogers2019shifting--25?**). However, little research has been conducted on how blueline tilefish will respond to climate change. The Climate Vulnerability Assessment for the Southeast US also predicted that the effect of climate change on blueline tilefish would be neutral (**burton2023climate--25?**).

25.6 Climate Effects on Abundance and Distribution

Little research has been conducted on the response of blueline tilefish to climate change. Rising water temperatures likely drove a rapid increase in landings in the Mid-Atlantic throughout the early 21st century as blueline tilefish distributions expanded north beyond their traditional South Atlantic boundary (**pinsky2018preparing--25?**; **rogers2019shifting--25?**). Blueline tilefish may exhibit a moderately-high risk to achieving optimal biomass and yield due to climate change on productivity (**mafmc2025update--25?**).

25.7 Life History Synopsis

Blueline tilefish have only recently been found on the Northeast U.S. Continental Shelf. They live in crevices or on ledges on the shelf, shelf break, and shelf slope from Georges Bank to Mexico and the Gulf of America (**schmidtke2017life--25?**). Blueline tilefish are multiple batch spawners and presumably spawn on/near the bottom (**low2023characterizing--25?**). Spawning occurs from March through October in the southern portion range, and May - November in the Mid-Atlantic Bight (**harris2004age--25?**; **schmidtke2017life--25?**). Eggs and larvae are estimated to remain pelagic for 4-6 weeks and depend on currents for transport. Eddies and gyres may provide local retention while the Gulf Stream transports eggs and larvae up the East Coast. It seems likely that substantial numbers of blueline larvae are transported out of the eastern Gulf of Mexico, to the South Atlantic, and from the South Atlantic into the Mid-Atlantic over the course of a few weeks (**sedar2025report--25?**). Juveniles settle to the bottom and occupy benthic habitats around offshore submarine canyons on the shelf edge, constructing burrows in seafloor sediments or within rocks and boulders (**schmidtke2017life--25?**). Juveniles and adults consume a diverse array of benthic invertebrates, including polychaetes, crustaceans, and decapods. Gastropods and copepods have also been observed and are likely consumed when diel vertical migrations bring them close to the seafloor (**bielsa1987food--25?**; **ross1982feeding--25?**). Age at maturity is up to 3.7 years in the Mid-Atlantic and 5 years in the South Atlantic (**harris2004age--25?**; **low2023characterizing--25?**). Maximum age is 37 years, although it is theorized

that individuals could possibly reach upwards of 50 years in age (**schmidtke2017life--25?**; **spanik2023validating--25?**). In the Northwest Atlantic, blueline tilefish are managed by the Mid-Atlantic Fishery Management Council under the Tilefish Fishery Management Plan. Council management of blueline tilefish in the Mid-Atlantic began as a response to increased commercial and recreational landings in the region. The Mid-Atlantic stock has not been assessed. Population levels and fishing status are currently unknown but management measures have been in place since 2017 (**mafmc2017amendment--25?**).

25.8 Figures

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26 Butterfish (*Peprilus triacanthus*)

26.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = Low

Data Quality Model Confidence Score =Life History Data Quality = 95%

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26.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	Low	Low	None
Directional Effect	Neutral	Positive	Decrease

26.3 Score Drivers

26.3.1 Vulnerability:

add text here

26.3.2 Climate Exposure:

add text here

26.3.3 Biological Sensitivity:

Butterfish received a **Low** sensitivity score. Certainty was 100%. No attributes had scores greater than 2.5. The only score greater than 2 was Other Stressors (2.1). Butterfish spawn in estuarine habitats like Long Island Sound and Chesapeake Bay that are impacted by urbanization impacts such as hypoxia, loss of seagrass beds, and changes to freshwater inputs (**guyant2022northeast--26?**).

These results are similar to the 2016 Assessment where all attribute scores were less than 2.5 and the overall sensitivity score was Low (**hare2016vulnerability--26?**).

26.4 Data Quality Metrics

26.4.1 Life History Data Quality:

95% of data quality scores were greater than or equal to 2, indicating excellent data quality. No data gaps (mean attribute data quality score < 2) were identified.

26.4.2 Model Confidence:

add text here

26.5 Additional Metrics

26.5.1 Distribution Shifts:

add text here

26.5.2 Directional Effect:

The effect of climate change on butterfish was predicted to be **neutral**, with 98% certainty in expert scores. Butterfish have expanded their range northward and into deeper waters (**suca2021environmental--26?**; **adams2022update--26?**). However, reliance on zooplankton and copepod prey in the spring may lead to mismatches that impact recruitment (**suca2018feeding--26?**). In the previous Assessment, experts predicted that butterfish would be positively impacted by climate change (**hare2016vulnerability--26?**).

26.6 Climate Effects on Abundance and Distribution

Butterfish distributions and abundance in the Mid-Atlantic Bight are strongly related to bottom temperature (**kohut2013toward--26?**). Butterfish have shifted their distribution northward in recent decades. (**collie2008long--26?**) documented an increase in butterfish abundance in Narragansett Bay in the latter half of the 20th century, and they recently have expanded their range into the Gulf of Maine (**suca2021environmental--26?**; **adams2022update--26?**). Their depth distribution has also shifted to inhabiting deeper waters (**suca2021environmental--26?**). An increase in butterfish prevalence in the Gulf of Maine has altered seabird diets, and is replacing preferred prey in common terns (**smith2023effects--26?**).

26.7 Life History Synopsis

Butterfish distribution ranges from Newfoundland in Canada to Atlantic Florida, but they are primarily found from the Gulf of Maine to Cape Hatteras. They are a schooling fish and shift distribution in response to changing water temperatures: moving north and inshore over the summer to spawn and south and offshore in the winter to seek warmer water (**cross1999essential--26?**). Butterfish are broadcast spawners and spawn primarily in the evening or at night, however no direct observations have been made (**horn1970systematics--26?**). In general, butterfish spawn starting in May through early fall, peaking in June-July, although timing is dependent on location (**cross1999essential--26?**). Egg duration and larval growth is temperature dependent (**colton1963eggs--26?**), and most eggs and larvae prefer temperatures 9-19°C (**cross1999essential--26?**). Butterfish eggs and larvae are widely distributed throughout their range from spring to fall as spawning occurs over their entire range. Larvae are associated with the Gulf Stream, although it is not clear how far they travel (**morse1987marmap--26?**). Later life stages (juveniles and adults) can inhabit surface waters to depth on the continental shelf and into coastal bays and estuaries (**schreiber1973fishes--26?**). Survival is reduced in juveniles and adults below 10°C (**cross1999essential--26?**). Juvenile and adult butterfish feed mainly on planktonic prey including thaliaceans (primarily *Larvacea* and *Hemimyaria*), mollusks (primarily squids), crustaceans, coelenterates, polychaetes, small fishes, and ctenophores (**horn1970systematics--26?**). Recent diet analysis on 2013-2015 trawl specimens illustrates that spring diets are dominated by *Calanus* copepods, amphipods, larvaceans, and fall diets are dominated by amphipods (**suca2018feeding--26?**). There is little information about larval predation; juvenile and adult butterfish are preyed upon by haddock, silver hake, goosefish, weakfish, bluefish, swordfish, sharks (hammerhead), and longfin inshore squid (**bigelow1953fishes_butterfish--26?**; **horn1970systematics--26?**). Butterfish age at maturity is 1-2 years; the maximum lifespan is 6 years although most butterfish live roughly 2-3 years (**horn1970systematics--26?**; **draganik1966rate--26?**). In the Northwest Atlantic, butterfish are managed by the Mid-Atlantic Fishery Management Council under the Mackerel, Squid, and Butterfish Fishery Management Plan. As of the 2024 stock

assessment, the stock is not overfished and overfishing is not occurring (**nefsc2024butterfish--26?**).

26.8 Figures

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27 Chub mackerel (*Scomber colias*)

27.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = Low **Data Quality**Model Confidence Score =Life History Data Quality = 68%

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27.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	Low	Not Available	-
Directional Effect	Neutral	Not Available	-

27.3 Score Drivers

27.3.1 Vulnerability:

add text here

27.3.2 Climate Exposure:

add text here

27.3.3 Biological Sensitivity:

Chub mackerel were given a **Low** sensitivity score, with 99% certainty according to bootstrap analysis. The only attribute that received a score greater than 2.5 was Stock Size. This is likely because no stock assessment for chub mackerel exists (**mafmc2025chub--27?**).

Chub mackerel were not assessed in the 2016 Climate Vulnerability Assessment (**hare2016vulnerability--27?**), nor any other recent Assessment.

27.4 Data Quality Metrics

27.4.1 Life History Data Quality:

Only 68% of data quality scores were greater than or equal to 2, indicating poor data quality. Data gaps (mean attribute data quality score < 2) included Complexity in Reproductive Strategy, Dispersal of Early Life Stages, Egg and Larval Survival and Settlement Requirements, Other Stressors, and Stock Size/Status.

27.4.2 Model Confidence:

add text here

27.5 Additional Metrics

27.5.1 Distribution Shifts:

add text here

27.5.2 Directional Effect:

Experts predicted that climate change would have a **neutral** effect on chub mackerel. Expert certainty was 87%. Variations in abundance have been observed and are likely due to oceanographic processes due to their pelagic lifestyle, but the exact drivers are unclear.

27.6 Climate Effects on Abundance and Distribution

Fluctuations in Atlantic chub mackerel abundance have been reported in the Mid-Atlantic and New England, and may result from environmental influences such as temperature and upwelling. Oceanographic processes likely influence recruitment and distribution, as all life stages of chub mackerel are fully pelagic (**hernandez2000synopsis--27?**). In the eastern Atlantic, chub mackerel distributions have shifted northward, with a delayed and prolonged spawning season with increasing latitude (**dominguez2022spatial--27?**). In the Alboran Sea, chub mackerel abundance is vulnerable to variations in sea surface temperature, salinity, currents, and wind patterns (**ayman2024environmental--27?**).

27.7 Life History Synopsis

Atlantic chub mackerel is a schooling pelagic species ranging from Nova Scotia to Argentina, including the Gulf of America, and are found on the continental shelf at depths of 250-300m (**mafmc2025chub--27?**). They are batch spawners, spawning January through April from North Carolina through the Gulf of America (**daley2018growth--27?**). Spawning generally occurs in waters greater than 10°C and is possibly triggered by water temperatures between 15- 20°C (**hernandez2000synopsis--27?**). Chub mackerel eggs develop for approximately 53-56 hrs at 18-20°C. Larval stages last between 16-24 days, developing faster at higher temperatures (**hernandez2000synopsis--27?**). Atlantic chub mackerel grow rapidly during the first year of life, reaching maturity around 2 years of age (**daley2018growth--27?**) and can live up to 13 years (**hernandez2000synopsis--27?**). Chub mackerel are generalist predators of zooplankton and other pelagic prey including mysids, euphausids, decapod larvae, copepods, amphipods, small fish, cephalopods. Cannibalism has been observed in this species (**hernandez2000synopsis--27?**). There is little data on chub mackerel predation in the Northwest Atlantic. The Mid-Atlantic Fishery Management Council funded a study on diet composition of marlins and tunas and found very few chub mackerel in the stomachs of yellowfin tuna and white marlin (**mafmc2025chub--27?**). In 2019, the Mid-Atlantic Fishery Management Council approved Amendment 21 to the Mackerel, Squid, and Butterfish Fishery Management Plan. This Amendment established conservation and management measures for Atlantic chub mackerel in the Northwest Atlantic (**mafmc2019amendment--27?**). There is currently no stock assessment of chub mackerel in the Mid-Atlantic. Biomass is assumed to be at a sustainable level based on stock productivity and low fishery capacity (**mafmc2025chub--27?**).

27.8 Figures

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28 Cobia (*Rachycentron canadum*)

28.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = Low

Data Quality Model Confidence Score =Life History Data Quality = 88%

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28.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	Low	Moderate (**burton2023climate--28?**)	Decrease
Directional Effect	Neutral	Neutral (**burton2023climate--28?**)	None

28.3 Score Drivers

28.3.1 Vulnerability:

add text here

28.3.2 Climate Exposure:

add text here

28.3.3 Biological Sensitivity:

Cobia received a **Low** sensitivity score. Certainty was 53%. The only attribute that received a score greater than 2.5 was Relationship to Calcified Organisms. Cobia primarily consume crustaceans, which make them vulnerable to ocean acidification (**meyer1996food--28?**; **burton2023climate--28?**).

Cobia was not assessed in the 2016 Climate Vulnerability Assessment for the Northeast Shelf, but was assessed in the recent assessment for the Southeast Shelf (**burton2023climate--28?**). In this assessment, it received a Moderate sensitivity score. This score was driven by Dispersal of Early Life Stages, Early Life History Survival and Settlement Requirements, Reproductive Complexity, Spawning Cycle, and Sensitivity to Ocean Acidification (**burton2023climate--28?**).

28.4 Data Quality Metrics

28.4.1 Life History Data Quality:

88% of data quality scores were greater than or equal to 2, indicating strong data quality. Egg and Larval Survival and Settlement Requirements and Juvenile Habitat Requirements were identified as data gaps (mean attribute data quality score < 2).

28.4.2 Model Confidence:

add text here

28.5 Additional Metrics

28.5.1 Distribution Shifts:

add text here

28.5.2 Directional Effect:

Experts predicted that climate change would have a **neutral** effect on Cobia. Certainty was 95%. Warmer temperatures could lead to northward shifts in distributions and earlier seasonal migrations (**crear2020contemporary--28?**). The recent Southeast US Climate Vulnerability Assessment also predicted that climate change would have a neutral effect on Cobia.

28.6 Climate Effects on Abundance and Distribution

Cobia off the eastern U.S. Atlantic coast currently overwinter offshore in waters south of North Carolina. Suitable habitat increases in the summer months with warming temperatures. Habitat distribution models suggest increased habitat suitability northward, shifting distributions poleward, and possible earlier seasonal migrations (**crear2020contemporary--28?**). This shift northward could prompt revision of management regulations as the fishery expands to new states or regions. Early life stage cobia may be somewhat resistant to projected rises in CO^2 levels (**ebanks2013influence--28?**), although juvenile cobia in a laboratory study exhibited poor growth performance and reduced feeding at elevated temperatures (**le2024impact--28?**).

28.7 Life History Synopsis

Cobia are a pelagic, continental shelf fish distributed from Massachusetts to Argentina but are most abundant from Virginia to the Gulf of America (**shaffer1989synopsis--28?**). Two stocks are recognized in the United States: one along the Atlantic coast south to Georgia and the other along the Atlantic coast of Florida into the Gulf (**sedar2020sedar58--28?**). Cobia are batch spawners, and move inshore to spawn when temperatures reach 20-25°C. They spawn over a period of 4-6 weeks, peaking in June-July in Virginia (**joseph1964spawning--28?**). Cobia eggs hatch within 36 hours of fertilization. Given the short time frame between fertilization and hatching, it is likely that eggs and larvae do not travel far from spawning grounds (**shaffer1989synopsis--28?**). Little is known about the feeding habits of larval and juvenile cobia. Adult cobia are generalist carnivores, feeding on benthic-epibenthic crustaceans (*Portunids* and *Callinectes* sp.), teleosts, cephalopods, and elasmobranchs (**smith1995life--28?**). Juveniles are pelagic and often associated with structure for feeding and shelter. Juveniles and adults can be found near shipwrecks, buoys, shellfish beds, and submerged aquatic vegetation (**sedar2020sedar58--28?**). Few predators of cobia in U.S. Atlantic waters are known, although dolphinfish (*Coryphaena hippurus*) have been reported to prey on small cobia (**shaffer1989synopsis--28?**). Over 50% of cobia are mature by age 2, and can live up to 15 years (**sedar2020sedar58--28?**). Adults are highly migratory. Cobia exhibit a general migratory trend of overwintering in south Florida and the Florida Keys waters, and then migrating north along the Atlantic and Gulf coasts in late winter-early spring. They likely

return to south Florida coastal waters in the late fall (**hendon2008seasonal--28?**). The Atlantic migratory cobia are managed by the Atlantic States Marine Fisheries Commission. The population is not overfished and overfishing is not occurring as of the last benchmark stock assessment in 2020 (**sedar2020sedar58--28?**).

28.8 Figures

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29 Eastern oyster (*Crassostrea virginica*)

29.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = Very High

Data Quality Model Confidence Score =Life History Data Quality = 90%

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29.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	Very High	High	Increase
Directional Effect	Neutral	Negative	Increase

29.3 Score Drivers

29.3.1 Vulnerability:

add text here

29.3.2 Climate Exposure:

add text here

29.3.3 Biological Sensitivity:

The Eastern oyster received a **Very High** sensitivity score, with 100% certainty according to bootstrap analysis. Five attributes had scores greater than 3.5: Adult Mobility (4), Other Stressors (4), Relationship to Calcified Organisms (3.9), Complexity in Reproductive Strategy (3.8), and Egg and Larval Survival and Settlement Requirements (3.5).

Eastern Oysters are sessile once they settle out of the plankton, and settle in estuarine habitats. As a result, they are exposed to a variety of stressors due to urbanization impacts such as hypoxia, loss of seagrass beds, dredging, and changes to freshwater inputs (**guyant2022northeast--29?**). Spawning is initiated by warmer temperatures. Settling out of the plankton also is dependent on temperature and the presence of conspecifics. Because recruitment is density dependent, population collapse can be sudden.

In the 2016 Assessment, Eastern Oysters received a High sensitivity score (**hare2016vulnerability--29?**). This was driven by Sensitivity to Ocean Acidification, Adult Mobility, and Other Stressors, similar to our results.

29.4 Data Quality Metrics

29.4.1 Life History Data Quality:

90% of data quality scores were greater than or equal to 2, indicating excellent data quality. Population Growth Rate and Stock Size/Status were identified as data gaps (mean attribute data quality score < 2).

29.4.2 Model Confidence:

add text here

29.5 Additional Metrics

29.5.1 Distribution Shifts:

add text here

29.5.2 Directional Effect:

Experts predicted that climate change would have a **neutral** effect on Eastern Oysters. Certainty was low at 60%. While oysters live in vulnerable estuarine habitats (**guyant2022northeast--29?**), spawning occurs throughout the year when temperatures are warm (**berringan1991oyster--29?**). Aquaculture programs also help contribute to population recovery (**mackenzie1981biotic--29?**). In the previous Assessment, experts predicted that climate change would have a negative impact on Eastern Oysters.

29.6 Climate Effects on Abundance and Distribution

Climate change, warmer temperatures, and perhaps different tidal ranges, may increase competition for resources with current competitors, like hooked mussel, invasive species such as *Didemnum* sp., or increase exposure to predators like veined rapa whelk (**white1996predators--29?**). Depletion of oyster beds will also likely have negative effects for many marine species and habitats, for they provide refuge and homes for many fish and invertebrates species and help prevent erosion (**eobrt2007status--29?**). Hurricanes can cover oysters in silt, wash them off beds, or impose extended "freshettes" on oyster beds to the degree that they are significantly impaired and need restoration. This may become more problematic, for hurricanes are thought to increase with increased warming temperatures and climate change (**perret1999oyster--29?**). Since adult oysters are sessile and inhabit mainly highly vulnerable habitats in areas with high climate exposure, their spawning habitat is also highly vulnerable (**farr2021assessment--29?**).

29.7 Life History Synopsis

Eastern oysters are found along the eastern U.S. from the Gulf of St. Lawrence to the Gulf of America. They inhabit brackish and salty waters, forming beds or reefs by attaching to each other or other hard substrate (**ozbay2014eastern--29?**). Eastern oysters are a keystone species, and provide valuable habitat for benthic and intertidal organisms. Oysters are sessile filter feeders, contributing to increased water quality through reduced sediment suspension and nutrient uptake and sequestration (**ozbay2014eastern--29?**). Eastern oyster populations in the Mid-Atlantic and Gulf declined in the 19th and 20th centuries, mainly from anthropogenic factors such as land use, habitat degradation, and overharvesting. Disease mortality from Dermo and MSX continues to diminish oyster recovery (**eobrt2007status--29?**). Aquaculture production of oysters contributes to population recovery and habitat restoration programs across the East Coast (**mackenzie1981biotic--29?**). Oysters spawn during the summer in the North and Mid-Atlantic and throughout most of the year in the South Atlantic when temperatures remain above 20°C and salinities above 10ppt (**berringan1991oyster--29?**). The first larval stage (trochophore) is formed four to six hours following fertilization. Larvae normally

settle two to three weeks after hatching, but can last up to 2 months (**eobrt2007status--29?**). Larvae move both by passive and active (as development progresses) transport, although larval transport in estuaries seems to be minimal. A balance is reached where the larvae can find suitable areas to settle while still being retained in estuaries (**eobrt2007status--29?**). Settlement ends at the end of the larval stage, and the larval duration is highly temperature dependent (**shumway1996natural--29?**). (**fritz1984selective--29?**) found that planktotrophic oyster larvae fed upon a variety of phytoplankton taxa (coccoids, centrate diatoms, pennate diatoms, dinoflagellates, and flagellates). (**baldwin1991omnivorous--29?**) found that the larvae are actually omnivorous, preying on both autotrophic and heterotrophic plankton. Predation on oyster larvae is extremely high by filter-feeders in the water column and in the benthos. Along the Atlantic coast, drills have been responsible for mortality on 33% of spat in Long Island Sound. Oysters can compete with other mussels and barnacles for shell and substrate spaces, resulting in increased mortality when competitors are abundant (**kennedy1996biology--29?**). Juveniles and adults feed primarily on phytoplankton and suspended detritus, relying on food-laden water being moved past them in order to extract food from the water column (**langdon1996digestion--29?**). Maturity occurs between 1 and 3 years of age (**fsuml_oyster_lifecycle--29?**). Several ages have been proposed, the most common being 20 years. Oysters in Maryland appear to survive at at least 10 years based on plantings at restoration sites which are not thought to have received natural spat sets. It is likely that the maximum longevity is substantially longer than 10 years (**mddnr2025stock--29?**). Wild and aquaculture harvest is managed by individual jurisdictions or states, and management guidelines vary (**ozbay2014eastern--29?**).

29.8 Figures

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30 Golden tilefish (*Lopholatilus chamaeleonticeps*)

30.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = Moderate **Data Quality**Model Confidence Score =Life History Data Quality = 93%

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30.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	Moderate	High (**burton2023climate--30?**)	Decrease
Directional Effect	Neutral	Neutral (**burton2023climate--30?**)	None

30.3 Score Drivers

30.3.1 Vulnerability:

add text here

30.3.2 Climate Exposure:

add text here

30.3.3 Biological Sensitivity:

Golden Tilefish received a **Moderate** sensitivity score with 62% certainty. Four attributes had scores greater than 2.5: Stock Size (3), Adult Mobility (3), Population Growth Rate (2.8), and Historical Variability in Temperature (2.7).

Golden Tilefish primarily occupy burrows on the seafloor. Die offs have been observed after meanders in the Gulf Stream, suggesting that they are very sensitive to temperature shifts (**able1982burrow--30?**). Golden tilefish mature at 2-4 years and maximum ages of 45 years have been documented (**steimle1999essential--30?**; **bigelow1953fishes--30?**), suggesting they are a long-lived species that may be susceptible to overfishing. Overfishing is currently occurring in the Mid-Atlantic stock (**nefsc2024golden--30?**).

Golden tilefish were not assessed along the Northeast Shelf in the 2016 Assessment, but were assessed in the recent Southeast US Climate Vulnerability Assessment. They received a High sensitivity score in that assessment. This was driven by two attributes: Habitat Specialization and Population Growth Rate (**burton2023climate--30?**), similar to our results.

30.4 Data Quality Metrics

30.4.1 Life History Data Quality:

93% of data quality scores were greater than or equal to 2, indicating excellent data quality. No data gaps (mean attribute data quality score < 2) were identified.

30.4.2 Model Confidence:

add text here

30.5 Additional Metrics

30.5.1 Distribution Shifts:

add text here

30.5.2 Directional Effect:

Experts predicted that climate change would have a **neutral** impact on Golden tilefish. Certainty was 92%. Golden Tilefish occupy the 'warm belt' where shelf and slope waters meet. Warming temperatures could increase the amount of preferred habitat. In the Southeast Climate Vulnerability Assessment, experts also predicted that climate change will have a neutral impact on Golden Tilefish, and hypothesized that warming temperatures could increase the amount of available habitat in the northeast (**burton2023climate--30?**).

30.6 Climate Effects on Abundance and Distribution

Golden tilefish prefer a very narrow temperature range (8-17°C); increased ocean acidification may have negative impacts on survival, growth, and abundance (**salois2024golden--30?**). (**nesslage2021environmental--30?**) identified a correlation between negative Atlantic Multidecadal Oscillation (AMO) and positive North Atlantic Oscillation (NAO) and golden tilefish recruitment class pulses. Catch per unit effort is influenced by temperature and oceanographic currents (Gulf Stream and Labrador Current) (**nesslage2021environmental--30?**). A large golden tilefish die-off and resulting fishery collapse in the spring of 1882 has been correlated with meanders in the Gulf Stream, suggesting that they are very sensitive to temperature changes (**fisher2014life--30?**). Changes to weather patterns and a rise in stormy conditions along with very cold bottom temperatures can drive golden tilefish off the shelf and cause a decrease in fishing (**salois2024golden--30?**).

30.7 Life History Synopsis

Golden tilefish inhabit continental shelf and slope waters along the entire U.S. East and Gulf Coasts, although they are most abundant from Nantucket to Cape May, New Jersey (**steimle1999essential--30?**). Two genetically-different tilefish stocks have been identified in United States waters: the Mid-Atlantic Bight and south of Cape Hatteras into the Gulf of America (**steimle1999essential--30?**). Individuals of many species off Cape Hatteras were smaller and less active than their conspecifics off Virginia. Tilefish are classified as serial or fractional spawners from March to November with a peak in activity in June (**idelberger1988reproductive--30?**). However, with serial or fractional spawners there is some doubt whether all of the eggs in the ovaries are released during a single seasonal spawning cycle. Residual eggs could be resorbed during the winter (**grimes1988reproductive--30?**). Artificially-fertilized eggs hatched in 40 hours at 22-24.6°C; it is likely that egg duration increases in colder temperatures (**fahay1981preliminary--30?**). The duration of the larval stage is unknown. Movements of water masses along the shelf break may play a role in recruitment success, including larval survival and retention near spawning grounds. Strong recruitment pulses have been linked to shifts in the Gulf Stream Index,

AMO, and NAO (**salois2024golden--30?**). Little is known about larval diets, but they likely prey on zooplankton, particularly *Calanus* (**steimle1999essential--30?**). Common predators of larval and juvenile tilefish are monkfish, spiny dogfish, and bottom-dwelling sharks (**steimle1999essential--30?**). Post-larval tilefish are primarily sedentary and non-migratory. They settle in waters approximately 8-17°C, known as the "warm belt" (where shelf and slope water meet) on the outer continental shelf and upper slope of the Northwest Atlantic (**able1982burrow--30?**). Juveniles and adults often occupy simple vertical shaft burrows in semi-lithified clay of the shelf break, submarine canyon walls, and flanks around 100-200m depth (**able1982burrow--30?**). Lobster pots, red crab traps, ship wrecks and other artificial structures have also been used by tilefish (**steimle1999essential--30?**). Diet shifts with size, with crustaceans, mollusks, and echinoderms dominating young/small tilefish diets. Adult tilefish are opportunistic benthic foragers, feeding on shrimp, crabs, snails, worms, anemones, sea cucumbers, smaller fish, squid, and bivalves. They are known to scavenge on fresh material and reject rancid bait. Adults will also consume anthropogenic material (**steimle1999essential--30?**). Age at maturity is typically 2-4 years (**steimle1999essential--30?**), with a maximum age up to 45 years (**bigelow1953fishes--30?**). The Northwest Atlantic stock is managed by the Mid-Atlantic Fishery Management Council under the Tilefish Fishery Management Plan. The most recent benchmark stock assessment occurred in 2024. The stock is not overfished, but overfishing is occurring and the Mid-Atlantic stock is reduced to end overfishing (**nefsc2024golden--30?**).

30.8 Figures

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31 Haddock (*Melanogrammus aeglefinus*)

31.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = Moderate **Data Quality**Model Confidence Score =Life History Data Quality = 97%

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31.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	Moderate (All Stocks)	Low (All Stocks)	Increase
Directional Effect	Negative (Georges Bank) - Neutral (Gulf of Maine)	Negative (All Stocks)	None (Georges Bank) - Increase (Gulf of Maine)

31.3 Score Drivers

31.3.1 Vulnerability:

add text here

31.3.2 Climate Exposure:

add text here

31.3.3 Biological Sensitivity:

Both the Georges Bank and Gulf of Maine stocks of Haddock received **Moderate** sensitivity scores. Certainties were 68% and 71%, respectively. For both stocks, Complexity in Reproductive Strategy (3) and Egg and Larval Survival and Settlement Requirements (3) had scores greater than 2.5. For the Gulf of Maine stock, Spawning Cycle (2.5) also had a score greater than 2.5.

Haddock spawn in aggregations. While bottom type conditions drive where haddock spawn, the timing of spawning has been correlated to temperatures in some studies. Retention of larvae is important, especially to the Georges Bank stock, as strong year-classes are present when circulation patterns promote retention on or near Georges Bank (**brodziak2005essential--31?**). In the Gulf of Maine there is evidence that tidal currents help promote the retention of larvae.

In the previous Climate Vulnerability Assessment for the Northeast Shelf, Haddock received a Low sensitivity score. No attributes scored above 2.5 in the previous assessment (**hare2016vulnerability--31?**).

31.4 Data Quality Metrics

31.4.1 Life History Data Quality:

97% of data quality scores were greater than or equal to 2 for both stocks, indicating excellent data quality. No data gaps (mean attribute data quality score < 2) were identified.

31.4.2 Model Confidence:

add text here

31.5 Additional Metrics

31.5.1 Distribution Shifts:

add text here

31.5.2 Directional Effect:

Experts predicted that the **Georges Bank** Haddock stock will be **negatively** impacted by climate change, while climate change will **neutrally** impact the **Gulf of Maine** stock. Certainties were 70% and 68%, respectively. Haddock abundance on Georges Bank has increased recently and coincides with rising ocean temperatures (**friedland2008does--31?**; **wang2024density--31?**), but increasing temperatures may also lead to increased predation by cod (**durant2020nonlinearity--31?**). Retention plays a large role in recruitment on Georges Bank, but not much is known about the role retention plays in the Gulf of Maine stock (**brodziak2005essential--31?**).

Experts predicted that haddock would be negatively impacted by climate change in the previous Assessment (**hare2016vulnerability--31?**).

31.6 Climate Effects on Abundance and Distribution

In the Northeast Atlantic, early life stage haddock and cod compete for food and later life stage cod prey upon haddock. Changing climate and the increasing number of warm years increases cod abundance, causing a decline in haddock (**durant2020nonlinearity--31?**). Haddock abundance on Georges Bank has increased significantly in recent years and may coincide with a rise in extreme ocean temperatures or phytoplankton blooms (**friedland2008does--31?**; **wang2024density--31?**). A large recruitment event in Georges Bank haddock in 2003 was hypothesized to be related to the timing of the fall bloom and enhanced adult condition, affecting reproductive output (**friedland2008does--31?**). Spatial distribution changes of haddock were modeled using bottom trawl survey data, and results showed range expansion and contraction in both spring and fall. Higher temperatures correspond with northward distribution shifts on Georges Bank (**wang2024density--31?**). Future warming may shift haddock's southern range boundary, making it difficult to define stock boundaries.

31.7 Life History Synopsis

In the Northwest Atlantic, haddock are found from Newfoundland to Cape May, New Jersey but are most abundant in the Gulf of Maine and on Georges Bank (**brodziak2005essential--31?**). The stock is managed in two subunits for the Gulf of Maine and Georges Bank (**nefsc2024gb--31?**; **nefsc2024gom--31?**). Haddock primarily spawn on the Northeast peak of Georges Bank and in the Great South Channel. Spawning occurs from January to July with a delay in peak spawning time with increasing latitude (**smith1985retention--31?**). Spawning distribution is influenced more by restrictive spawning area and bottom type conditions than by temperature variation (**page1989spawning--31?**). At water temperatures typical of Georges Bank, haddock eggs hatch in about 15 days. Larvae metamorphose into

juveniles in roughly 30-42 days at lengths of 2-3 cm (**page1989spawning--31?**). Small juveniles initially live and feed in the epipelagic zone and remain in the upper part of the water column for 3-5 months (**fahay1983guide--31?**). The retention of haddock larvae in suitable nursery areas is an important factor in determining year class strength of Georges Bank haddock. The clockwise gyre around the main portion of Georges Bank provides a physical mechanism to retain haddock larvae on the Bank. There is limited information on retention of larval haddock in the Gulf of Maine. Larvae can potentially be advected as far south as the Mid-Atlantic Bight, but local retention or transport to nursery habitats to the southwest of Georges Bank are important to strong recruitment (**brodziak2005essential--31?**). Haddock larvae are passive foragers on less motile prey: invertebrate eggs, copepods, and phytoplankton (**marak1960food--31?**). Planktonic prey declines after juveniles become demersal, feeding on ophiuroids and polychaetes instead of copepods and pteropods (**brodziak2005essential--31?**). Common predators of juvenile haddock include spiny dogfish, skates, and other groundfish. Gray seals also prey on adult haddock (**brodziak2005essential--31?**). Adult haddock are indiscriminate consumers of benthic, sedentary or slow-moving invertebrates: crustaceans, annelids, polychaetes, mollusks, and echinoderms. Prey consumption varies by area and prey availability indicating adaptability (**tam2016trans--31?**). Juvenile and adult haddock are demersal and usually found between 40-150m depth. Preferred bottom types include gravel, pebbles, clay and smooth hard sand, particularly smooth areas between rocky patches (**kleinmacphee2002haddock--31?**). Age at maturity ranges from 1-4 years with a maximum age of 22 years (**nefsc2014saw59--31?**). The Georges Bank and Gulf of Maine haddock stocks are managed by the New England Fishery Management Council under the Northeast Multispecies (Groundfish) Fishery Management Plan. Both stocks are not overfished and overfishing is not occurring (**nefsc2024gb--31?**; **nefsc2024gom--31?**).

31.8 Figures

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32 Longfin squid (*Doryteuthis (Amerigo) pealeii*)

32.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = Low

Data Quality Model Confidence Score =Life History Data Quality = 93%

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32.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	Low	Low	None
Directional Effect	Neutral	Positive	Decrease

32.3 Score Drivers

32.3.1 Vulnerability:

add text here

32.3.2 Climate Exposure:

add text here

32.3.3 Biological Sensitivity:

Long-finned squid received a **Low** sensitivity score. Certainty was 98% according to bootstrap results. No attributes scored greater than 2.5. Complexity in Reproductive Strategy received the highest score (2.4). Eggs are cemented to bottom substrate, which can include algae and boulders that may be impacted by dredging activities in nearshore spawning grounds (**jacobson2005essential--32?**; **guyant2022northeast--32?**).

Long-finned squid also received a Low sensitivity score in the previous Climate Vulnerability Assessment for the Northeast Shelf. Similar to our results, this previous score was driven by Early Life History Requirements due to the benthic egg masses of this species (**hare2016vulnerability--32?**).

32.4 Data Quality Metrics

32.4.1 Life History Data Quality:

93% of data quality scores were greater than or equal to 2, indicating excellent data quality. No data gaps (mean attribute data quality score < 2) were identified.

32.4.2 Model Confidence:

add text here

32.5 Additional Metrics

32.5.1 Distribution Shifts:

add text here

32.5.2 Directional Effect:

Experts predicted that climate change would have **neutral** impacts on long-finned squid. Certainty was 95%. The short life-span of long-finned squid can make it challenging to understand the effects of climate change. In warmer temperatures, squid may grow faster and mature earlier, but they may also be negatively impacted by ocean acidification (**pecl2008potential--32?**; **arkhipkin2015world--32?**) . In the previous assessment along the Northeast shelf, experts predicted that climate change would have a positive impact on long-finned squid, with a low amount of certainty (**hare2016vulnerability--32?**).

32.6 Climate Effects on Abundance and Distribution

Cephalopods have been shown to be sensitive to environmental variability and climate change, due to their short lifespan (**moustahfid2021ecological--32?**). Inshore squids, such as longfin, may have complex responses to climate change. As temperatures increase, squids will likely hatch smaller and earlier, grow faster, mature at a smaller size, and have shorter life spans. They may also require more food and oxygen, but a reduced capacity to cope without food (**pecl2008potential--32?**). Longfin squid recruitment may negatively be affected by impacts of increasing ocean acidification on the paralarval stage (**arkhipkin2015world--32?**). Longfin squid raised under elevated pCO^2 increased hatching time (~24 hours longer) and had shorter mantle lengths as well as significantly reduced statolith surface area (by 25%) and abnormally shaped and porous statoliths (**kaplan2013adverse--32?**). Abundance and distribution changes have been linked to the North Atlantic Oscillation, Gulf Stream, and surface fronts on the outer continental shelf (**dawe2007ocean--32?**; **hendrickson2023cyclical--32?**; **manderson2011ocean--32?**).

32.7 Life History Synopsis

In the Northwest Atlantic, longfin squid are most abundant from Georges Bank to Cape Hatteras although their range expands from Newfoundland to the Gulf of Venezuela (**jacobson2005essential--32?**). The North Atlantic stock undertakes seasonal migrations: moving offshore in the fall to overwinter along the shelf edge and slope and migrating inshore in the spring following warming water temperatures to spawn (**hatfield2002geographic--32?**). More recent studies after the development of statolith aging techniques highlight a peak in spawning from late spring to summer (**hatfield2002geographic--32?**; **macy2001seasonal--32?**) and hatching in late summer (**jacobson2005essential--32?**). Some spawning occurs throughout the year both inshore and offshore depending on hatching month and length-at-age (**macy1994application--32?**; **macy2001seasonal--32?**), with winter spawning thought to occur primarily south of Cape Hatteras (**hatfield2002geographic--32?**). Eggs incubate for 11 - 27 days on sandy or muddy bottoms attached to solid substrate, with faster development times are warmer temperatures. The duration of the larval stage is unclear, but they occupy coastal surface waters. Once juveniles, they transition from inhabiting surface waters to a demersal lifestyle (**jacobson2005essential--32?**). The Gulf Stream and warm core rings may facilitate the recruitment transport of larval and juvenile squid northward, but potential for inputs to the population from the Southeast U.S. Continental Shelf and offshore are difficult to quantify (**hatfield2002geographic--32?**). Juveniles and adults are generally demersal and inhabit the continental shelf and upper slope to depths of 400 m. This varies seasonally, with adults occupying shallower habitats in summer and fall, and deeper habitats in the winter. Diel vertical migrations have been reported (**jacobson2005essential--32?**). Juveniles consume planktonic organisms, primarily copepods. Their preferred prey type varies with size (**jacobson2005essential--32?**). Larger individuals feed on

crustaceans and small fish. Diets shift seasonally from crustacean-dominated in the spring to fish-dominated in the fall. Cannibalism is observed in individuals larger than 5 cm (**jacobson2005essential--32?**). Longfin squid are prey for a variety of finfish, marine mammals, and birds (**jacobson2005essential--32?**). Most statolith-aging studies suggest that longfin squid have a short lifespan of around 1 year or less, reaching maturity shortly before spawning and senescence (**macy2001seasonal--32?**). Longfin squid are managed by the Mid-Atlantic Fishery Management Council under the Mackerel, Squid, and Butterfish Fishery Management Plan. The stock is not overfished, and overfishing reference points are not available (**mafmc2025longfin--32?**).

32.8 Figures

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33 Monkfish (*Lophius americanus*)

33.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = Low

Data Quality Model Confidence Score =Life History Data Quality = 88%

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33.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	Low (Both Stocks)	Low (Both Stocks)	None (Both Stocks)
Directional Effect	Negative (Both Stocks)	Neutral (Both Stocks)	Decrease (Both Stocks)

33.3 Score Drivers

33.3.1 Vulnerability:

add text here

33.3.2 Climate Exposure:

add text here

33.3.3 Biological Sensitivity:

Both the **Northern** and **Southern** Stocks of Monkfish received a **Low** sensitivity score. Certainties were 99% and 98% respectively. For both stocks, the only attribute that received a score greater than 2.5 was Population Growth Rate (2.6 for both stocks). Monkfish mature slowly, reaching maturity at 4-5 years (**richards2008population--33?**). Maximum ages range from 7 years for males to 14 years for females (**nefsc2010fifty--33?**).

Monkfish also received a Low sensitivity score in the 2016 Climate Vulnerability Assessment for the Northeast Shelf. Like in this assessment, this result was driven by Population Growth Rate (**hare2016vulnerability--33?**).

33.4 Data Quality Metrics

33.4.1 Life History Data Quality:

88% of data quality scores were greater than or equal to 2, indicating strong data quality. No data gaps (mean attribute data quality score < 2) were identified.

33.4.2 Model Confidence:

add text here

33.5 Additional Metrics

33.5.1 Distribution Shifts:

add text here

33.5.2 Directional Effect:

Climate change is predicted to **negatively** impact both the **northern** and **southern** stocks of Monkfish. Certainties were 94% and 87% respectively. Monkfish are most abundant on Georges Bank when bottom temperatures are between 5-14 C, suggesting that warming temperatures would decrease the amount of preferred habitat (**siemann2018summering--33?**). Experts in the previous Assessment predicted that climate change would have a neutral impact on Monkfish with a high degree of uncertainty (**hare2016vulnerability--33?**).

33.6 Climate Effects on Abundance and Distribution

In Icelandic waters, monkfish distribution changes and increased stock size have been correlated with rising water temperatures and expanding suitable habitat (**solmundsson2007recent--33?**). Monkfish movement and relative abundance on Georges Bank is driven by seasonal changes in bottom temperature. Their distribution may shift in response to future changes in thermal habitat, highlighting the need for further management strategies (**siemann2018summering--33?**). Maximum size of monkfish in the Northwest Atlantic has decreased, and is related to seasonality and bottom temperature (**charbonneau2020pervasive--33?**).

33.7 Life History Synopsis

Monkfish are a bottom dwelling fish ranging from the northern Gulf of St. Lawrence to Cape Hatteras. The commercial fishery operates from Maine to North Carolina (**steimle1999monkfish--33?**). Spawning takes place from spring through early autumn, progressing from south to north over a 6 month period (**richards2008population--33?**). Pelagic larvae hatch from egg veils (up 12 m long and 1.5 m wide) 1-3 weeks after being laid, depending on temperature. Larvae remain in the egg veil 2-3 days after hatching. Monkfish larvae feed on zooplankton, including copepods, crustacean larvae, and chaetognaths (**bigelow1953fishes--33?**). Larval to juvenile transition occurs at 5-10 cm TL (total length) when the elongate fins and body gradually assume the adult form (**nefsc2010fifty--33?**). Once larvae transition to juveniles, they move to the benthos. Juveniles and adults settle on mud, gravelly sand, algae, and rock substrates (**steimle1999monkfish--33?**). Once moved to the bottom, juveniles can eat sand lance, but invertebrates such as red shrimp and squid make up the majority of their diet. The amount of invertebrates eaten by juveniles decreases as they get older (**steimle1999monkfish--33?**). Monkfish are ambush predators and opportunistic feeders; prey found in their stomachs include a variety of benthic and pelagic species. Monkfish are also cannibalistic and can eat sea birds and diving ducks. Adults spend most of their time resting on the bottom, but have been reported at the surface after a storm event, preying on seabirds (**richards2008population--33?**). Large monkfish have few predators, but smaller monkfish are common prey for swordfish, sharks, and skates (**steimle1999monkfish--33?**). Monkfish move by slow swimming or by using their sturdy pectoral fins to "walk." Despite slow speeds, juveniles and adults migrate in response to thermal conditions, spawning, or food availability (**steimle1999monkfish--33?**). Monkfish were found on Georges Bank in summer and early fall, and found in deeper waters/on the edge of the bank in winter months, suggesting that movements may be influenced by temperature (**rountree2006extraction--33?**). Age at maturity is 4-5 years, longer for females and for the Mid-Atlantic stock (**richards2008population--33?**). Maximum age is listed at 7 years for males and 12-14 years for females (**nefsc2010fifty--33?**). Monkfish in the Northwest Atlantic are divided into two stocks: the Gulf of Maine/Northern Georges Bank and Southern

Georges Bank/Mid-Atlantic. Both stocks do not have stock assessments that provide indices on overfishing and overfished status, although an index based assessment is used for catch advice (**deroba2022draft--33?**).

33.8 Figures

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34 Northern Quahog (*Mercenaria mercenaria*)

34.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = High

Data Quality Model Confidence Score =Life History Data Quality = 88%

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34.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	High	Very High	Decrease
Directional Effect	Negative	Negative	None

34.3 Score Drivers

34.3.1 Vulnerability:

add text here

34.3.2 Climate Exposure:

add text here

34.3.3 Biological Sensitivity:

The Northern Quahog received a **High** Sensitivity score, with 100% certainty. Three attributes received scores greater than 3: Adult Mobility (4), Relationship to Calcified Organisms (3.8), and Population Growth Rate (3.1).

Northern Quahogs, like other bivalves, are mostly sessile, only capable of small-scale movements (**stanley1983species--34?**). Similarly, they also have calcium carbonate shells that put them at risk of ocean acidification (**stanley1983species--34?**). They are long-lived, slowly maturing species, reaching maturity at 1-3 years and maximum ages usually around 12-20 years, although 46 year old quahogs have been observed (**bricelj1992aspects--34?**; **ridgway2011new--34?**).

In the previous Climate Vulnerability Assessment for the Northeast Shelf, Northern Quahogs received a Very High sensitivity score. This was driven by Sensitivity to Ocean Acidification, Adult Mobility, and Other Stressors (**hare2016vulnerability--34?**). In our analysis, Other Stressors received a score of 2.9.

34.4 Data Quality Metrics

34.4.1 Life History Data Quality:

88% of data quality scores were greater than or equal to 2, indicating strong data quality. Only Stock Size/Status was considered a data gap (mean attribute data quality < 2).

34.4.2 Model Confidence:

add text here

34.5 Additional Metrics

34.5.1 Distribution Shifts:

add text here

34.5.2 Directional Effect:

Experts predict that the Northern Quahog will be **negatively** impacted by climate change, with 70% certainty. Temperature and salinity are important to reproductive success and settlement – increasing temperatures outside the preferred temperature range could inhibit spawning and increased freshwater intrusions could inhibit metamorphosis (**stanley1983species--34?**; **kennish1975effects--34?**). Eutrophication and ocean acidification could also impact stock fitness (**carmichael2004changes--34?**; **ivanina2013interactive--34?**). In the previous Climate Vulnerability Assessment for the Northeast shelf, experts also predicted that Northern Quahogs would be negatively impacted by climate change (**hare2016vulnerability--34?**).

34.6 Climate Effects on Abundance and Distribution

Temperature is the decisive factor for final gamete maturation in northern quahogs. (**kennish1975effects--34?**) cited 21 to 25°C as the required or preferred temperature range for spawning. Metamorphosis is inhibited at salinities below 17.5 to 20 ppt (**stanley1983species--34?**). Increased freshwater intrusions (man-made or otherwise) have resulted in clam die offs. Eutrophication of habitat has resulted in reduced growth and health of stocks (**carmichael2004changes--34?**). Recent research suggests that quahogs are relatively resilient to suboxic conditions, which may help protect them from predators that are more sensitive to these conditions (**altieri2006local--34?**). Northern quahogs exposed to higher temperatures and elevated pCO^2 led to decreased shell hardness and energy deficiency (**ivanina2013interactive--34?**). Alternatively, northern quahogs exposed to marine heat wave conditions exhibited increased absorption efficiency, respiration rate, excretion rate, and scope for growth, indicating their ability to recover from persistent or repeated marine heat waves (**liang2024hard--34?**).

34.7 Life History Synopsis

The northern quahog, or hard clam, is a hard shell clam inhabiting intertidal and subtidal waters from the Gulf of St. Lawrence south to Florida (**kraeuter2001biology--34?**). Spawning occurs from May through August, depending on latitude and temperature. Female hard clams require 2 - 2.5 months to spawn out completely, but the greatest release of eggs is during the initial spawning of the season (**stanley1983species--34?**). Eggs hatch within 12-14 hours. Larvae are planktonic until metamorphosis, which occurs within 5-7 days at 30°C, or 16-24 days at 18°C (**stanley1983species--34?**; **rice1992review--34?**). Eggs may be carried by currents up to 25km from spawning sites. Larvae feed on phytoplankton, primarily dinoflagellates and use physical transport mechanisms to travel to the entrance of estuaries (**stanley1983species--34?**). The clams prefer to settle on sandy substrate with a few small rocks and/or shells, and set more densely in sand than mud (**mackenzie1979management--34?**). Juveniles likely

consume phytoplankton or small zooplankton. The adult hard clam feeds by filtering out plankton and microorganisms that are carried along the bottom by currents (**stanley1983species--34?**), burrowing into the sediment and leaving their siphons exposed to feed. However, adults cease filter feeding at 6°C (**kraeuter2001biology--34?**). They are preyed upon by gastropods, crabs, and starfish (**stanley1983species--34?**). These clams have slow growth rates, reaching maturity at 1-3 years with a maximum lifespan of 12-20 years (**bricelj1992aspects--34?**), although they have been found up to 46 years (**ridgway2011new--34?**). There are numerous reports of isolated incidents of hard clam disease causing high mortality rates. Disease may be related to a parasite (QPX - Quahog parasite unknown). The prevalence of QPX appears to be increasing (**smolowitz1998observations--34?**; **dove2004histological--34?**). Quahogs occur in areas of harmful algal blooms (HABs) and may act to reduce prevalence and length of HABs due to filtration. Negative effects of HABs on quahog have been noted, specifically the presence of HABs have been shown to impact growth rates (**cerrato2004effect--34?**). Because these clams are primarily estuarine and inhabit specific bays and coastal zones, the fishery is managed on a localized, state-by-state basis rather than by federal fishery councils. Primary management tools include minimum legal size limits (typically 50 mm), gear restrictions, and daily bag limits (**ncdmf2025north--34?**; **njdep2014inventory--34?**).

34.8 Figures

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35 Ocean pout (*Zoarces americanus*)

35.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = High

Data Quality Model Confidence Score =Life History Data Quality = 83%

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35.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	High	High	None
Directional Effect	Negative	Negative	None

35.3 Score Drivers

35.3.1 Vulnerability:

add text here

35.3.2 Climate Exposure:

add text here

35.3.3 Biological Sensitivity:

The ocean pout received a **High** sensitivity score, with a certainty of 100%. Two attributes had scores greater than 3: Stock Size (4) and Dispersal of Early Life Stages (3.6). The stock is currently overfished, but overfishing is not occurring (**nefsc2022ocean--35?**). Eggs masses and larvae are both demersal, with larvae quickly associating with benthic habitats after hatching (**steimle1999_pout--35?**).

In the 2016 Climate Vulnerability Assessment, ocean pout also received a High sensitivity score. This was driven by Dispersal and Early Life History, Stock Status, and Population Growth Rates (**hare2016vulnerability--35?**). In our assessment, Population Growth Rate received a score of 2.8.

35.4 Data Quality Metrics

35.4.1 Life History Data Quality:

83% of data quality scores were greater than or equal to 2, indicating strong data quality. Adult Mobility and Egg and Larval Survival and Settlement Requirements were considered data gaps (mean attribute data quality score < 2).

35.4.2 Model Confidence:

add text here

35.5 Additional Metrics

35.5.1 Distribution Shifts:

add text here

35.5.2 Directional Effect:

Experts predict that ocean pout will be **negatively** impacted by climate change with 99% certainty. Ocean pout prefer colder waters and are sluggish movers (**steimle1999_pout--35?**), which may make it difficult to avoid stressful conditions such as hypoxia ((**azarovitz1976effects--35?**). In the previous Climate Vulnerability Assessment, experts made a similar prediction, but certainty was lower (**hare2016vulnerability--35?**).

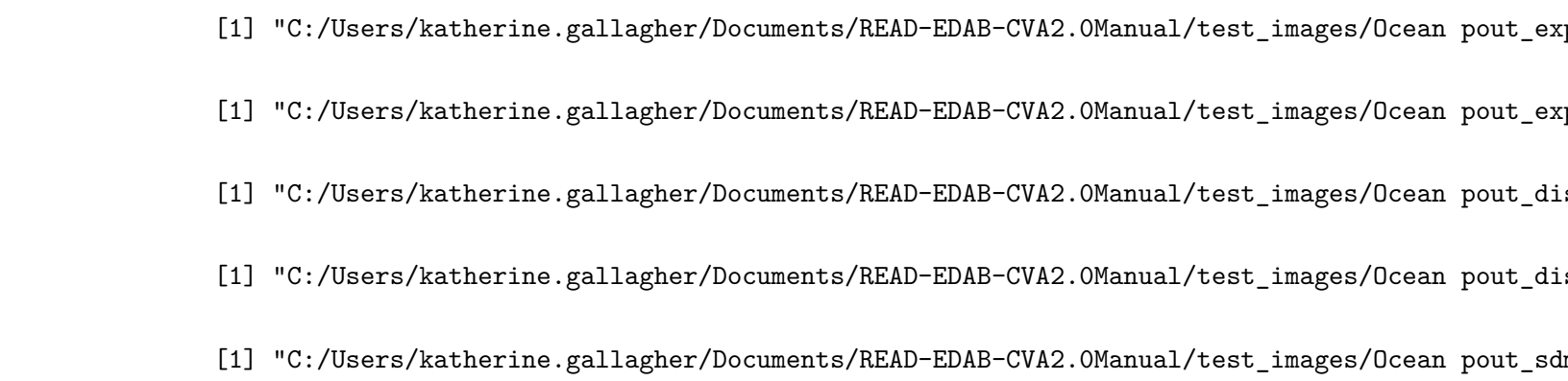
35.6 Climate Effects on Abundance and Distribution

(**nye2009changing--35?**) identified an increase in depth of occurrence of ocean pout in the Gulf of Maine. Although ocean pout have been reported to be quite hardy, they appear to have limited tolerance of hypoxic and anoxic conditions; they are common victims of these conditions which occasionally occur in the New York Bight (**azarovitz1976effects--35?**). Ocean pout are sluggish swimmers, which can explain their inability to avoid stressful conditions, such as hypoxia (**steimle1999_pout--35?**). (**killen2008lack--35?**) found that young ocean pout from Newfoundland do not show thermal compensation in response to long-term temperature changes.

35.7 Life History Synopsis

Ocean pout are a non-migratory, bottom dwelling fish found along the Atlantic coast from Labrador to Delaware (**steimle1999_pout--35?**). Spawning occurs in the late summer through early winter (peak in September-October) with earlier peaks (August-October) in the south (**wilk1979annual--35?**). Eggs are laid in gelatinous masses in a sheltered place on the bottom, such as rocky crevices, where they are guarded either by one or both parents until hatching (**steimle1999_pout--35?**). Eggs develop in 2-3 months, temperature dependent (**olsen1946studies--35?**). Larvae are about 30 mm long at hatching (**clark1982ocean--35?**) and live at the bottom at temperatures below 10°C (**steimle1999_pout--35?**). Juvenile and adult ocean pout use both open and rough habitats. (**sedberry1983food--35?**) reported that juveniles were common during the summer on the sandy, mid- to outer-continental shelf (approximately 35-95 m). In the Gulf of Maine, juveniles occur in shallow coastal waters around rocks and attached algae, and in rivers with saline bottom waters (**auster1985some--35?**). (**macdonald1983laboratory--35?**) reported that ocean pout feed by sorting mouthfuls of sediments for infauna and do not use visual cues to stimulate feeding; they maintained that ocean pout were not very effective at capturing moving prey and rarely left the bottom to feed. (**auster1985some--35?**) and (**auster1995patterns--35?**), however, considered ocean pout an "ambush predator" that waits in sediment depressions for prey to approach or drift by in the current. Juveniles feed primarily on gammarid amphipods and polychaetes (**sedberry1983food--35?**). Adult ocean pout feed on a variety of benthic invertebrates, including polychaetes, mollusks, crustaceans, and echinoderms (**steimle1999_pout--35?**). Adults reach maturity at 2+ years (**steimle1999_pout--35?**), with a maximum age of 18 years (**hoenig1983empirical--35?**). Although they do not migrate, ocean pout move seasonally within a region: offshore in late summer and inshore in the spring in the Gulf of Maine, and offshore to cooler waters in the summer returning inshore in the fall in Southern New England and Georges Bank (**steimle1999_pout--35?**). Ocean pout are managed by the New England Fishery Management Council under the Northeast Multispecies Fishery Management Plan. As of the 2022 management track assessment, the stock is not overfished and overfishing is not occurring (**nefsc2022ocean--35?**).

35.8 Figures



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36 Ocean quahog (*Arctica islandica*)

36.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = Very High

Data Quality Model Confidence Score =Life History Data Quality = 87%

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36.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	Very High	Very High	None
Directional Effect	Negative	Negative	None

36.3 Score Drivers

36.3.1 Vulnerability:

add text here

36.3.2 Climate Exposure:

add text here

36.3.3 Biological Sensitivity:

The ocean quahog received a **Very High** sensitivity score, with a certainty of 71%. Three attributes had scores greater than 3.5: Adult Mobility (4), Population Growth Rate (4), and Relationship to Calcified Organisms (3.6).

Similar to other bivalves, ocean quahogs have calcium carbonate shells and are sessile (**cargnelli1999_oq--36?**). They are also one of the longest-lived bivalve species, with maximum ages high as 507 years old documented, and ages over 300 years old common in the Northwest Atlantic (**sower2023examination--36?**). They mature at 13 years of age (**cargnelli1999_oq--36?**).

Ocean quahogs also received a Very High sensitivity score in the previous Climate Vulnerability Assessment for the Northeast Shelf. The same attributes: Adult Mobility, Population Growth Rate, and Sensitivity to Ocean Acidification, drove the score (**hare2016vulnerability--36?**).

36.4 Data Quality Metrics

36.4.1 Life History Data Quality:

87% of data quality scores were greater than or equal to 2, indicating strong data quality. No data gaps (mean attribute data quality score < 2) were identified.

36.4.2 Model Confidence:

add text here

36.5 Additional Metrics

36.5.1 Distribution Shifts:

add text here

36.5.2 Directional Effect:

Experts predicted that Ocean Quahog would be **negatively** impacted by climate change. Certainty was 100%. Warming conditions are expected to shift the species' distribution north and further offshore, possibly leading to competition between surfclams and ocean quahogs (**weinberg2002influence--36?**; **stromp2023evaluation--36?**; **spencer2024climate--36?**). Ocean acidification will also negatively impact the species. The 2016 Climate Vulnerability Assessment also suggested that Ocean Quahogs would be negatively impacted by climate change (**hare2016vulnerability--36?**).

36.6 Climate Effects on Abundance and Distribution

Radiocarbon aging on ocean quahog shells suggest past (200 years) ocean quahog distributions inshore of their current range, indicating a range shift coinciding with the regression of the Cold Pool after the Little Ice Age (**leclaire2022historical--36?**). With settlement occurring at the end of the larval stage, it may be presumed that temperature impacts settlement rate. (**weinberg2002influence--36?**) note that an increase in temperature would shift the species offshore and north. Future warming conditions are predicted to result in range shifts in Atlantic surfclams that would lead to potential competition for food and space between surfclams and ocean quahogs (**stromp2023evaluation--36?**; **spencer2024climate--36?**).

36.7 Life History Synopsis

Ocean quahogs are bivalve mollusks that inhabit the Northeast U.S. Continental Shelf from Newfoundland to Cape Hatteras (**cargnelli1999_oq--36?**). The ocean quahog is among the longest-lived and slowest growing of marine bivalves, reaching maturity around 13 years of age and harvestable size at age 20 (**cargnelli1999_oq--36?**). Maximum ages have been documented as high as 507 years old, with ages over 300 years old more common in the Northwest Atlantic (**sower2023examination--36?**). Spawning occurs either in summer or fall, and can extend for months, releasing eggs and sperm into the water column at low levels over time (**cargnelli1999_oq--36?**). Multiple annual spawnings may occur at the individual and population levels (**mann1982seasonal--36?**). Eggs and larvae drift with currents until metamorphosis, then settle to the bottom (**mafmc1997_oq--36?**). Larval development, at minimum, lasts for 55 days and decreases with increasing temperature (**lutz1981experimental--36?**). Larvae go through three stages of development (trochophores, veligers and pediveligers) with the duration of each stage being temperature dependent (**cargnelli1999_oq--36?**). (**vandewater1997biological--36?**) noted that wind-forced circulation may play a significant role in planktonic and benthic recruitment on Georges Bank and that interannual differences in these winds can yield very different abundance patterns over the bank. Once settled to the bottom, juvenile ocean quahogs are found offshore in sandy substrates, but may survive in

muddy intertidal environments if protected from predators (**kraus1989growth--36?**). Cross-shelf distributions of juveniles and adults remain within a 10°C optimal temperature range, as temperatures above 16°C limit distribution (**cargnelli1999_oq--36?**). Ocean quahogs are suspension feeders on phytoplankton, using their relatively short siphons which are extended above the surface of the substrate to pump in water (**cargnelli1999_oq--36?**). Juvenile and small ocean quahogs are commonly preyed upon by crustaceans (rock crabs, sea stars) and fish (haddock, cod) (**cargnelli1999_oq--36?**). Ocean quahogs are managed by the Mid-Atlantic Fishery Management Council under the Atlantic Surfclam-Ocean Quahog Fishery Management Plan. The species is mainly harvested off of Maine and Long Island. The last stock assessment was conducted in 2019, where the stock was not overfished and overfishing was not occurring (**mafmc2025_oq--36?**).

36.8 Figures

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37 Pollock (*Pollachius virens*)

37.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = High

Data Quality Model Confidence Score =Life History Data Quality = 85%

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37.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	High	Moderate	Increase
Directional Effect	Negative	Negative	None

37.3 Score Drivers

37.3.1 Vulnerability:

add text here

37.3.2 Climate Exposure:

add text here

37.3.3 Biological Sensitivity:

Pollock received a **High** sensitivity score, with 100% certainty according to bootstrap analysis. Two attributes had scores greater than 3: Population Growth Rate (3.6) and Complexity in Reproductive Strategy (3.4). Pollock are long-lived and relatively slow to mature, reaching maturity at 4-7 years of age and living for up to 24 years (**nefsc201355th--37?**). Spawning occurs at a narrow range of temperatures (**cargnelli1999_pollock--37?**; **collette2002bigelow--37?**). Sound production, and conversely the ability to hear sounds, may play a role in reproduction (**looby2022quantitative--37?**; **hawkins2019importance--37?**).

In the 2016 Climate Vulnerability Assessment, Pollock received a Moderate sensitivity score. This was driven by Population Growth Rate, similar to our study, and Early Life History Requirements, because juveniles rely on inshore habitats (**hare2016vulnerability--37?**). Our corresponding attribute, Egg and Larval Survival and Settlement Requirements, received a score of 3 in this assessment, up slightly from the previous assessment.

37.4 Data Quality Metrics

37.4.1 Life History Data Quality:

85% of data quality scores were greater than or equal to 2, indicating strong data quality. Egg and Larval Survival and Settlement Requirements and Relationship to Calcified Organisms were considered data gaps (mean attribute data quality score < 2).

37.4.2 Model Confidence:

add text here

37.5 Additional Metrics

37.5.1 Distribution Shifts:

add text here

37.5.2 Directional Effect:

Experts predicted that Pollock would be **negatively** impacted by climate change. Certainty was 70%. This is similar, albeit with lower certainty, than the results from the 2016 Assessment (**hare2016vulnerability--37?**). Pollock spawn at a narrow range of temperatures and prefer temperatures between 3 and 11 C (**cargnelli1999_pollock--37?**; **collette2002bigelow--37?**). Shifts to deeper waters have already been observed (**nye2009changing--37?**; **murawski1993climate--37?**).

37.6 Climate Effects on Abundance and Distribution

Because of the narrow range for when spawning occurs (4.5-6°C) (**cargnelli1999_pollock--37?**; **collette2002bigelow--37?**), increase in temperatures and climate change could have implications on the stock's success. The center of biomass for Northwest Atlantic pollock has not shifted and distribution has not moved northward, although they have been recently found at greater depths (**nye2009changing--37?**). (**murawski1993climate--37?**) noted a range contraction, increased depth of occurrence, and increased temperature of occurrence in pollock and other deep-water fish in the Gulf of Maine.

37.7 Life History Synopsis

Atlantic pollock are found in the Northwest Atlantic from the Grand Banks and Gulf of St. Lawrence to the Delmarva Peninsula, with highest abundances on the Scotian Shelf and in the Gulf of Maine (**cargnelli1999_pollock--37?**). Spawning takes place from September to April on the Scotian Shelf and November to February in the Gulf of Maine (**cargnelli1999_pollock--37?**). Optimal egg development occurs between 2.2 and 8.9°C (**cargnelli1999_pollock--37?**). Eggs are pelagic and free-floating, usually found in water 50-250 m in depth. Larvae are normally occurring from the shore out to the 200 m depth contour, but have been reported in waters as deep as 1550 m (**hardy1978development--37?**). Eggs and larvae undergo significant movement away from the spawning area as they drift with currents. For example, larvae from the northern Gulf of Maine are thought to originate from spawning grounds in the southern Gulf of Maine (**hardy1978development--37?**). Larvae are pelagic, consuming generally the most abundant copepods in their preferred size range, and metamorphose into juveniles at 3-4 months of age (**cargnelli1999_pollock--37?**). Juveniles have been reported over a wide variety of substrates, including sand, mud, or rocky bottom and vegetation (**ojeda1990diversity--37?**). Inshore subtidal and intertidal zones are utilized by age 0-1 juveniles and serve as important nursery areas. Age 2+ juveniles move offshore, inhabiting depths of 130-150 m (**hardy1978development--37?**). The primary prey of juvenile pollock are crustaceans. Euphausiids, in particular *Meganyctiphanes norvegica*, are the most important crustacean prey of juveniles (**tyler1972food--37?**). Fish and mollusks

make up a smaller proportion of the juvenile diet; however, in some cases fish may play a more important role in the diet (**collette2002bigelow--37?**). Pollock reach maturity at 3-7 years of age, longer for females, and reach a maximum age of 24 years (**nefsc201355th--37?**). Adults inhabit a wide range of depths, 35-365 m, but most occur within the 137 m depth contour and depths of 100-125 m are preferred (**scott1982depth--37?**). They prefer temperatures 0-14°C , although they tend to avoid temperatures > 11°C and < 3°C (**scott1982depth--37?**). For most of the year, adult pollock undergo minor inshore-offshore movements linked to temperature. Pollock migration rates vary with stock areas, but there is considerable movement across the Scotian Shelf and Gulf of Maine (**mayo1989patterns--37?**). The diet of adults consists of, in order of decreasing importance, euphausiids, fish and mollusks. *M. norvegica* is the single most important prey item and Atlantic herring is the most important fish species (**steele1963pollock--37?**). The primary predator of adult pollock are monkfish, although spiny dogfish and other pollock also prey on adults (**cargnelli1999_pollock--37?**). Atlantic pollock are managed by the New England Fishery Management Council under the Northeast Multispecies Fishery Management Plan. The stock is not currently overfished and overfishing is not occurring (**nefsc2024pollock--37?**).

37.8 Figures

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38 Red drum (*Sciaenops ocellatus*)

38.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = Moderate

Data Quality Model Confidence Score =Life History Data Quality = 95%

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38.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	Moderate	Moderate	None
Directional Effect	Neutral	Positive	Increase

38.3 Score Drivers

38.3.1 Vulnerability:

add text here

38.3.2 Climate Exposure:

add text here

38.3.3 Biological Sensitivity:

Red Drum received a **Moderate** sensitivity score with 50% certainty. This was driven by five attributes with scores over 2.5: Complexity in Reproductive Strategy (3.3), Population Growth Rate (3), Other Stressors (2.9), Egg and Larval Survival and Settlement Requirements (2.6), and Stock Size (2.6).

Red Drum utilize spawning aggregations and spawn in estuaries, which are known vulnerable habitats (**lowerre2016assessing--38?**; **guyant2022northeast--38?**). It is hypothesized that red tides impact spawning aggregations in the Gulf of America (**lowerre2016assessing--38?**). Males also use sounds to attract females to spawning locations (**looby2022quantitative--38?**; **montie2016long--38?**). They mature at 3-4 years of age, and can live up to 62 years, making them a very long-lived species (**asmfc2024reddrum--38?**). Larvae may rely on currents for transport into nursery habitats, which include seagrass beds.

Seagrass beds may contain an olfactory cue for settlement, and are highly vulnerable habitats in the Northeast US (**asmfc2024reddrum--38?**; **havel2016settlement--38?**; **comyns1989age--38?**; **guyant2022northeast--38?**). Adults are also dependent on shellfish reef habitats and estuarine water column habitats that are highly vulnerable (**guyant2022northeast--38?**). Invasive species may also displace the native plant and prey species that red drum rely on (**asmfc2024reddrum--38?**).

While the stock within the Northeast US is not overfished and overfishing is not occurring, models for the northern stock had high uncertainty and instability and could not be used to determine stock status in the most recent assessment (**asmfc2024reddrum--38?**).

In the previous Climate Vulnerability Assessment for the Northeast Shelf, Red Drum also received a Moderate sensitivity score. This score was driven by similar attributes to our score: Population Growth Rate, Spawning Cycle, and Early Life History Requirements (**hare2016vulnerability--38?**).

38.4 Data Quality Metrics

38.4.1 Life History Data Quality:

95% of data quality scores were greater than or equal to 2, indicating excellent data quality. No data gaps (mean attribute data quality score < 2) were identified.

38.4.2 Model Confidence:

add text here

38.5 Additional Metrics

38.5.1 Distribution Shifts:

add text here

38.5.2 Directional Effect:

Experts predict that climate change will have a **neutral** effect on Red Drum. Certainty was 85%. Higher temperatures have been correlated with strong recruitment events (**goldberg2022analysis--38?**). However, Red Drum occupy vulnerable estuarine habitats, and rely on highly vulnerable habitats such as seagrass beds and shellfish reef habitats (**guyant2022northeast--38?**). In the previous Assessment for the Northeast Shelf, experts predicted Red Drum would be positively impacted by climate change, albeit with low certainty (**hare2016vulnerability--38?**).

38.6 Climate Effects on Abundance and Distribution

Red drum recruitment has been correlated with wind directions and sea surface temperature (**asmfc2024reddrum--38?**). In North Carolina, the duration and magnitude of N and NE winds in August was twice as strong in high recruitment years. Elevated nearshore sea surface temperatures and coastal productivity (high chlorophyll a) were positively correlated with strong year classes (**goldberg2022analysis--38?**). Red drum acclimated to higher temperatures had a higher standard metabolic rate, decreased critical oxygen threshold, and reduced ability to recover from exercise. Crude oil exposure (hypoxic conditions) also resulted in lower metabolic rates and reduced ability to recover from exercise, and combined with climate-induced increases in temperature, are likely to impact red drum respiratory performance and fitness (**ackerly2021effects--38?**). Laboratory experiments on early life stage red drum resulted in decreased survival and increased heart rate when exposed to elevated pCO^2 ; however the effects were less severe on red drum compared to similar studies on other species (**lonthair2017early--38?**).

38.7 Life History Synopsis

Red drum, or redfish, are a popular recreational fish species found in estuarine and inshore waters from Massachusetts to Florida. The stock is divided into northern and southern subunits from New Jersey to North Carolina and South Carolina to Florida, respectively (**asmfc2024reddrum--38?**). Spawning occurs within 45-60 days between August and October and where currents can move larvae/eggs into nearby estuaries and nursery habitats

(**asmfc2024reddrum--38?**). Optimal spawning temperatures in laboratory studies ranged between 22 - 25°C, with spawning observed from 22 - 30°C (**asmfc2024reddrum--38?**). Eggs hatch within 24-36 hours (**peters1987early--38?**). Larvae settle at approximately 17 days old in Florida estuaries (**rooker1998vulnerability--38?**). Temperatures of 25°C and salinities around 30 are optimal for hatching success and larval survival. Larvae growth rates were also correlated with ocean temperatures in the Gulf of America (**asmfc2024reddrum--38?**; **comyns1989age--38?**). Larvae in Florida consume abundant copepods, whereas juveniles consume abundance amphipods, shrimp, and mysids, switching to crabs and fish as they grow (**peters1987early--38?**). Juveniles rely on structural nursery habitats such as seagrasses for protection from predators. Juveniles are most abundant in estuarine waters and inlets, while fish older than age four inhabit deeper waters (**rooker1998vulnerability--38?**). Adults in the Gulf of America (TX) consume white shrimp, gulf menhaden, and blue crab. Dominant prey species shift seasonally (shrimp in fall, menhaden in spring) while some prey species are consistent throughout the year (crab) (**scharf2000feeding--38?**). Age at maturity in red drum off North Carolina is 3-4 years, with a maximum age of 62 years (**asmfc2024reddrum--38?**). Red drum move out of estuaries into coastal waters as they mature. Seasonal movements with seasonal site fidelity have been observed in Georgia but some populations (particularly in the southern stock) are resident year round. Seasonal migrations between Virginia and North Carolina have been observed between May and September (**asmfc2024reddrum--38?**). The northern stock is managed by the Atlantic States Marine Fisheries Commission. The northern stock is not overfished and overfishing is not occurring, however the southern stock is overfished and overfishing is currently occurring (**asmfc2024reddrum--38?**).

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39 Scup (*Stenotomus chrysops*)

39.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = Moderate

Data Quality Model Confidence Score =Life History Data Quality = 98%

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39.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	Moderate	Low	Increase
Directional Effect	Neutral	Positive	Increase

39.3 Score Drivers

39.3.1 Vulnerability:

add text here

39.3.2 Climate Exposure:

add text here

39.3.3 Biological Sensitivity:

Scup received a **Moderate** sensitivity score with a certainty of 59%. A total of four attributes had scores greater than 2.5: Spawning Cycle (3), Complexity in Reproductive Strategy (2.9), Dispersal of Early Life Stages (2.9), and Other Stressors (2.8).

Scup only spawn once per year between May and August (**steimle1999_scup--39?**). Spawning occurs in inshore waters once they are warmer than 10C and rely on highly vulnerable habitats such as submerged aquatic vegetation and estuarine habitats (**steimle1999_scup--39?**; **guyant2022northeast--39?**). Spawning may also rely on soniferous behaviors (**looby2022quantitative--39?**). Eggs and larvae do not disperse far from spawning locations (**steimle1999_scup--39?**). Adults utilize benthic habitats on the continental shelf and in estuaries. As a result, they are exposed to a variety of stressors due to urbanization impacts such as hypoxia, changes to freshwater inputs, and dredging (**guyant2022northeast--39?**).

In the 2016 Climate Vulnerability Assessment, Scup received a Low sensitivity score. Similar to our analysis, this score was driven by Spawning Cycle and Early Life History Requirements, as well as Population Growth Rate (**hare2016vulnerability--39?**). In our analysis, Population Growth Rate received a score of 2.4.

39.4 Data Quality Metrics

39.4.1 Life History Data Quality:

98% of data quality scores were greater than or equal to 2, indicating excellent data quality. No data gaps (mean attribute data quality score < 2) were identified.

39.4.2 Model Confidence:

add text here

39.5 Additional Metrics

39.5.1 Distribution Shifts:

add text here

39.5.2 Directional Effect:

Experts predicted that climate change will have a **neutral** effect on Scup. Certainty was 93%. Scup distributions have shifted northward and offshore (**bell2015disentangling--39?**; **palacios2023quantifying--39?**), likely in search of preferred temperatures, which range between 9-24C (**steimle1999_scup--39?**), but their center of distribution has not changed (**roday2026recreational--39?**). Juveniles and adults both utilize vulnerable estuarine habitats (**guyant2022northeast--39?**). In the previous assessment, experts predicted that Scup would be positively impacted by climate change, however, uncertainty was high (**hare2016vulnerability--39?**).

39.6 Climate Effects on Abundance and Distribution

Scup distributions in the spring and fall have shifted northward and offshore, which may necessitate new management strategies (**bell2015disentangling--39?**; **palacios2023quantifying--39?**). (**roday2026recreational--39?**) analyzed fish distribution shifts using recreational catch and fishery independent data. Scup distributions showed a moderate range expansion but little to no change to center of distribution (**roday2026recreational--39?**). Juvenile scup exposed to elevated pCO^2 in a laboratory setting had no significant effect on growth, survival, or otolith condition. Scup spawning areas in Long Island Sound are already exposed to high pCO^2, indicating some tolerance to ocean acidification (**perry2015effect--39?**).

39.7 Life History Synopsis

Scup are found in the Northwest Atlantic from Cape Cod, Massachusetts to Cape Hatteras, North Carolina (**steimle1999_scup--39?**). There are two U.S. stocks: Mid-Atlantic/New England (Atlantic coast) and South Atlantic (**nefsc2025scup--39?**). Spawning occurs once per year in coastal bays and estuaries in May-August, beginning with lower latitudes (**steimle1999_scup--39?**). Eggs take 2-3 days to hatch depending on temperature. The larval stage lasts from 2 mm TL to 18-19 TL; the duration of this period is unclear. Larval distributions are highest in inshore bays and estuaries, suggesting that eggs and larvae do not travel far during planktonic stages (**steimle1999_scup--39?**). The biggest cue for larval settlement is size (~25 mm), with size in part being a function of temperature - most common at 15-20°C (**steimle1999_scup--39?**). Larval diets are poorly understood, but larvae have been successfully reared on zooplankton in lab experiments (**bowman1987food--39?**). Early juveniles consume copepods, so larval scup may also consume copepods (**michelman1988biology--39?**). Larval predators include plantivorous medusae, crustaceans, and fishes (**steimle1999_scup--39?**). Juveniles have been caught in intertidal and subtidal habitats, over sand, silty-sand, shell, mud, mussel beds and eelgrass (*Zosteria marina*). Offshore winter-spring distributions indicate that they occur in

habitats ranging from relatively flat, open, sandy-silty bottoms to the head of submarine canyons, and other areas with topographical relief and varying sediments. Juveniles can use biogenic depressions, sand wave troughs, and possibly mollusk shell fields for shelter in winter, suggesting structure may be important to juveniles (**able1998first--39?**). Juveniles primarily feed on polychaetes, epibenthic amphipods and other small crustaceans, mollusks, and fish eggs and larvae (**steimle1999_scup--39?**). Prey size increases with individual size, with copepods and mysids being important to early juveniles and bivalve mollusks consumed by larger fish (**bowman1987food--39?**). Adult scup are also benthic feeders and forage on a variety of prey, including small crustaceans (including zooplankton), polychaetes, mollusks, small squid, vegetable detritus, insect larvae, hydroids, sand dollars, amphipods, and small fish (**steimle1999_scup--39?**). Similar to juveniles, their prey size scales with individual size. Diets can vary based on location, and season. Feeding slows or stops entirely in winter (**byron2010stability--39?**). Adults are primarily benthic and utilize both estuarine and shelf benthos (**steimle1999_scup--39?**). Age at maturity is around 2 years of age, with a maximum age between 15-20 years (**nefsc2015sixtieth--39?**). Scup undertake extensive inshore/offshore migrations driven by temperature preferences. They migrate south and offshore in the fall and return to inshore bays and estuaries in the spring to spawn. Scup can become inactive or dormant if caught in cold pockets < 6°C during their inshore spring migration and prefer temperatures 9-24°C (**steimle1999_scup--39?**). The North Atlantic scup fishery is managed jointly by the Mid-Atlantic Fishery Management Council and Atlantic States Marine Fisheries Commission. The 2025 Management Track Assessment reported the Atlantic coast stock as not overfished and overfishing not occurring (**nefsc2025scup--39?**).

39.8 Figures

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40 Shortfin squid (*Illex illecebrosus*)

40.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = Low

Data Quality Model Confidence Score =Life History Data Quality = 85%

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40.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	Low	Low	None
Directional Effect	Neutral	Positive	Decrease

40.3 Score Drivers

40.3.1 Vulnerability:

add text here

40.3.2 Climate Exposure:

add text here

40.3.3 Biological Sensitivity:

Short-finned squid received a **Low** sensitivity score, with a certainty of 89%. Only one attribute – Egg and Larval Survival and Settlement Requirements (2.9) – received a score greater than 2.5. Short-finned squid rely on the Gulf Stream for transport between southern spawning habitats and northern feeding grounds. Therefore, recruitment has been linked to climate oscillations such as the North Atlantic Oscillation that drive shifts in the Gulf Stream (**dawe2000environmental--40?**; **odor2013chapter--40?**; **hendrickson2004population--40?**).

In the previous Climate Vulnerability Assessment for the Northeast Shelf, Short-finned squid also received a Low sensitivity score. In that analysis, no attributes had a score greater than 2.5 (**hare2016vulnerability--40?**).

40.4 Data Quality Metrics

40.4.1 Life History Data Quality:

85% of data quality scores were greater than or equal to 2, indicating strong data quality. Only Stock Size/Status was considered a data gap (mean attribute data quality score < 2).

40.4.2 Model Confidence:

add text here

40.5 Additional Metrics

40.5.1 Distribution Shifts:

add text here

40.5.2 Directional Effect:

Experts predicted that climate change will have a **neutral** effect on short-finned squid. Certainty was 96%. Recruitment is highly dependent on transport mechanisms such as the Gulf Stream and catch per unit effort is correlated with cooler bottom temperatures (**dawe2000environmental--40?**; **salois2023shelf--40?**; **odor2013chapter--40?**; **hendrickson2004population--40?**). However, short-finned squid are able to adapt to the available niche space in response to predators, competitors, and prey due to their short life

cycle (**dawe2000environmental--40?**). In the previous Assessment, short-finned squid were predicted to be positively impacted by climate change, although with high uncertainty (**hare2016vulnerability--40?**).

40.6 Climate Effects on Abundance and Distribution

Distribution and abundance of shortfin squid on the Northeast U.S. Continental Shelf is highly related to oceanographic conditions such as shelf-slope dynamics and positions of the shelf-slope front and Gulf Stream (**dawe2007ocean--40?**). (**dawe2000environmental--40?**) theorized based on catch data from Newfoundland that squid abundance may co-vary with that of its most important predators, competitors, and prey in an adaptive manner such that the population increases and expands when niche space becomes available. These shifts appear to result from high water temperatures, a negative North Atlantic Oscillation (NAO) index, and a southward shift in the Gulf Stream position. Greater catch per unit effort (CPUE) of shortfin squid has been associated with cooler bottom temperatures, timing and location of warm core rings in the winter and spring, and slope water/shelf slope front dynamics (**salois2023shelf--40?**). These features may contribute to areas of high primary productivity, food sources, and transport mechanisms for larvae.

40.7 Life History Synopsis

The northern shortfin squid inhabits Northwest Atlantic shelf and slope waters from the Labrador Sea to the Florida Straits (**hendrickson2004essential--40?**). Spawning occurs year round, but peaks seasonally from October to June. Shortfin squid are semelparous; females release multiple egg capsules during a single spawning season, after which they die (**hendrickson2004essential--40?**). Laboratory studies indicate that hatching occurs in 16 days at 13°C, 12 days at 16°C, and 8 days at 21°C. In the laboratory, paralarvae enter a transitional stage at approximately 5.0 mm ML, followed by a juvenile stage at about 7.0 mm ML (**odor1982embryonic--40?**). The Gulf Stream and associated eddies are hypothesized to be the primary mechanisms for transport of larvae to northern latitudes and then inshore via the eddies (**trites1983physical--40?**). Larvae have small yolk-sacs, so they rely on rapid transportation to food-rich habitats (**hendrickson2004population--40?**). Larvae feed on a variety of prey items (fish, nematodes, euphausiids). Larvae and juveniles are abundant in the slope water of the Gulf Stream and in the adjacent upwelling regions at times of maximum production such as the spring plankton bloom (**hendrickson2004essential--40?**). As the initial biomass of the bloom is transferred up through various trophic levels, the squid grow at a pace which allows them to feed continuously at the level where biomass is peaking (**odor2013chapter--40?**). Paralarvae, juveniles and adults of Illex exhibit diel vertical migrations to feed at night, and remain near the seabed during the day (**arkhipkin1986diel--40?**). An ontogenetic shift in diet from a predominance of crustaceans to a predominance

of fish and squid is evident as the squid mature (**dawe1988length--40?**). Larger, older Northern shortfin squid feed primarily on fish, but cannibalism of small individuals (most likely males) by larger females also occurs, particularly during autumn. Fish prey consists of the early life history stages of Atlantic cod, Arctic cod, redfish, sand lance, mackerel, Atlantic herring, haddock, and sculpin (**dawe1988length--40?**). Shortfin squid are important prey for seabirds and many fish species, including tunas, hakes, spiny dogfish, monkfish, and Atlantic cod (**hendrickson2004essential--40?**). Juveniles and adults make seasonal migrations off the continental shelf in the fall and back onto the shelf to spawning grounds in the Mid-Atlantic in the spring (**hendrickson2004essential--40?**). Shortfin squid are a short lived species, reaching maturity at 40-60 days and a maximum lifespan of less than one year (**hendrickson2004population--40?**). Adults are found at bottom temperatures between 3.5-15°C and surface temperatures greater than 20°C (**hendrickson2004essential--40?**). Shortfin squid in the Northwest Atlantic are managed by the Mid-Atlantic Fishery Management Council under the Mackerel, Squid, and Butterfish Fishery Management Plan. The most recent stock assessment occurred in 2025; management measures are in place but stock status is unknown as there are no reference points for this stock (**nefsc2025northern--40?**).

40.8 Figures

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41 Smooth dogfish (*Mustelus canis*)

41.1 Summary

Scores Vulnerability = | Climate Exposure = | Biological Sensitivity = Moderate

Data Quality Model Confidence Score = | Life History Data Quality = 95%

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41.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	Moderate	Low (**hare2016vulnerability--41?**; **loughran2025climate--41?**)	Increase
Directional Effect	Neutral	Neutral (**hare2016vulnerability--41?**; **loughran2025climate--41?**)	None

41.3 Score Drivers

41.3.1 Vulnerability:

add text here

41.3.2 Climate Exposure:

add text here

41.3.3 Biological Sensitivity:

Smooth dogfish received a **Moderate** sensitivity score, with a certainty of 69% according to bootstrap analysis. Two attributes had scores greater than 2.5: Other Stressors (2.7) and Juvenile Habitat Requirements (2.6).

Smooth dogfish rely on coastal habitats throughout their life-cycles, which are impacted by eutrophication and hypoxia, changes in freshwater input, and increases in turbidity due to activities such as dredging (**plumlee2018community--41?**). Juveniles in particular rely on nursery estuary and seagrass bed habitats, which are vulnerable habitats (**guyant2022northeast--41?**; **farr2021assessment--41?**).

In the previous Northeast Climate Vulnerability Assessment, smooth dogfish received a Low sensitivity score. This was driven by Population Growth Rate, which only received a score of 2.2 in our assessment (**hare2016vulnerability--41?**). The Atlantic stock of Smooth dogfish were also assessed in the recent Highly Migratory Species Climate Vulnerability Assessment, where it also received a Low score because no attributes scored higher than 2.5 (**loughran2025climate--41?**).

41.4 Data Quality Metrics

41.4.1 Life History Data Quality:

95% of data quality scores were greater than or equal to 2, indicating excellent data quality. No data gaps (mean attribute data quality score < 2) were identified.

41.4.2 Model Confidence:

add text here

41.5 Additional Metrics

41.5.1 Distribution Shifts:

add text here

41.5.2 Directional Effect:

Experts predict that climate change will have a **neutral** impact on smooth dogfish with 96% certainty. While smooth dogfish occupy estuarine habitats that are more likely to be degraded and are highly vulnerable (**guyant2022northeast--41?**; **plumlee2018community--41?**), smooth dogfish may be tolerant of future warming and could possibly expand their range (**bockus2020thermal--41?**). Both the previous Climate Vulnerability Assessment in the Northeast and the Highly Migratory Species Assessment also predicted that climate change would have a neutral impact on climate change (**hare2016vulnerability--41?**; **loughran2025climate--41?**).

41.6 Climate Effects on Abundance and Distribution

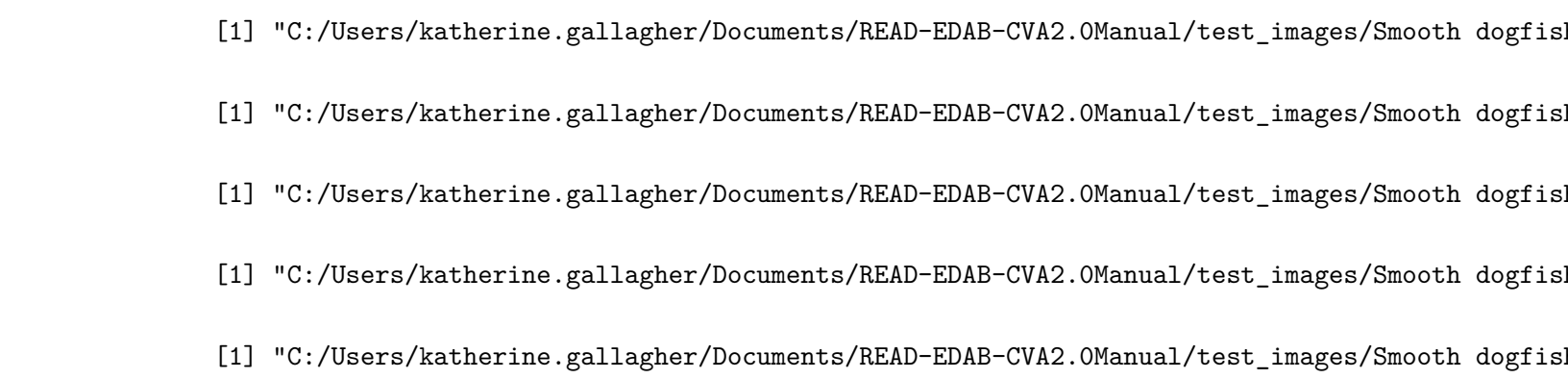
Little information exists on the potential effects of climate on smooth dogfish abundance and distribution. Coastal areas, including estuaries and smooth dogfish nursing grounds, are affected by anthropogenic activities that degrade a portion of smooth dogfish habitat. These areas are also more likely to be impacted by hypoxia and changes to freshwater input/increases in turbidity (**plumlee2018community--41?**). Smooth dogfish exposed to higher temperatures (increase of 6°C) did not result in a thermal stress response, indicating a possible tolerance and ability for range expansion (**bockus2020thermal--41?**).

41.7 Life History Synopsis

Smooth dogfish are a demersal, highly migratory shark species (HMS). The Northwest Atlantic population of smooth dogfish ranges from Cape Cod to Florida and the northern Gulf of America (**conrath2002age--41?**). The species undergoes temperature-driven seasonal migrations north in the summer and south in the winter. They overwinter in Chesapeake Bay to South Carolina from January to May, migrating north to Southern New England beginning in early spring to give birth to pups (**castro2010sharks--41?**; **dellapa2018hierarchical--41?**). Females ovulate between June and July, with gestation lasting 10 months. Birth occurs in southern New England between early May to mid-July (**castro2010sharks--41?**). Once females have given birth in nearshore or estuarine areas,, young of year and immature age classes may remain in nursery grounds while females migrate north to suitable temperatures (**bangley2017habitat--41?**). Smooth dogfish prefer temperatures 9-32°C. They begin appearing in the mid-Atlantic and New England when temperatures are greater than 6-7°C, and are most abundant along the East Coast between 13-15°C (**bockus2020thermal--41?**). Young of year individuals use shallow shoals and marsh creek habitats particularly during night hours. Juveniles are found throughout temperate coastal estuaries, creeks, and seagrass beds, which may serve as nursery grounds (**castro2010sharks--41?**). Juveniles prey on small shrimp, polychaete worms, and crabs, and may fill an important predatory role during times of

year when other piscivores are rare (**rountree1996seasonal_smooth--41?**). Adult smooth dogfish are bottom feeders, primarily consuming crustaceans and mainly feeling at night. They are described as a trophic specialist due to the morphological and functional modifications that allow them to feed on hard-shelled/exoskeleton-bearing prey (**castro2010sharks--41?**; **bangley2017habitat--41?**). Smooth dogfish are prey items for a variety of larger sharks and fishes (**gelsleichter1999food--41?**). Age at maturity is 2-3 years for males and 4-7 years for females, with a maximum age of 11 years for males and 13 years for females (**conrath2002age--41?**; **giresi2013morphological--41?**). In state waters, smooth dogfish are managed by the Atlantic States Marine Fisheries Commission's Interstate Fishery Management Plan for Coastal Sharks (**asmfc2024_smooth--41?**). NOAA Fisheries produces an annual Stock Assessment and Fisheries Evaluation (SAFE) Report detailing the status of HMS in federal waters (**nmfs2023safe--41?**). The current stock status is not overfished and overfishing is not occurring (**sedar2015smoothdogfish--41?**).

41.8 Figures



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42 Spanish mackerel (*Scomberomorus maculatus*)

42.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = Low

Data Quality Model Confidence Score =Life History Data Quality = 85%

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42.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	Low	Low	None
Directional Effect	Neutral	Positive	Decrease

42.3 Score Drivers

42.3.1 Vulnerability:

add text here

42.3.2 Climate Exposure:

add text here

42.3.3 Biological Sensitivity:

Spanish mackerel received a **Low** sensitivity score. Certainty was 100% according to bootstrap analysis. No attributes scored higher than 2.5. The highest attribute score was for Other Stressors (2.4). Spanish mackerel adults use estuarine water column habitats, which are classified as highly vulnerable due to impacts of stratification, hypoxia, increased fresh water runoff, and habitat loss (**guyant2022northeast--42?**).

Similarly, in the previous Climate Vulnerability Assessment, Spanish Mackerel was also given a Low sensitivity score and no attributes scored above 2.5 (**hare2016vulnerability--42?**).

42.4 Data Quality Metrics

42.4.1 Life History Data Quality:

85% of data quality scores were greater than or equal to 2, indicating strong data quality. No data gaps (mean attribute data quality score < 2) were identified.

42.4.2 Model Confidence:

add text here

42.5 Additional Metrics

42.5.1 Distribution Shifts:

add text here

42.5.2 Directional Effect:

Experts predict that climate change will have a **neutral** effect on Spanish mackerel. Certainty was 92%. Spanish mackerel prefer temperatures between 21-27 C, so warming temperatures will likely expand suitable habitat (**berrien1977biological--42?**). Spawning, however, occurs in coastal areas and estuaries such as Chesapeake Bay (**cooksey1996reproductive--42?**; **mceachran1980distribution--42?**). These habitats are vulnerable due to anthropogenic impacts and may impact reproductive success (**guyant2022northeast--42?**). The previous Climate Vulnerability Assessment predicted that Spanish mackerel would be positively impacted by climate change (**hare2016vulnerability--42?**).

42.6 Climate Effects on Abundance and Distribution

Little work exists on the effects of climate on Spanish mackerel populations in the Northwest Atlantic. Temperature likely limits movement; they are not found in waters colder than 18°C (**berrien1977biological--42?**). Rising ocean temperatures may expand suitable habitat in the Mid-Atlantic and north to New England. Larval Spanish mackerel were largely resistant to contaminants from the Deepwater Horizon oil spill, and larval abundance did not differ before v. after the spill (**ransom2016condition--42?**).

42.7 Life History Synopsis

Spanish mackerel are a migratory, schooling fish found along the eastern United States and Gulf from New York to Florida (**godcharles1986species--42?**). There are two Spanish mackerel stocks, one in the South Atlantic and one in the Gulf of America (**sedar2022sedar78--42?**; **sedar2023sedar81--42?**). The Atlantic stock overwinters off the Florida coast and migrates northward to the Mid-Atlantic in the spring and early summer. They return south to Florida once the Mid-Atlantic cools in the fall (**berrien1977biological--42?**). Spanish mackerel prefer temperatures 21-27°C and are not seen below 18°C (**berrien1977biological--42?**). Spawning occurs from May to September along the southeast Atlantic coast, with lower latitudes spawning earlier than higher latitudes (**collins1987larval--42?**). In Chesapeake Bay, spawning occurs from June to August, peaking in June (**cooksey1996reproductive--42?**). Females are batch spawners, releasing as many as 1.5 million eggs over the spawning season (**collins1987larval--42?**). Larval collection data indicate that spawning occurs at depths of 12-35 meters over the inner continental shelf including estuarine and coastal areas (**mceachran1980distribution--42?**). Hatching occurs approximately 25 hours after fertilization at water temperatures averaging 26°C. Larvae and early juveniles grow 1.9 mm per day for approximately the first 23 days of life. From 23-40 days, growth is accelerated, with young fishes growing as much as 5 mm per day. Thereafter, growth slows to approximately 2.1 mm per day (**schmidt1993age--42?**). Larvae are found in surface waters between 19.6–29.8°C with a salinity of 28.3–37.4 ppt. It is unclear if there is a dependency on currents for transport to nursery habitats (**godcharles1986species--42?**). Larvae feed on other larval fishes such as carngids, clupeids, and engraulids and crustaceans (**berrien1977biological--42?**). Juveniles and adults are voracious, opportunistic, carnivores. Food consists mostly of small fishes with lesser quantities of pandalid and penaeoid shrimps and squids. Clupeoids such as round herring, thread herring, menhaden, and alewives, along with anchovies, are important forage species. They seldom feed on bottom invertebrates (**bowman2000food--42?**; **berrien1977biological--42?**). Spanish mackerel reach maturity around 1-1.5 years of age (**juan2013life--42?**) with a maximum lifespan of around 12 years (**sedar2022sedar78--42?**). They are largely preyed upon by dolphins, sharks, and tunas (**berrien1977biological--42?**). Management of Spanish mackerel in state waters is coordinated by the Atlantic States Marine

Fisheries Commission (ASMFC) Coastal Pelagics Management Board. Both stocks are not overfished or subject to overfishing (**sedar2022sedar78--42?**; **sedar2023sedar81--42?**).

42.8 Figures

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43 Spiny Dogfish (*Squalus acanthias*)

43.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = Low **Data Quality**Model Confidence Score =Life History Data Quality = 90%

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43.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	Low	Low	None
Directional Effect	Neutral	Neutral	None

43.3 Score Drivers

43.3.1 Vulnerability:

add text here

43.3.2 Climate Exposure:

add text here

43.3.3 Biological Sensitivity:

Spiny dogfish received a **Low** sensitivity score, with a certainty of 100%. The only attribute with a score greater than 2.5 was Population Growth Rate (3.6). They are a long-lived, slowly maturing species. Age at maturity ranges from 6 years for males and 12 years for females, with maximum lifespans of 35-40 years (**stehlik2007--43?**). Gestation periods last from 18-24 months (**stehlik2007--43?**).

Spiny dogfish also received a Low sensitivity score in the previous Climate Vulnerability Assessment. This score was also driven by Population Growth Rates (**hare2016vulnerability--43?**).

43.4 Data Quality Metrics

43.4.1 Life History Data Quality:

90% of data quality scores were greater than or equal to 2, indicating excellent data quality. No data gaps (mean attribute data quality score < 2) were identified.

43.4.2 Model Confidence:

add text here

43.5 Additional Metrics

43.5.1 Distribution Shifts:

add text here

43.5.2 Directional Effect:

Experts predict that climate change will have a **neutral** effect on spiny dogfish. Certainty was 91%. Occurrence of spiny dogfish is correlated with warmer temperatures (**sagarese2014a--43?**). They are highly migratory and will move north-south to follow waters within their preferred temperature range (6-13C) (**stehlik2007--43?**; **carlson2014--43?**). They may benefit from the loss of cod as they occupy similar feeding niches (**neuenhoff2019--43?**; **ono2019--43?**). The previous Assessment also predicted that climate change would have a neutral impact on spiny dogfish (**hare2016vulnerability--43?**).

43.6 Climate Effects on Abundance and Distribution

(**nye2009--43?**) observed a range expansion and southward shift in center of biomass in spiny dogfish, although changes in biomass over the study period may have influenced these shifts. Seasonal occurrence of spiny dogfish is highly correlated with depth and bottom temperature, with a higher probability of occurrence in warmer years versus colder years (**sagarese2014a--43?**). Spiny dogfish shift their distribution seasonally; variation in environmental drivers may impact population and fishery dynamics as temperatures continue to increase. Climate-inducing seasonal distribution shifts influencing habitat availability for spiny dogfish may also lead to vulnerability of the inshore fishery in the fall (**sagarese2014b--43?**).

43.7 Life History Synopsis

Atlantic spiny dogfish are a schooling, highly migratory shark species (HMS) inhabiting inshore and offshore waters from Labrador to Florida, although they are most abundant from Nova Scotia and Cape Hatteras. Spiny dogfish perform diel vertical migrations; most schools are found around 40-50 m depth, but can descend upwards of 500 m (**stehlik2007--43?**; **carlson2014--43?**). There are populations that occupy the continental shelves of Europe, Asia, and the northeast Pacific, but these populations do not mix (**stehlik2007--43?**). Spiny dogfish perform north-south migrations and move between inshore-offshore waters in response to water temperature, migrating north in the spring and summer and south in the fall and winter to remain in waters between 6-13°C (**stehlik2007--43?**; **carlson2014--43?**). Age at maturity is 6 years for males and 12 years for females, with a lifespan of 35 years for males and 40 years for females. Spawning occurs offshore in the winter, with a gestation period of 18-24 months (**stehlik2007--43?**). Proportions of mature females in catch are correlated with bottom temperatures, with more mature females caught at relatively higher bottom temperatures. However, how temperature impacts reproduction is still unclear (**sagarese2014b--43?**). Spiny dogfish are opportunistic feeders, preying upon a variety of fishes (herring, mackerel, hakes, sand lance, menhaden), squid, ctenophores, polychaetes, crustaceans, bivalves. Diets can shift based on location and availability (**stehlik2007--43?**; **devries2025--43?**). Predators of spiny dogfish include sharks, seals, cod, goosefish, dolphins, and larger fish (**stehlik2007--43?**). Spiny dogfish occupy a similar niche to that of Atlantic cod, whose populations have been impacted by increasing seal populations in the Northwest Atlantic and Canada. It has been hypothesized that the loss of cod could be a benefit to spiny dogfish, since they occupy similar feeding niches (**neuenhoff2019--43?**; **ono2019--43?**). The Mid-Atlantic Fishery Management Council (MAFMC) leads the joint management of the Spiny Dogfish Fishery Management Plan, with support from the New England Fishery Management Council (NEFMC) and Atlantic States Marine Fisheries Commission (ASMFC) (**mafmc2025_spinydog--43?**). The ASMFC manages spiny dogfish in state waters via the Interstate Fishery Management Plan for Spiny Dogfish (**asmfc2002_spinydog--43?**). As of 2023, spiny dogfish in the Northwest Atlantic are not overfished and overfishing is not occurring (**nefsc2023_spinydog--43?**).

43.8 Figures

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44 Striped bass (*Morone saxatilis*)

44.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = High

Data Quality Model Confidence Score =Life History Data Quality = 98%

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44.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	High	High	None
Directional Effect	Negative	Neutral	Decrease

44.3 Score Drivers

44.3.1 Vulnerability:

add text here

44.3.2 Climate Exposure:

add text here

44.3.3 Biological Sensitivity:

Striped Bass received a **High** sensitivity score. Certainty was 85%. A total of seven attributes had scores greater than 3: Egg and Larval Survival and Settlement Requirements (3.7), Complexity in Reproductive Strategy (3.4), Other Stressors (3.4), Spawning Cycle (3.4), Population Growth Rate (3.2), Stock Size (3.1), and Dispersal of Early Life Stages (3.1).

Striped Bass are anadromous, spawning in fresh water once a year (**bigelow1953striped--44?**). Estuarine and river water column habitats used by spawning adults are considered highly vulnerable (**guyant2022northeast--44?**). Eggs and larvae are semi-buoyant drift downstream away from spawning sites and require relatively fast current speeds (0.3 m/s) to avoid sinking and being smothered (**fay1983species--44?**). Predation and temperature play a large role in larval survival (**nefsc2019saw66--44?**; **fay1983species--44?**; **smith1981predation--44?**).

Striped bass are long lived, reaching maturity between 2-8 years of age and living up to 30 years (**nefsc2019saw66--44?**). The stock is currently overfished, but overfishing is not occurring (**asmfc2024assessment_stripedbass--44?**).

In the previous Climate Vulnerability Assessment, striped bass received a High sensitivity score. Similar to our results, Spawning Cycle and Early Life History Requirements drove this score (**hare2016vulnerability--44?**).

44.4 Data Quality Metrics

44.4.1 Life History Data Quality:

98% of data quality scores were greater than or equal to 2, indicating excellent data quality. No data gaps (mean attribute data quality score < 2) were identified.

44.4.2 Model Confidence:

add text here

44.5 Additional Metrics

44.5.1 Distribution Shifts:

add text here

44.5.2 Directional Effect:

Experts predicted that striped bass would be **negatively** impacted by climate change. Certainty amongst expert scores was 100%. Spawning and recruitment success is driven by freshwater outputs and subnormal winter temperatures (**oconnor2012findings--44?**; **dixon2026climate--44?**; **heinle1977contribution--44?**). Estuarine and river spawning habitats are also highly vulnerable (**guyant2022northeast--44?**). The previous Climate Vulnerability Assessment predicted that climate change would have a neutral effect on striped bass (**hare2016vulnerability--44?**).

44.6 Climate Effects on Abundance and Distribution

(**coutant1990temperature--44?**) analyzed habitat suitability for striped bass on the U.S. east coast with predicted temperature increases and declines in precipitation. In freshwater areas, high temperatures, altered streamflow, and low oxygen may constrain suitable habitat in the summer. Habitat loss in the Chesapeake Bay is predicted, with a possible range expansion into Nova Scotia and further into the Gulf of St. Lawrence (**coutant1990temperature--44?**). Onset and cessation of the spawning season in two spawning grounds of the Hudson River Estuary have occurred earlier in recent years. Spawning sites in the upper estuary are more vulnerable to climate change compared to sites in the lower estuary (**pan2023spatial--44?**). Larval striped bass abundance in the Hudson River Estuary is positively correlated with high freshwater flows, which may decrease under predicted climate scenarios (**oconnor2012findings--44?**). Striped bass recruitment has been linked to winter North Atlantic Oscillation (NAO) and marine heatwaves (**dixon2026climate--44?**). Several studies support a general conclusion that dominant year classes through recent history have always been preceded by a colder than normal winter (i.e. subnormal winter temperatures). One proposed mechanism of the severe winter-strong year class association is through the estuarine food chain (**heinle1975carbon--44?**; **heinle1977contribution--44?**).

44.7 Life History Synopsis

Atlantic striped bass inhabit freshwater coastal and estuarine waters from Canada to Florida, including the Gulf of America from Florida to Louisiana (**nefsc2019saw66--44?**). Striped bass are anadromous, spawning once a year in fresh or nearly fresh water. Along the Atlantic coast, the spawning period ranges from mid-February in Florida to June and July in the Gulf of St. Lawrence. In the mid-Atlantic, spawning is concentrated during the months of April - June (**bigelow1953striped--44?**). Spawning peaks are apparently triggered by a noticeable increase in water temperature, generally beginning at temperatures of at least 14°C (**johnson1975production--44?**). Eggs hatch within 30 - 74 hrs, occurring faster at higher temperatures. Larval stages last up until age 1 when juveniles migrate out of rivers and

into estuaries (**nefsc2019saw66--44?**). Larvae and eggs may drift down stream in rivers, away from spawning sites. Eggs are semi-buoyant, requiring a current of 30 cm/sec in order to keep the eggs afloat so they don't settle to the bottom and die (**fay1983species--44?**). Egg and larval mortality, as well as larval growth rates and yolk utilization are all linked to water temperatures greater than 18°C (**nefsc2019saw66--44?**). Predation and competition by American shad are important factors in spawning and larval locations (**fay1983species--44?**). In addition, adult and juvenile white perch and copepod *Cyclops bicuspidatus* probably consume large numbers of striped bass larvae (**smith1981predation--44?**). Larval striped bass feed on zooplankton; juveniles eat small shrimps and other crustaceans, annelid worms, bay anchovies, gobies, and insects (**bason1971ecology--44?**). Most juveniles remain in the river and estuarine areas where they were spawned. During the first summer, young-of-the-year striped bass generally move downstream and shoreward (**fay1983species--44?**). A variety of environmental factors influence the timing of this migration, including temperature, pH, and salinity (**nefsc2019saw66--44?**). Males reach maturity at age 2-4 and females from age 4-8. Striped bass have a fairly long lifespan, up to 30 years (**nefsc2019saw66--44?**). Adults are generalists, consuming a variety of fish and invertebrates, depending on fish size, time of year, and habitat (**nefsc2019saw66--44?**). Adults have few predators, with the exception of some seals and sharks (**nefsc2019saw66--44?**). The Atlantic States Marine Fisheries Commission manages striped bass in coastal and estuarine waters from Maine to Virginia, and coastal areas of North Carolina under Amendment 7 of the Interstate Fishery Management Plan (**asmfc2022amendment7--44?**). Estuarine waters of North Carolina are managed by the state. The stock is currently overfished, with management measures and a rebuilding plan in place, but not subject to overfishing (**asmfc2024assessment_stripedbass--44?**).

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45 Summer flounder (*Paralichthys dentatus*)

45.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = Low

Data Quality Model Confidence Score =Life History Data Quality = 100%

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45.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	Low	Low	None
Directional Effect	Positive	Neutral	Increase

45.3 Score Drivers

45.3.1 Vulnerability:

add text here

45.3.2 Climate Exposure:

add text here

45.3.3 Biological Sensitivity:

Summary flounder received a **Low** sensitivity score. Certainty was 75% according to bootstrap analysis. The only attribute that received a score greater than 2.5 was Juvenile Habitat Requirements (2.7). Juvenile summer flounder rely on estuary habitats, including marshes, oyster reefs, and seagrass beds for important nursery habitats. Many of these are considered highly vulnerable habitats (**guyant2022northeast--45?**).

In the 2016 Climate Vulnerability Assessment, summer flounder also received a Low sensitivity score. In that assessment, no attributes scored above 2.5 (**hare2016vulnerability--45?**).

45.4 Data Quality Metrics

45.4.1 Life History Data Quality:

100% of data quality scores were greater than or equal to 2, indicating excellent data quality. No data gaps (mean attribute data quality score < 2) were identified.

45.4.2 Model Confidence:

add text here

45.5 Additional Metrics

45.5.1 Distribution Shifts:

add text here

45.5.2 Directional Effect:

Experts predict that summer flounder will be **positively** impacted by climate change. Certainty was 65%. Spatio-temporal models have suggested that summer flounder abundance will increase with increasing temperatures (**deen2025estimating--45?**). Increasing winter water temperatures may help summer flounder expand their distributions northward (**bell2014externally--45?**). In the previous Assessment, experts predicted that the impacts of climate change on summer flounder would be neutral; however, certainty was low (**hare2016vulnerability--45?**).

45.6 Climate Effects on Abundance and Distribution

Summer flounder distributions are correlated with depth, bottom temperature, sea surface temperature (SST), and bottom salinity (**deen2025estimating--45?**). Their distribution in the Northwest Atlantic may shifting north due to increasing winter temperatures, but could also be affected by reductions in fishing pressure or range expansion (**bell2014externally--45?**). Large-scale climate drivers, such as the Gulf Stream Index, may affect summer flounder recruitment and distribution (**oleary2019understanding--45?**). Spatio-temporal modeling results from (**deen2025estimating--45?**) report increased abundance of summer flounder in the fall by 10-15% with a 1°C increase in SST, 12% with a 1°C increase in bottom temperature and 3-4% increase with each meter in depth. In the spring, abundance increased by 30% with a 1°C increase in SST. Summer flounder embryos exhibit reduced survival when exposed to elevated CO^2. Larvae exposed to elevated CO^2 had tissue damage and initiated metamorphosis earlier and at smaller sizes (**chambers2014effects--45?**).

45.7 Life History Synopsis

Summer flounder are found both inshore and offshore in waters from Nova Scotia to Florida (**packer1999_sf--45?**). Spawning season lasts from September through February/March, with lower latitudes starting sooner. Spawning adults migrate to overwintering habitats on the continental shelf edge. In the late spring, they move back inshore into shallow waters and estuaries (**packer1999_sf--45?**). The timing of peak spawning in October/November coincides with the breakdown of thermal stratification on the continental shelf and the maximum production of autumn plankton which is characteristic of temperate ocean waters of the northern hemisphere (**packer1999_sf--45?**; **morse1981reproduction--45?**). Egg development takes anywhere from 48-142 hours, depending on water temperatures. At 14°C, larvae metamorphosize into juveniles within approximately 65 days of hatching. Time to metamorphosis also was dependent on temperatures, with metamorphosis completing as fast as 46.5 days at 14.5°C (**smith1973distribution--45?**). Larvae do not drift far from spawning regions; however, tidal stream transport and vertical migration behavior may be important to determining larval distributions (**packer1999_sf--45?**; **hare2006sinking--45?**). Larvae prefer waters with warmer temperatures and intermediate salinities. Temperature affects the rate of larvae yolk sac absorption (which impacts when larvae have to depend on exogenous feeding), development and growth (**watanabe1998light--45?**). Studies have also shown that larval size increases with greater light intensity (presumably affecting activity and energy metabolism) and higher salinities (**watanabe1998light--45?**; **burke1998influence--45?**). Larvae are preyed upon by a variety of predators, including blue crabs and sea robins. The type and abundance of predators (ex: *Crangon septemspinosa* or pelagic mummichugs) could determine whether a metamorphic summer flounder stays in the substrate or the water column (*C. septemspinosa* would force them to the water column, and vice versa for

mummichugs) (**keefe1994contributions--45?**). Larvae preferentially settle on sandy substrate; bottom type, presence of predators, and water temperatures may all impact settlement (**keefe1994contributions--45?**). Young-of-the-year can occur in a variety of habitats, including shallow, mud-bottomed marsh creeks, shallow sand substrates (including seagrass beds), deep sand substrate, and deep fine-sand substrates (**szedlmayer1993ultrasonic--45?**). Juveniles are opportunistic feeders and differences in diet are often related to the availability of prey. There also appears to be ontogenetic changes in diet. Smaller individuals eat polychaetes and crustaceans. Metamorphic stage individuals reflect calanoid copepod diets (pelagic), while late metamorphosis stage individual and juveniles' diets reflected shrimp, polychaetes, harpactoid copepods, amphipods, small fishes, small gastropod mollusks, and plant material (benthic) (**grover1998feeding--45?**). Adults are demersal, with highest abundance in waters deeper than 30m and at temperatures 2-27°C (**packer1999_sf--45?**). Adults are also opportunistic feeders. Fish and crustaceans make up the majority of their diet, but dominant prey items shift following prey availability (**cernadas2021composition--45?**). Adult summer flounder are preyed upon by sharks, rays, and monkfish (**packer1999_sf--45?**). Summer flounder reach maturity at approximately 13-17 months. Males can reach 12 years of age and females 14 years, although individuals living upwards of 20 years have been reported (**nefsc2019assessment_sf--45?**). The summer flounder stock is cooperatively managed by the Mid-Atlantic Fishery Management Council and Atlantic States Marine Fisheries Commission under the Summer Founder, Scup, and Black Sea Bass Fishery Management Plan. The stock is not overfished and overfishing is not occurring (**nefsc2025assessment_sf--45?**).

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46 Tautog (*Tautoga onitis*)

46.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = Moderate **Data Quality**Model Confidence Score =Life History Data Quality = 92-93%

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46.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	Moderate (All Stocks)	High	Decrease (All Stocks)
Directional Effect	Negative (DelMarVa)	Neutral	Decrease (DelMarVa)
	Neutral (LIS, NY/NJ, MA/RI)		None (LIS, NY/NJ, MA/RI)

46.3 Score Drivers

46.3.1 Vulnerability:

add text here

46.3.2 Climate Exposure:

add text here

46.3.3 Biological Sensitivity:

All four stocks of Tautog (**DelMarVa, Long Island Sound, NJ/NY, and MA/RI**) received **Moderate** sensitivity scores. Certainty for all stocks was 94%. These scores were driven by Population Growth Rate (3.5) and Juvenile Habitat Requirements (2.8) for all stocks.

Tautog are long-lived, slowly maturing species. They mature at 3-4 years of age and maximum ages range between 25-30 years (**steimle1999tautog--46?**). Juveniles prefer areas with high amounts of cover, including algal habitats, which are considered highly vulnerable (**guyant2022northeast--46?**).

In the 2016 Climate Vulnerability Assessment, Tautog received a High sensitivity score. This was driven by Adult Mobility and Population Growth Rate (**hare2016vulnerability--46?**).

46.4 Data Quality Metrics

46.4.1 Life History Data Quality:

Between 92 - 93% of data quality scores were greater than or equal to 2 across all four Tautog stocks, indicating excellent data quality. Stock Size/Status was identified as a data gap (mean attribute data quality score < 2) for all stocks.

46.4.2 Model Confidence:

add text here

46.5 Additional Metrics

46.5.1 Distribution Shifts:

add text here

46.5.2 Directional Effect:

Experts predicted different directional effects for different Tautog stocks. The DelMarVa stock were predicted to be **negatively** impacted by climate change, whereas experts predicted that climate change would **neutrally** impact the Long Island Sound, NJ/NY, and MA/RI stocks. Certainty was greatest for the MA/RI stocks (96%) followed by the Long Island Sound (88%), NJ/NY (79%) and DelMarVa stocks (57%). Temperature has been linked to recruitment, migration timing, spawning, and feeding behaviors (**steimle1999tautog--46?**; **moran2023impacts--46?**). Tautog are not tolerant of anoxic conditions (**howell1994abundance--46?**). In the previous Climate Vulnerability Assessment for the Northeast Shelf, the impacts of climate change on tautog were predicted to be neutral, with a high degree of uncertainty (**hare2016vulnerability--46?**).

46.6 Climate Effects on Abundance and Distribution

Temperature is linked to recruitment, spawning, and migration timing in tautog (**steimle1999tautog--46?**), but it is unclear how warming temperatures will affect phenology. Tautog exhibit some site fidelity, which may present a limited ability to expand or shift their range in response to warming temperatures (**langan2021climate--46?**). (**moran2023impacts--46?**) observed differences in prey type and feeding behavior in tautog exposed to varying temperatures, highlighting the need for further research on the relationship between increasing temperature and feeding behavior. (**howell1994abundance--46?**) reported that tautog are not tolerant of dissolved oxygen (DO) concentrations below 2 mg/l, and prefer DO levels above 3 mg/l. Tautog mortalities were reported by commercial fishers and were observed on the beach during a major anoxia-hypoxia episode off New Jersey in 1976.

46.7 Life History Synopsis

Tautog are a slow-growing, long-lived coastal species inhabiting the outer coast of Nova Scotia to Georgia, although they are most abundant from Cape Cod to Chesapeake Bay (**steimle1999tautog--46?**). They do not reach maturity until age 3 (males) or 4 (females) and can live up to 25-30 years (**steimle1999tautog--46?**). Tautog spawn in pairs, or in small groups with a single female and several males. The method depends on sex-ratios, the presence of a patriarchy, and availability of shelter and food (**steimle1999tautog--46?**). They spawn from April to August; occurring earlier (April) in the southern portion of the range. Peak spawning occurs in June/July in northern portions of habitat. They perform inshore/offshore migrations in order to spawn. In the north, tautog overwinter offshore and migrate inshore in the spring to spawn and feed before moving back offshore as temperatures decline (below 11°C)

in the fall and winter. Southern populations may undertake shorter seasonal migrations or remain either inshore or offshore for the entire year (**steimle1999tautog--46?**). Tautog enter a torpor state in waters between 7-10°C and prefer temperatures 10-22°C (**olla1974daily--46?**). In a laboratory setting, eggs hatch in 42-48 hours at 20-22°C with longer development times at lower temperatures (**perry1994artificial--46?**). Larval duration lasts about 20-25 days, depending on location (**sogard1992variability--46?**). Larvae consume small fish and copepods, such as *Acartia tonsa* (**simpson1989population--46?**). Larvae survival is dependent on temperatures for both incubation period and recruitment. There was no direct evidence in the literature to support reliance on specific hydrographic conditions, however, spawning locations near the mouths of nursery habitats,and possible offshore advection suggest that local current patterns may play a role in larval distribution (**steimle1999tautog--46?**). Juveniles prefer areas with high amounts of cover that they can hide in/around. However, they can utilize a variety of vertical habitats, including piers, artificial and real seagrass/algal beds, shells, and marine debris. Larger juvenile and adult tautog are closely associated with hard- surface, reef-like habitats, remaining close to their preferred home location (**sogard1992variability--46?**). Juvenile tautog forage on benthic, as well as planktonic, prey. As they grow, juvenile tautog rely less on planktonic food resources and feed primarily on benthic prey, although the prey remain primarily crustaceans, such as decapods (**dorf1994ecology--46?**). Adult tautog are durophagous and feed chiefly on blue mussels and other shellfish throughout the year (**steimle1999tautog--46?**). Seabirds, particularly cormorants, prey on juvenile tautog. Hakes, smooth and spiny dogfish, skates, and sea ravens have been reported to consume adult tautog (**steimle1999tautog--46?**). Tautog is managed by the Atlantic States Marine Fisheries Commission. The Tautog Fishery Management Plan divides the stock into four regions: Massachusetts-Rhode Island (MARI), Long Island Sound (LIS), New Jersey-New York Bight (NJ-NYB), and Delaware-Maryland-Virginia (DMV) (**asmfc2017tautog--46?**). The 2025 stock assessment update indicated tautog are not overfished in MARI, LIS, or NJ-NYB, but overfished in DMV. Overfishing is not occurring in MARI or LIS, but the NJ-NYB and DMV regions are experiencing overfishing (**asmfc2025tautog--46?**).

46.8 Figures

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47 White hake (*Urophycis tenuis*)

47.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = Moderate **Data Quality**Model Confidence Score =Life History Data Quality = 85%

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47.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	Moderate	Moderate	None
Directional Effect	Negative	Negative	None

47.3 Score Drivers

47.3.1 Vulnerability:

add text here

47.3.2 Climate Exposure:

add text here

47.3.3 Biological Sensitivity:

White hake received a **Moderate** sensitivity score, with a certainty of 59%. Four attributes had scores greater than 2.5: Stock Size (3), Juvenile Habitat Requirements (2.9), Population Growth Rate (2.8), and Spawning Cycle (2.8).

Spawning occurs in the spring months along the continental shelf near Georges Bank and the Mid Atlantic Bight (Lang et al 1996). Juveniles transition from pelagic to demersal at approximately two months, where eelgrass becomes an important habitat. These habitats are highly vulnerable (**guyant2022northeast--47?**). White hake reach maturity at 2.5-2.8 years and have lifespans of approximately 20 years (**nefsc2013saw56--47?**). The stock is under a rebuilding plan, but is not overfished nor is overfishing occurring (**nefsc2025whitehake--47?**).

White hake also received a Moderate sensitivity score in the 2016 Climate Vulnerability Assessment (**hare2016vulnerability--47?**). This was driven by Stock Status, Population Growth Rate, and Spawning Cycle, similar to our results.

47.4 Data Quality Metrics

47.4.1 Life History Data Quality:

85% of data quality scores were greater than or equal to 2, indicating strong data quality. Egg and Larval Survival and Settlement Requirements and Other Stressors were identified as data gaps (mean attribute data quality score < 2).

47.4.2 Model Confidence:

add text here

47.5 Additional Metrics

47.5.1 Distribution Shifts:

add text here

47.5.2 Directional Effect:

Experts predicted that white hake will be **negatively** impacted by climate change. Certainty was 70%. This is similar to the results of the previous Assessment (**hare2016vulnerability--47?**). White hake have a narrow temperature preference (9-13C) and as a result, distributions have been shifting northward as bottom waters warm (**lafreniere2023what--47?**). Recruitment is strongly linked to circulation patterns, which may be impacted by future change (**han2008dispersion--47?**; **ings2008episodic--47?**).

47.6 Climate Effects on Abundance and Distribution

White hake are a common groundfish species in the Gulf of Maine, which is warming faster than most of the world's oceans (**pershing2021climate--47?**). In response to warming in the Gulf of Maine, white hake show a preference for narrow temperature (9-13°C) and depth (55-100 m) ranges and display a gradual northward movement in their center of gravity (**lafreniere2023what--47?**). (**nye2009changing--47?**) observed a range contraction in white hake in the Gulf of Maine. White hake may be affected by changes in circulation patterns. Small white hake larvae have been collected in early summer in a Gulf Stream warm core ring seaward of southern Scotian Shelf and southeast of Georges Bank. Patterns of strong recruitment have been linked to the strength of the Labrador Current on the Great Banks and strong onshore transport and downwelling conditions in coastal Newfoundland (**han2008dispersion--47?**; **ings2008episodic--47?**).

47.7 Life History Synopsis

White hake are a groundfish species inhabiting deep water from Newfoundland to the Mid-Atlantic Bight (**chang1999essential--47?**). Males reach maturity at 2.5 years and females at 2.8 years; both males and females have a lifespan of around 20 years of age (**nefsc2013saw56--47?**). In the Gulf of Maine, white hake of all ages tend to migrate inshore in warmer months and disperse into deeper waters in colder months, while hake in the south move offshore to spawn in the spring; fall movements may be linked to prey movements (**langton1994selected--47?**). Based on the distribution and abundance of pelagic juveniles, as well as circulation patterns throughout the region, (**fahay1989white--47?**) suggested that the southern portion of the stock (below the Gulf of St. Lawrence) spawns in early spring (April-May) in deep waters along the continental slope, primarily off southern Georges Bank and the Mid-Atlantic Bight. White hake eggs are buoyant, and hatch in the water column in 3-7 days (**fahay1983guide--47?**). Larvae may be affected by oceanographic conditions like warm core rings, the Gulf Stream, and Labrador Current for transport (**han2008dispersion--47?**; **ings2008episodic--47?**). White hake prefer temperatures between 7-20°C (**chang1999essential--47?**). Eggs, larvae, and early juveniles are pelagic; older juveniles and adults are demersal. Most catches

occur in Canadian waters at depths between 175 to 375 m with no differences between mature and immature hake. In the Gulf of Maine, depth preferences are 55-100 m (**fahay1989white--47?**; **nozeres2015review--47?**). Juvenile white hake settle to the bottom at around 2 months of age. Pelagic juveniles are in the upper water column during the daytime. Demersal juveniles inhabit muddy and fine grained sandy bottom sediment (**chang1999essential--47?**; **fahay1989white--47?**). Demersal juveniles feed primarily on polychaetes, shrimps, and other crustaceans (**bowman1981food--47?**). Adults feed on fish, including juveniles of their own and similar species (silver hake), Atlantic herring, Gulf Stream flounder, and possibly alewife, shrimps, krill, and other crustaceans (**langton1994selected--47?**; **nozeres2015review--47?**). The white hake fishery is managed as a single stock by the New England Fisheries Management Council, under the Northeast Multispecies (Small-Mesh) Fishery Management Plan. The stock is under a rebuilding plan, but is not overfished and overfishing is not occurring (**nefsc2025whitehake--47?**).

47.8 Figures

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48 Windowpane flounder (*Scophthalmus aquosus*)

48.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = Low - Moderate

Data Quality Model Confidence Score =Life History Data Quality = 75%

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48.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	Moderate (Gulf of Maine/Georges Bank) - Low (Southern New England/Mid Atlantic)	Low	Increase (Gulf of Maine/Georges Bank) - None (Southern New England/Mid Atlantic)
Directional Effect	Neutral (Gulf of Maine/Georges Bank) - Negative (Southern New England/Mid Atlantic)	Neutral	None (Gulf of Maine/Georges Bank) - Decrease (Southern New England/Mid Atlantic)

48.3 Score Drivers

48.3.1 Vulnerability:

add text here

48.3.2 Climate Exposure:

add text here

48.3.3 Biological Sensitivity:

The **Gulf of Maine/Georges Bank** stock of Windowpane flounder received a **Moderate** sensitivity score, with a certainty of 69%. This score was driven by Stock Size (3.5) and Complexity in Reproductive Strategy (2.6).

The **Southern New England/Mid Atlantic** stock received a **Low** sensitivity score, with a certainty of 74%. No attributes received a score greater than 2.5 for this stock. Stock Size and Complexity in Reproductive Strategy both received scores of 2.4.

Spawning occurs throughout most of the year (**AbleFahay2010--48?**; **Cadrin2021--48?**), but relies on cold bottom temperature (8.5-13.5 C) (**Chang1999_windowpane--48?**). Spawning occurs in estuaries and nearshore environments; estuarine water column habitats are considered highly vulnerable (**guyant2022northeast--48?**). The stock status of the Gulf of Maine/Georges Bank stock is unknown. The southern stock is not overfished and overfishing is not occurring (**NEFSC2023a_windowpane--48?**; **NEFSC2023b_windowpane--48?**).

In the previous Climate Vulnerability Assessment, Windowpane flounder received a Low sensitivity score. This was driven by Stock Status (**hare2016vulnerability--48?**).

48.4 Data Quality Metrics

48.4.1 Life History Data Quality:

75% of data quality scores were greater than or equal to 2, indicating good data quality. Across both stocks, Complexity in Reproductive Strategy, Dispersal of Early Life Stages, Egg and Larval Survival and Settlement Requirements, Population Growth Rate, and Relationship to Calcified Organisms were all identified as data gaps (mean attribute data quality score < 2).

48.4.2 Model Confidence:

add text here

48.5 Additional Metrics

48.5.1 Distribution Shifts:

add text here

48.5.2 Directional Effect:

Experts predicted that effects of climate change would be **neutral** for the Gulf of Maine/Georges Bank stock, with 73% certainty. For the Southern New England/Mid-Atlantic stock, the effects were predicted to be **negative**, with 58% certainty. Increasing temperatures have been shown to displace windowpane flounder, and extreme cold winters have increased juvenile mortality (**Bell2021--48?**; **Wilber2016--48?**). In the previous Climate Vulnerability Assessment, experts predicted that impacts of climate change on windowpane flounder would be neutral (**hare2016vulnerability--48?**).

48.6 Climate Effects on Abundance and Distribution

Percent biomass of windowpane flounder in the Northwest Atlantic increases with changes in seasonality, from cold springs, to warm springs, to fall (**Dunn2016--48?**). Increasing temperatures ($>$ 7-10°C) have been shown to displace windowpane flounder (**Bell2021--48?**). In New York/New Jersey Harbor, cold winter temperatures decreased windowpane abundance and overwinter mortality in small juveniles (**Wilber2016--48?**). Overwintering success appears to be driven in part by temperatures, with extreme cold conditions resulting in high overwinter mortality of age-0 individuals. This likely has a larger impact on fall-spawned juveniles since resilience to extreme cold temperatures is size-dependent and fall-spawned individuals have less time to grow (**Neuman2009_windowpane--48?**). Individual windowpane have also demonstrated avoiding hypoxic conditions (**Cadrin2021--48?**).

48.7 Life History Synopsis

Windowpane flounder inhabit shallow waters from the Gulf of St. Lawrence to Florida, but are most abundant from Georges Bank to Chesapeake Bay (**Chang1999_windowpane--48?**). They reach maturity around 1.5 years of age and can live up to 8 years (**Cadrin2021--48?**). Windowpane spawn from February to November on the inner continental shelf, Georges Bank, and within estuaries. Spawning occurs in the Mid-Atlantic Bight in both spring and fall, whereas spawning in Georges Bank peaks in summer (**AbleFahay2010--48?**) (**Cadrin2021--48?**). Windowpane spawn in the evening or at night on or near bottom at temperatures ranging from 6-21°C (**Chang1999_windowpane--48?**). Eggs hatch after 8 days. The duration of the larval stage is unknown, but eggs and larvae remain in the water column feeding on suspended copepods (**Chang1999_windowpane--48?**). Surveys from Canadian waters suggest that small (< 10 cm TL) windowpane consume bay shrimp and crab larvae, which may be a transition point from zooplankton prey to larger prey items (**Hanson2014--48?**). Larvae settle to the bottom at approximately 10 mm TL. However, individuals collected on Georges Bank may be planktonic up to 20 mm. There is no indication that settlement is driven by environmental factors (**Morse1995_windowpane--48?**). Juveniles that settle in shallow inshore waters move to deeper offshore waters as they grow. Juveniles inhabiting Georges Bank (< 60 m) undergo seasonal movements to deeper waters along the southern flank of the Bank occur during late autumn, as bottom temperatures drop, and overwintering occurs in deeper areas until late spring (**KleinMacPhee2002_windowpane--48?**). From Massachusetts to the Gulf of Maine, they can be found in estuaries and on the shelf (**Chang1999_windowpane--48?**). Adults are found on the benthos in waters < 100 m and tolerate a wide range of temperatures (0-26.8°C) (**Chang1999_windowpane--48?**). Adult windowpane may travel along the coast for considerable distances. Juveniles and adults may migrate to nearshore or estuarine habitats in the Mid-Atlantic Bight. Seasonal movements into offshore overwintering habitats have also been documented (**Chang1999_windowpane--48?**). Stomach content data collected during Northeast Fisheries Science Center (NEFSC) bottom trawl surveys indicate windowpane feed on small crustaceans (e.g., mysids and decapod shrimp) and various fish larvae including hakes and tomcod, as well as their own species (**Langton1981--48?**; **Chang1999_windowpane--48?**). Windowpane flounder are managed by the New England Fishery Management Council under the Northeast Multispecies (Groundfish) Fishery Management Plan. The fishery is divided into two stocks: Northern Gulf of Maine/Georges Bank and Southern New England/Mid-Atlantic. Overfishing and overfished status of the northern stock is unknown, as the assessment has not estimated reference points since 2020 (**NEFSC2023a_windowpane--48?**). The southern stock is not overfished and overfishing is not occurring (**NEFSC2023b_windowpane--48?**).

48.8 Figures

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49 Winter flounder (*Pseudopleuronectes americanus*)

49.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = Moderate - High

Data Quality Model Confidence Score =Life History Data Quality = 95 - 100%

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49.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	Moderate (Gulf of Maine/Georges Bank) - High (Southern New England/Mid Atlantic)	High	Decrease (Gulf of Maine/Georges Bank) - None (Southern New England/Mid Atlantic)
Directional Effect	Negative (All Stocks)	Negative	None

49.3 Score Drivers

49.3.1 Vulnerability:

add text here

49.3.2 Climate Exposure:

add text here

49.3.3 Biological Sensitivity:

The **Georges Bank** and **Gulf of Maine** stocks of Winter Flounder both received **Moderate** sensitivity scores. Certainty for both stocks was 58%. For these stocks, five attributes had scores greater than 2.5: Complexity in Reproductive Strategy (3.3 and 3.4), Egg and Larval Survival and Settlement Requirements (2.7 and 2.9), Other Stressors (2.6 and 2.9), Spawning Cycle (2.8), and Stock Size (3 and 2.7).

The **Southern New England/Mid-Atlantic** stock received a **High** sensitivity score, with a certainty of 91%. Three attributes had scores greater than 3: Complexity in Reproductive Strategy (3.5), Egg and Larval Survival and Settlement Requirements (3.2), and Other Stressors (3.1).

Winter flounder spawn in winter and early spring before waters reach temperatures of 6C (**pereira1999essential--49?**). Spawning occurs in nearshore and estuarine habitats, including muddy and sandy substrates, and native salt marsh habitats which are considered highly vulnerable (**guyant2022northeast--49?**). Larvae are dependent on bottom currents to help retain them in and around spawning areas and nursery habitats. Adults depend on estuarine habitats, which may be impacted by hypoxia and changes to freshwater inputs (**guyant2022northeast--49?**). Predators include striped bass, spiny dogfish, and seals, which have all experienced distribution shifts that could result in increased predation pressure (**pereira1999essential--49?**).

These results are similar to those from the 2016 Climate Vulnerability Assessment where winter flounder were given a High sensitivity score. This score was also driven by Stock Status and Early Life History Requirements (**hare2016vulnerability--49?**).

49.4 Data Quality Metrics

49.4.1 Life History Data Quality:

Between 95 and 100% of data quality scores were greater than or equal to 2 for both stocks of winter flounder, indicating excellent data quality. The only data gap (mean attribute data quality score < 2) identified was Stock Size/Status for the Gulf of Maine stock.

49.4.2 Model Confidence:

add text here

49.5 Additional Metrics

49.5.1 Distribution Shifts:

add text here

49.5.2 Directional Effect:

Climate change was predicted to **negatively** impact all three stocks of winter flounder. Certainties ranged from 58% - 100% across stocks, with the lowest certainty for the Gulf of Maine stock and the highest for the Southern New England/Mid-Atlantic stock. Increasing temperatures have been linked to poor spring recruitment and higher summer time mortality, in addition to shifts in migration timing (**able2014temporal--49?**; **langan2021role--49?**; **klein2017effects--49?**). The Southern New England/Mid Atlantic stock has also shifted norward (**able2014temporal--49?**; **langan2021role--49?**). The previous Assessment also predicted that winter flounder would be negatively impacted by climate change (**hare2016vulnerability--49?**).

49.6 Climate Effects on Abundance and Distribution

A number of studies describe evidence of changes in winter flounder growth, abundance, and distribution as a result of climate change. The Southern New England/Mid-Atlantic Bight stock declined sharply in the 1980s from overfishing, has not recovered, and is shifting northward (**able2014temporal--49?**; **langan2021role--49?**). Warm spring temperatures have resulted in poor recruitment (**able2014temporal--49?**), and declining productivity due to warming estuarine waters has blocked recovery efforts (**bell2014externally--49?**). Elevated water temperatures can cause faster development rates of early life history winter flounder; length of incubation time is crucial as a nearshore and estuarine species as these areas are more susceptible to anthropogenic impacts (**klein2017effects--49?**). Increasing temperatures may alter predation rates, affecting survival of larval winter flounder, as well as migration and distribution patterns (**klein2017effects--49?**). In Narragansett Bay, warmer summer temperatures were linked to increasing mortality in young-of-the-year winter flounder. Greater abundance of mature winter flounder in the lower bay during spring may also indicate a potential shift in spawning migration timing (**langan2021role--49?**).

49.7 Life History Synopsis

Winter flounder are found in estuaries and on the continental shelf from Nova Scotia to North Carolina, and are most common in the Mid-Atlantic north of Delaware Bay (**pereira1999essential--49?**). They prefer temperatures between 0-20°C and depths less than 30m inshore and 100m offshore (**pereira1999essential--49?**). The fishery is divided into 3 stocks: Gulf of Maine, Georges Bank, and Southern New England/Mid-Atlantic. They reach maturity at 2- 3 years, varying across stocks with southern populations maturing younger than northern ones, and live to a maximum age of 15 years (**nefsc2011assessment_winter--49?**). Winter flounder spawn inshore in the winter through spring, peaking in February/March in the Mid-Atlantic and Southern New England and continuing into May in the Gulf of Maine (**pereira1999essential--49?**). Timing of spawning aggregations is linked to inshore migration timing, which is complex and possibly related to time of year, temperature, lunar cycle, and photoperiod, as well as varying across years. Individuals migrate inshore in the fall, and migrate offshore in the spring. The Georges Bank population does not migrate (**pearcy1962ecology--49?**). Females deposit eggs at night and on bottom substrates, usually sand or algal mats, and can deposit multiple batches of eggs per spawning season (**ziegler2019migration--49?**). Eggs are demersal and hatch within 2-3 weeks, depending on temperature. Larvae are initially planktonic but become more demersal as they metamorphosize, which starts 5-6 weeks after hatching. Larvae rely on bottom currents to keep them in and around spawning areas in inshore bays and estuaries. They are tolerant to a wide range of temperatures, salinities, and dissolved oxygen concentrations (**pearcy1962ecology--49?**). Metamorphosis is complete about 8 weeks after hatching, and is temperature dependent (**pereira1999essential--49?**). Size, rather than age, tends to drive metamorphosis and settlement (**manderson2003winter--49?**). Larvae consume a variety of planktonic prey including nauplii, invertebrate eggs, calanoids, and polychaetes, with larger larvae consuming larger prey. Copepods and other zooplankton are important during metamorphosis (**stehlik2000diet--49?**). Small medusae are important predators to larvae. Larvae can also be preyed upon by amphipods and Atlantic mackerel (**pereira1999essential--49?**). Juveniles utilize marine and estuarine rocky, sand, and mud substrates, particularly fine sand and gravel as well as marine and estuarine water columns. They also occupy submerged aquatic vegetation, marshes, oyster reefs, and macroalgae habitats (**bigelow1953fishes--49?**). Juveniles consume a variety of prey including copepods, amphipods, and polychaetes. The diversity of prey increases and the importance of smaller zooplankton decreases with size (**pearcy1962ecology--49?**; **pereira1999essential--49?**). Adults are omnivorous opportunistic feeders. They feed during the day as they depend on sight to locate prey. Their diet is limited by mouth size; diets primarily consist of polychaetes and crustaceans (mostly amphipods), but also include bivalves, fish eggs, and fish. Diets can shift based on available prey (**stehlik2000diet--49?**; **pereira1999essential--49?**). Predators of adult winter flounder include striped bass, spiny dogfish, and seals (**pereira1999essential--49?**). The New England Fishery Management Council manages winter flounder in federal waters under the Northeast Multispecies (Groundfish) Fishery Management Plan. The Atlantic

States Marine Fisheries Commission coordinates management in state waters. None of the 3 stocks are subject to overfishing. The Georges Bank and Southern New England/Mid-Atlantic stocks are not overfished; the Gulf of Maine stock has not been assessed and population abundance is unknown (**nefsc2025gom_winter--49?**; **nefsc2025snema_winter--49?**; **nefsc2025gb_winter--49?**).

49.8 Figures

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50 Witch flounder (*Glyptocephalus cynoglossus*)

50.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = High

Data Quality Model Confidence Score =Life History Data Quality = 78%

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50.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	High	High	None
Directional Effect	Negative	Negative	None

50.3 Score Drivers

50.3.1 Vulnerability:

add text here

50.3.2 Climate Exposure:

add text here

50.3.3 Biological Sensitivity:

Witch flounder received a **High** sensitivity score, with 100% certainty. Two attributes had scores greater than 3: Stock Size (3.9) and Population Growth Rate (3.5).

Winter flounder are a long-lived, slowly maturing species, reaching maturity at 4.5-6 years and living up to 30 year (**burnett1992several--50?**; **cargnelli1999_witch--50?**). The stock status is poorly understood; the stock is overfished but it is unclear if overfishing is occurring due to lack of data (**nefsc2024witch--50?**).

Witch flounder also received a High sensitivity score in the 2016 Climate Vulnerability Assessment. This was also driven by Stock Size and Population Growth Rate (**hare2016vulnerability--50?**).

50.4 Data Quality Metrics

50.4.1 Life History Data Quality:

78% of data quality scores were greater than or equal to 2, indicating good data quality. Both Adult Mobility and Egg and Larval Survival and Settlement Requirements were identified as data gaps (mean attribute data quality score < 2).

50.4.2 Model Confidence:

add text here

50.5 Additional Metrics

50.5.1 Distribution Shifts:

add text here

50.5.2 Directional Effect:

Experts predicted that witch flounder will be **negatively** impacted by climate change, with certainty equal to 95%. This is similar, but with higher certainty, to the 2016 Climate Vulnerability Assessment, which also predicted negative impacts from climate change on witch flounder (**hare2016vulnerability--50?**). While there are few direct studies of the impacts of climate change on witch flounder, adults are known to prefer relatively cold temperatures between 4-11 C (**cargnelli1999_witch--50?**). A recent study suggests that distributions are shifting southward (**mills2024multispecies--50?**).

50.6 Climate Effects on Abundance and Distribution

There have been few direct studies on the effect of climate change on witch flounder. Larval growth and survival has been linked to temperature in laboratory conditions (**bidwell2001effect--50?**). (**murawski1993climate--50?**) included witch flounder in a list of species that exhibit low variation in seasonal temperature or depth of occurrence. (**mills2024multispecies--50?**) found that witch flounder distribution has actually shifted significantly toward more southern latitudes.

50.7 Life History Synopsis

Witch flounder are a long-lived flatfish species inhabiting deep, cold waters in the Northwest Atlantic from Labrador to North Carolina (**cargnelli1999_witch--50?**). They are slow-growing, reaching maturity at 4.5 years for males and 6 years for females in Georges Bank and the Gulf of Maine (**burnett1992several--50?**). Witch flounder can live up to 30 years, reaching a maximum length of 78 cm (**cargnelli1999_witch--50?**). Witch flounder spawn from March to November, with peak spawning occurring in summer. The general trend is for spawning to occur progressively later from south to north. In the Gulf of Maine-Georges Bank region, spawning occurs from April to November, and peaks from May to August (**martin1978development--50?**; **burnett1992several--50?**). (**powles1970depth--50?**) suggested that witch flounder occupy three largely discrete ecological niches, resulting in the different life stages being geographically separated from each other. The pelagic egg and larval stages are spent in the water column over deep water. Egg hatching occurs 7 to 8 days after spawning, with the yolk sac absorbed 10 days after hatching. Larvae are pelagic, and the larval stage lasts from 4-6 months to 1 year (**cargnelli1999_witch--50?**). Eggs generally drift southward, and shelf currents may transport some larvae. Length may be the biggest cue for settlement (**cargnelli1999_witch--50?**). The juvenile stage occurs in very deep water when metamorphosed fish settle to the bottom. They are found

over mud, clay, silt, or muddy sand substrates (**cargnelli1999_witch--50?**). Juveniles remain separated from the adult population, occupying deeper areas until the onset of sexual maturity. (**walsh1987habitat--50?**) suggests however that while the juveniles and adults are in different areas, the niche separation may not be true. Witch flounder are capable of inhabiting depths up to 1500 m, although the majority are found at 90-300 m (**martin1978development--50?**; **kleinmacphee2002witch--50?**). The main food items in the witch flounder diet are polychaetes and crustaceans, although mollusks and echinoderms are also important. Overall, polychaetes were by far the most important food item, accounting for greater than 70% of the diet (**maurer1975food--50?**). However, there is a distinct ontogenetic shift in diet, with polychaetes increasing in importance and crustaceans decreasing in importance with age (**cargnelli1999_witch--50?**). Adult witch flounder generally do not migrate, although they prefer to stay within temperatures of 4-11°C. Some subpopulations may move from shallower waters in warmer months to deeper overwintering waters. Migration was not observed in the Gulf of Maine (**cargnelli1999_witch--50?**). Predators of witch flounder include monkfish, spiny dogfish, harp seals, Atlantic halibut, white hake, and some skates (**kleinmacphee2002witch--50?**). Witch flounder are managed as a single stock unit by the New England Fishery Management Council under the Northeast Multispecies (Groundfish) Fishery Management Plan. In the last Management Track Assessment, the stock status could not be determined due to a lack of biological reference points. The stock condition appears to be improving but is poor relative to historical levels (**nefsc2024witch--50?**).

50.8 Figures

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51 Yellowtail flounder (*Limanda ferruginea*)

51.1 Summary

ScoresVulnerability =Climate Exposure =Biological Sensitivity = Low - Moderate

Data Quality Model Confidence Score =Life History Data Quality = 93-97%

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51.2 Changes from Previous Assessment(s)

Metric	Current Assessment	Previous Assessment(s)	Change
Vulnerability			
Exposure			
Sensitivity	Low (Gulf of Maine/Cape Cod) - Moderate (Southern New England/Mid Atlantic & Georges Bank)	Low	None (Gulf of Maine/Cape Cod) - Increase (Southern New England/Mid Atlantic & Georges Bank)
Directional Effect	Neutral (Gulf of Maine/Cape Cod) - Negative (Southern New England/Mid Atlantic & Georges Bank)	Negative	None (Southern New England/Mid Atlantic & Georges Bank) - Increase (Gulf of Maine/Cape Cod)

51.3 Score Drivers

51.3.1 Vulnerability:

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51.3.2 Climate Exposure:

add text here

51.3.3 Biological Sensitivity:

The **Southern New England/Mid-Atlantic** and **Georges Bank** stocks of Yellowtail flounder both received **Moderate** sensitivity scores. Both scores had 88% certainty. These scores were driven by Stock Size (4 and 3.6), Spawning Cycle (2.7 and 2.6) and Egg and Larval Survival and Settlement Requirements (2.6). Adult Mobility (2.5) also played a role in the score for the Georges Bank stock.

The **Gulf of Maine/Cape Cod** stock of Yellowtail flounder received a **Low** sensitivity score, with 56% certainty. For this stock, only the Stock Size (2.8) attribute had a score greater than 2.5.

Spawning occurs from mid-February to July on Georges Bank and in nearshore habitats in Southern New England and the Mid-Atlantic (**MartinDrewry1978--51?**). Larvae feed on zooplankton prey and zooplankton abundance has been linked to recruitment success. Although it is also worth noting that studies directly testing for mismatches did not find evidence of mismatches between larvae and zooplankton overlap and recruitment (**Kittel2025--51?**; **Johnsonetal1999--51?**). Larvae perform vertical migrations to help increase retention in and around nursery locations. Recruitment has also been correlated with circulation patterns and the Mid Atlantic Cold Pool. Both the Georges Bank and Southern New England stocks are under rebuilding plans (**NEFSC2025c_yellowtail--51?**).

In the 2016 Climate Vulnerability Assessment, yellowtail flounder also received a Low sensitivity score. Similar to the Gulf of Maine/Cape Cod stock, the only attribute that scored above 2.5 in the 2016 Assessment was Stock Status (**hare2016vulnerability--51?**).

51.4 Data Quality Metrics

51.4.1 Life History Data Quality:

Between 93 - 97% of data quality scores were greater than or equal to 2 across both stocks, indicating excellent data quality. Complexity in Reproductive Strategy was identified as a data gap (mean attribute data quality score < 2) for the Southern New England/Mid-Atlantic stock.

51.4.2 Model Confidence:

add text here

51.5 Additional Metrics

51.5.1 Distribution Shifts:

add text here

51.5.2 Directional Effect:

Experts predicted that the effects of climate change on Yellowtail flounder would be **negative** for both the Georges Bank and Southern New England/Mid-Atlantic stocks. Certainty for both stocks was high at 98% and 100% respectively. In the 2016 Climate Vulnerability Assessment, experts also predicted that yellowtail would be negatively impacted by climate change (**hare2016vulnerability--51?**). For the Gulf of Maine/Cape Cod stock, the effect was predicted to be **neutral**, but with lower certainty (86%). Increased temperatures have been linked to decreased productivity, increased larval mortality, and reduced settlement rates (**Kitteletal2026--51?**; **Kleinetal2017--51?**). Recruitment has also been linked to the Mid-Atlantic Cold Pool (**Kittel2025--51?**).

51.6 Climate Effects on Abundance and Distribution

Yellowtail flounder abundance, distribution, and recruitment have been associated with a number of environmental and climate indices. Rapid warming of the Northwest Atlantic has resulted in decreased productivity in yellow flounder (**Kitteletal2026--51?**). Higher temperatures can increase mortality in yellowtail flounder larvae, and faster rates of warming causing earlier stratification may reduce settlement rates in juveniles (**Kleinetal2017--51?**).

Yellowtail recruitment in Southern New England is highly correlated with the North Atlantic Oscillation (NAO) and air temperature (**Sullivanetal2005--51?**). Low recruitment since 1990 was associated with unfavorable environmental conditions caused by a northward shift in the Gulf Stream (**Xu2016--51?**; **Kittel2025--51?**). The Mid-Atlantic Cold Pool, bottom temperature, and the Atlantic Multidecadal Oscillation (AMO) are also linked to recruitment (**Kittel2025--51?**). As warming continues, joint management between the United States and Canada may be more challenging as distributions change (**Sumailaetal2020--51?**).

51.7 Life History Synopsis

Yellowtail flounder are found in the Northwest Atlantic from Newfoundland to Chesapeake Bay, inhabiting sandy bottom waters between 130-230 m depth (**Johnsonetal1999--51?**) and at temperatures of -1 to 18°C (**OverholtzCadrin1998--51?**). Previous research has suggested that fish live up to 17 years, possibly as high as 20 years, although fish older than 7 years is unusual (**NEFSC2012_yellowtail--51?**). The fishery is divided into 3 stocks: Gulf of Maine/Cape Cod, Georges Bank, and Southern New England/Mid-Atlantic (**NEFSC2025a_yellowtail--51?**; **NEFSC2025b_yellowtail--51?**; **NEFSC2025c_yellowtail--51?**). Most females are mature by ages 2-3 (**NEFSC2012_yellowtail--51?**). Spawning occurs from mid-February to July along the continental shelf waters of Georges Bank, northwest of Cape Cod, Southern New England and nearshore along New Jersey and southern Long Island (**MartinDrewry1978--51?**). Spawning peaks from April to June in southern New England and late June to July in Canada (**BerrienSibunka1999--51?**). Laboratory studies show eggs begin to hatch 6-7 days after fertilization (**BenoitPepin1999--51?**). At sea hatching times range from 14.5 days at 4 C to 4.5 days at 14 C (**WalshMorgan2004--51?**). Eggs are pelagic (most at 10-90m depth), with movement limited to currents in spawning locations (**Smithetal1975--51?**). Larvae move from the surface down to about 20 m, with larger larvae making longer migrations. This habit may assist in the dispersal of larvae because circulation at the surface transports the larvae further and is likely associated with feeding behavior (**Smithetal1978--51?**). Larvae feed on zooplankton, particularly *Calanus* (**Kittel2025--51?**). Once larvae reach metamorphosis size (11.6-16 mm) bottom temperature (4-8°C) and depth (41-70m) are the most important parameters for determining settlement (**ColletteKleinMacPhee2002--51?**). Juveniles utilize marine and estuarine sandy and muddy substrates, as well as estuarine and marine water columns (**ColletteKleinMacPhee2002--51?**). Rotifers, copepods, and brine shrimp nauplii were fed to juveniles in laboratory studies (**BenoitPepin1999--51?**). *Calanus* copepods are considered a critical food source for larvae/juveniles (**Purchaseetal2000--51?**). Amphipods (41% by weight) and polychaetes (38.5%) dominate the diet of adult yellowtail, but they also can consume cnidarians, small crabs and shrimps, mysids, cumaceans, isopods, univalve and bivalve mollusks, echinoderms, and sometimes small fish (**Bowmanetal1987--51?**; **Bowmanetal2000--51?**). Similar to other groundfish, yellowtail may be impacted by increasing grey seal and dogfish populations and resulting increased predation pressures

(**Kittel2025--51?**). Other predators of juvenile and adult yellowtail include spiny dogfish, skates, cod, hakes, and monkfish (**Johnsonetal1999--51?**). Minimal migration patterns show that Southern New England yellowtail traveled eastward during spring and summer and westward in the fall and winter as a result of seasonal changes in temperature (**Smithetal1975--51?**). Recent tagging studies, however, show that individuals generally don't move between stocks (**Goetheletal2015--51?**). Yellowtail flounder are managed by the New England Fishery Management Council through the Northeast Multispecies (Groundfish) Fishery Management Plan. The Georges Bank stock is jointly managed by the Council and Canada, as the stock extends beyond the international boundary. The Cape Cod/Gulf of Maine stock is not overfished and not subject to overfishing (**NEFSC2025a_yellowtail--51?**). The Georges Bank stock is overfished but under a rebuilding plan; overfishing status is unknown (**NEFSC2025b_yellowtail--51?**). The Southern New England/Mid-Atlantic stock is overfished but under a rebuilding plan, but not subject to overfishing (**NEFSC2025c_yellowtail--51?**).

51.8 Figures

51.9 References

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- Maine-New Hampshire Inshore Trawl Survey
- Massachusetts Division of Marine Fisheries Bottom Trawl Survey
- Long Island Sound Trawl Survey
- New York Nearshore Trawl Survey
- New Jersey Ocean Stock Assessment Survey
- Delaware Trawl Surveys
- [National Observer Program](https://www.fisheries.noaa.gov/national/fisheries-observers/national-observer-program
- Gillnet Observer Program
- Pelagic Observer Program
- Southeast Fisheries Science Center Logbook Program
- National Marine Fisheries Services' Large Pelagics Survey

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